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MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE AND ROBOTICS

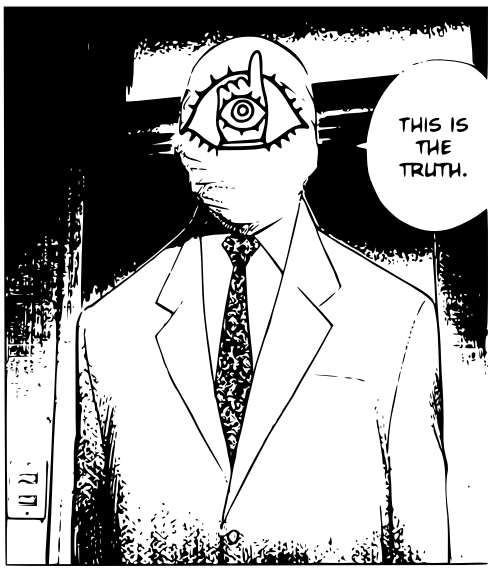
Autonomous And Mobile Robotics

Lecture notes

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This document summarizes and presents the topics for the Autonomous And Mobile Roboticscourse for the Master's degree in Artificial Intelligence and Robotics at Sapienza University of Rome. The document is free for any use. If the reader notices any typos, they are kindly requested to report them to the author.



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CHAPTER

1

WHEELED MOBILE ROBOTS

1.1 Configuration Space

We can define a robot as a collection of n_b bodies moving in an operative space, denoted W and called the **workspace**. W can be $\mathbb{R}^2$ (if the robot moves on a plane, like a planar 2R) as well as $\mathbb{R}^3$, if the robot moves in the space. For example, an elbow-type spatial manipulator is a system of $n_b = 3$ rigid bodies moving in $W = \mathbb{R}^3$. A quadrotor is a single body ($n_b = 1$) moving in the same workspace $W = \mathbb{R}^3$.

To describe the complete pose of a planar 2R manipulator, we might use the position vector and roll-pitch-yaw angles for the two bodies, which requires a total of 12 parameters for the whole system. Despite this, we know that 2 variables are enough to describe the whole configuration of the system.

Definition 1.1.1 : Configuration

The configuration of a robot is a **minimum** set of parameters that allows one to specify the posture[a] of the robot in the workspace.

[a] by posture we mean the position of all the points of the robot.

The configuration of a robot can be defined in several ways, and a choice of a set of parameters is not unique; however, each robot has a *minimum number* of parameters. We usually denote $\mathbf{q}$ the configuration vector:

$$\mathbf{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \tag{1.1}$$

in our context, it usually describes the Cartesian coordinates of a single body robot, or the angular coordinates of the body parts. The Cartesian coordinates have values in a Euclidean space, as $\mathbb{R}^2$ or $\mathbb{R}^3$, while the angular coordinates take values in a non-Euclidean space, such as the special orthogonal group $SO(3)$, the group of all the orthogonal 3×3 matrices with determinant $+1$, this group describes the orientation of a body in a 3D Euclidean space.

Definition 1.1.2 : Configuration Space

The configuration space, usually denoted $\mathcal{C}$, is the set of all possible values of the configuration of a robot. The *dimension* of $\mathcal{C}$ is the number n of parameters (generalized coordinates) in $\mathbf{q}$.

Let's see an example: consider a disk robot that moves on a plane, that can also rotate about its vertical axis to point towards a desired direction, shown in Figure 1.1.

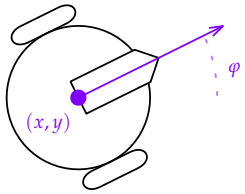


Figure 1.1: Disk robot with a tick in $\mathbb{R}^2$.

Its configuration is given by two Cartesian coordinates (x, y) and a value $\varphi \in S^1$ representing the angle between the tick and the horizontal axis. Hence, its configuration space is:

$$\mathcal{C} = \mathbb{R}^2 \times SO(2) = SE(2) \tag{1.2}$$

and it's not Euclidean, despite being called *special Euclidean group*. A polyhedral robot that can freely move in space is described by three Cartesian coordinates (x, y, z) and three roll-pitch-yaw angles (α, β, γ) describing its orientation, hence the configuration space is $SE(3)$:

$$\mathcal{C} = \mathbb{R}^3 \times SO(3) = SE(3). \tag{1.3}$$

We have seen in previous courses of robotics that the number of generalized coordinates of an n-link serial manipulator (with single-degree-of-freedom joints) is exactly n.

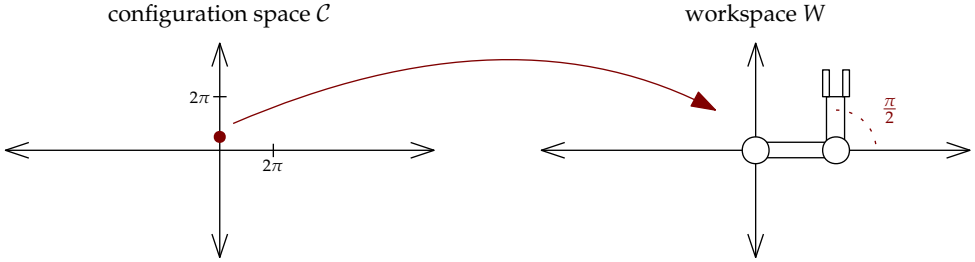
1.1.1 Topology of a Space

We are interested in describing the configuration space of a given manipulator, since it's not Euclidean (in most cases), we would like to have an idea of how to navigate this space. Before defining what a *manifold* is, we give some examples.

A 2R planar manipulator can be described by two angles, hence two values in $SO(2)$, so its configuration space is

$$\mathcal{C} = SO(2) \times SO(2) = SO(2)^2 \tag{1.4}$$

Let's try to represent graphically $\mathcal{C}$. We might represent a configuration as a point in the Cartesian plane:



The problem of this representation is that different points of the plane (angles differing by multiples of 2π) represent the same configuration $\mathbf{q} \in \mathcal{C}$, and hence map to the same posture of the robot in W, as shown in Figure 1.2.

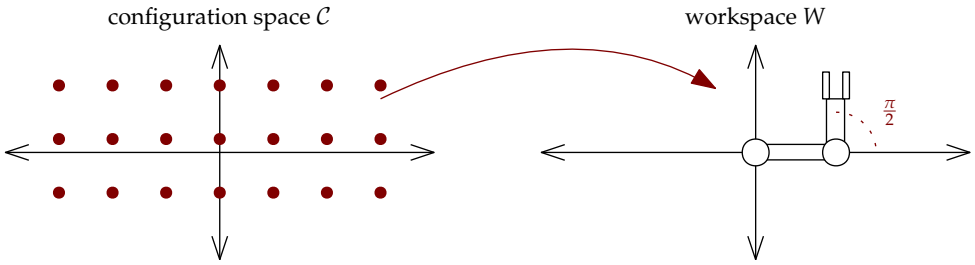
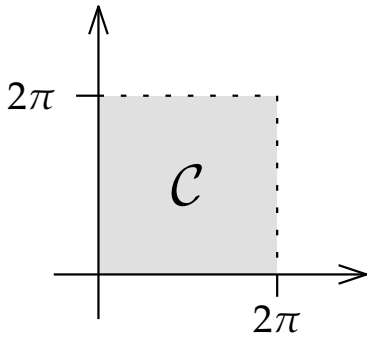
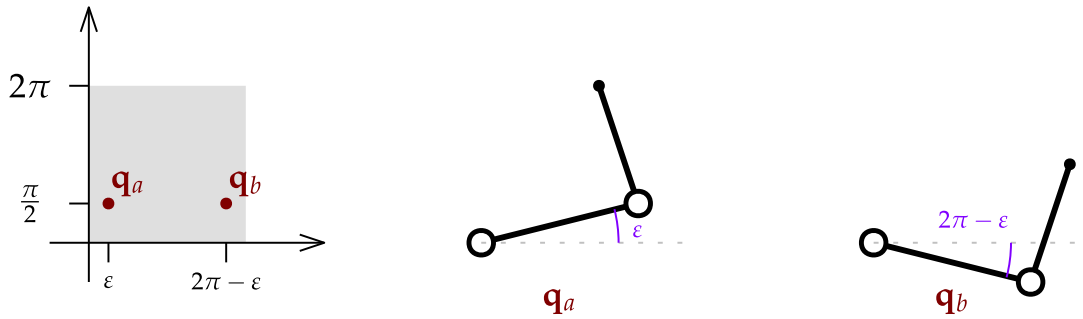


Figure 1.2: Different points of the plane map to the same posture.

To overcome this problem, we might try to represent the configuration space as a square in the plane $[0, 2\pi) \times [0, 2\pi)$:



This representation also presents a problem: the Euclidean distance between two points of the square does not represent the true distance between two postures in the workspace. Two points $\mathbf{q}_a$ and $\mathbf{q}_b$ that are very distant in the square can be mapped to two very close postures in W (for example, angles close to 0 and angles close to 2π):



This representation is said to be **not topologically correct**. To give a correct representation of $\mathcal{C}$ we need to introduce the concept of **manifold**. We will not give a formal and complete definition (as given in topology or differential geometry courses).

Definition 1.1.3 : Manifold (loose definition)

A **manifold** is a topological space that is locally homeomorphic to $\mathbb{R}^n$.

As we should know from previous robotics courses, the configuration space of a 2R is a torus, a manifold that is locally homeomorphic to $\mathbb{R}^2$, and can be visualized in the 3D space through an embedding as shown in Figure 1.3.

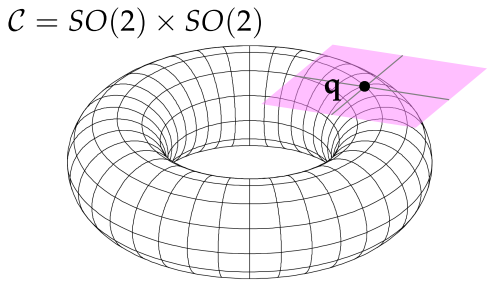


Figure 1.3: 3D representation of a torus.

1.1.2 Distance in the Configuration Space

Definition 1.1.4 : Distance in the Configuration Space

Given a configuration space $\mathcal{C}$, the **distance** is a function $d : \mathcal{C} \times \mathcal{C} \to \mathbb{R}$ such that, for each $\mathbf{q}_a, \mathbf{q}_b, \mathbf{q}_c$ in $\mathcal{C}$ we have:

$$d(\mathbf{q}_a, \mathbf{q}_b) \geq 0 \tag{1.5}$$
$$d(\mathbf{q}_a, \mathbf{q}_b) = 0 \iff \mathbf{q}_a = \mathbf{q}_b \tag{1.6}$$
$$d(\mathbf{q}_a, \mathbf{q}_b) = d(\mathbf{q}_b, \mathbf{q}_a) \tag{1.7}$$
$$d(\mathbf{q}_a, \mathbf{q}_b) \leq d(\mathbf{q}_a, \mathbf{q}_c) + d(\mathbf{q}_c, \mathbf{q}_b). \tag{1.8}$$

We know that the distance between two vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ is given by $\|\mathbf{u} - \mathbf{v}\|$. The distance between two points is the length of the *shortest path* between them, for a general manifold, this path is not always a straight line as for the Euclidean space.

Definition 1.1.5 : Geodesic

For a general manifold, the shortest path between two points is called **geodesic**, and the distance between the points is the length of their geodesic.

For example, consider the shortest path between two points on the surface of a sphere (a 2-manifold): the geodesic lies on the great circle (the circle with the biggest radius that passes through the two points), as shown in Figure 1.4. In general, for an arbitrary manifold, it is hard to determine the geodesic between two points.

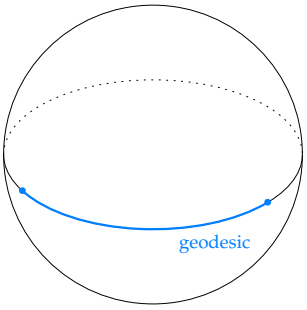


Figure 1.4: Geodesic on a sphere.

In the robotics context, we want to define the distance between two postures of a robot, this distance is defined in the workspace W, and to define that we don't need to specify directly the distance between their respective configurations in $\mathcal{C}$ (it would be harder to find the geodesic on $\mathcal{C}$).

Given a robot and its current configuration $\mathbf{q}$, we can define two functions:

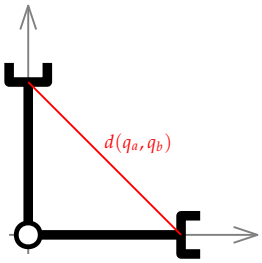
- $B(\mathbf{q})$ is a function that assigns to $\mathbf{q}$ the volume (exact region) of the workspace that the robot occupies when it is in the configuration $\mathbf{q}$, so it assigns to each configuration $\mathbf{q} \in \mathcal{C}$ a set of points in $\mathbb{R}^3$.
- $\mathbf{p}(\mathbf{q})$ is a function that assigns to $\mathbf{q}$ the exact position of a certain point of the robot when it's in the configuration $\mathbf{q}$.

For example, we might let $\mathbf{p}$ be the center of mass of the manipulator, and observe how it changes by varying $\mathbf{q} \in \mathcal{C}$. One simple way to define a distance metric is to consider the maximum Euclidean distance between two points of the manipulator in two different configurations:

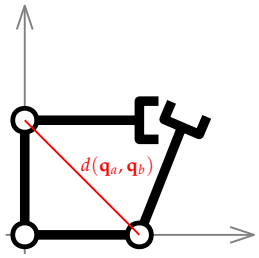
$$d(\mathbf{q}_a, \mathbf{q}_b) = \max_{\mathbf{p} \in \text{robot}} \|\mathbf{p}(\mathbf{q}_a) - \mathbf{p}(\mathbf{q}_b)\| \tag{1.9}$$

This distance is called **displacement metric**. Notice how in the definition we are writing $\mathbf{p} \in$ robot, where robot denotes the set of all the (material) points of the structure of the manipulator, whose positions clearly depend on the configuration. This kind of metric works under the assumption that, if the region occupied by the robot is similar in two different postures, then the distance between their respective configurations is small:

$$d(\mathbf{q}_a, \mathbf{q}_b) \simeq 0 \implies B(\mathbf{q}_a) \simeq B(\mathbf{q}_b). \tag{1.10}$$



1R manipulator



2R manipulator

To compute the distance in Equation (1.9), we should check each point of the robot, and in general this is an uncountably infinite set. If the links are polyhedral shapes, the problem becomes finite: for a rigid link, the difference $\mathbf{p}(\mathbf{q}_a) - \mathbf{p}(\mathbf{q}_b)$ is an affine function of the position of the point in the link frame, hence its norm is a convex function and its maximum over a polyhedron is attained at one of its (finitely many) vertices.

A more computationally feasible approach is to check only for a finite set of points in the robot structure, called **control points**. Let

$$P = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_c\} \tag{1.11}$$

be a set of c points of the robot (in general, they all depend on the configuration $\mathbf{q}$). We compute the distance between two postures in the following way:

$$d(\mathbf{q}_a, \mathbf{q}_b) = \max_{1 \le i \le c} \|\mathbf{p}_i(\mathbf{q}_a) - \mathbf{p}_i(\mathbf{q}_b)\|. \tag{1.12}$$

Not all the points of a robot are feasible control points, for example, the base of a manipulator usually doesn't change with the configuration. P is a set of *proper control points* if the function d as defined in Equation (1.12) satisfies the properties of a distance given in Definition 1.1.4.

Observation 1.1.1 : Approximation for Local Motion

Since a smooth manifold is locally diffeomorphic to $\mathbb{R}^n$, for sufficiently close configurations we can approximate the distance on the manifold (assuming it is smooth) with the Euclidean distance.

Observation 1.1.2 : Prismatic Joints

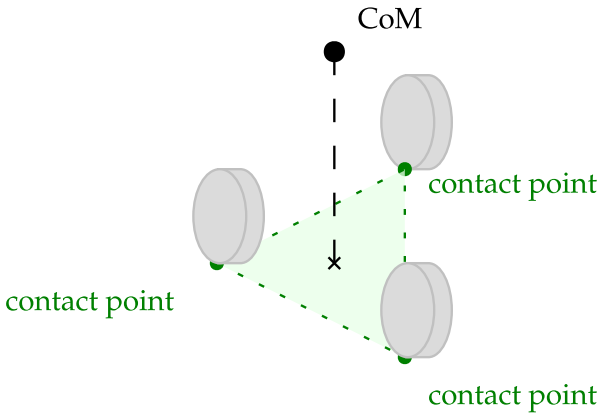
The configuration space of a robot with *only* prismatic joints is Euclidean.

1.2 Mechanics of Mobile Robots

The main sources of movement for a mobile robot are:

- Wheels: wheeled robots usually consist of one single rigid body, with the wheels in contact with the ground.
- Feet: legged robots, several rigid bodies articulated through joints.

The wheels are in contact with the ground, and this introduces some constraints. Each wheel defines a point of contact with the ground, we call the **support polygon** the convex hull of all the contact points. A wheeled robot is in **static balance** if the projection on the ground of its center of mass falls inside the support polygon. Of course a robot must be balanced in order not to fall to the ground.



The convex hull of two points is a line segment, hence a robot with only two wheels can't be easily balanced in a static fashion.

There are three main types of wheels:

- **Fixed Wheels**: They have a fixed orientation: Can roll but can't steer about their vertical axis (see Figure 1.5). They are usually actuated, if passive, they are used for balance and offer a contact point.
- **Steerable Wheels**: They can steer and are used for orientation, hence they are typically active (e.g. the front wheels of a car).
- **Caster Wheels**: They have a variable orientation but they are typically not actuated. Their axes of orientation are not aligned (the steering axis does not pass through the contact point), this property makes the wheels align automatically with the direction of motion, hence they are "invisible" kinematically, but offer a contact point.

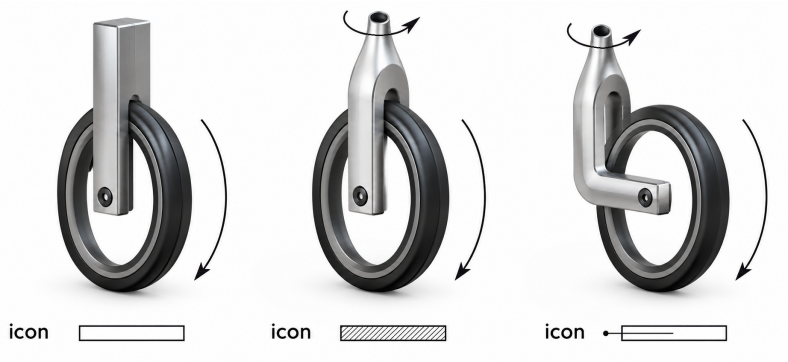
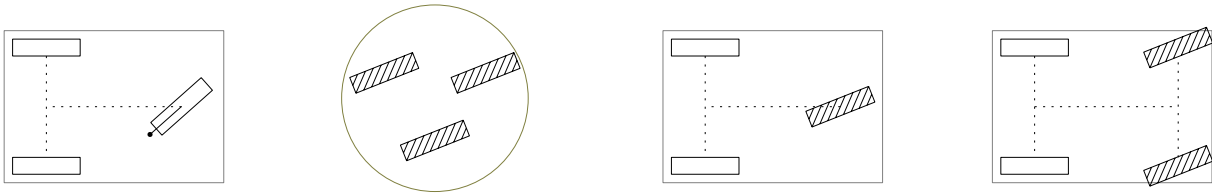


Figure 1.5: Fixed, steerable and caster wheels.

1.2.1 Main Kinematic Structures

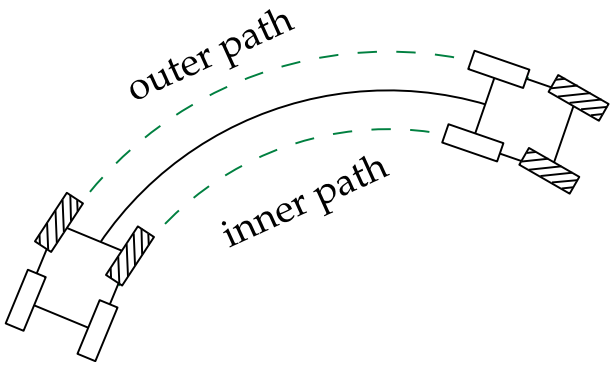
The simplest wheeled robot that we can have is the **differential drive** robot: it uses two aligned fixed wheels that are independently actuated, and one caster wheel added for balance. The **synchro-drive** mobile robots have three steerable wheels and two actuators: one controls simultaneously the orientation of all the wheels, and the other rolls them together. The base over the wheels hence will never change orientation. This type of configuration is preferable when the body to move is particularly heavy.



differential synchro-drive tricycle car-like

The tricycle has two fixed wheels in the back and one actuated steerable wheel in the front, the car-like is similar, but with two steerable wheels.

Consider the following path taken by a car-like robot:



In the same time interval, the wheels on the outer path travel a longer distance with respect to the wheels on the inner path, hence their rotation speed must be higher. In order to avoid using two distinct actuators for the two driving wheels we use a device called **differential**.

Definition 1.2.1 : Differential

A mechanical differential is a gear assembly (usually using bevel or planetary gears) connected to a single drive motor. It splits power to two wheels on the same axle while allowing them to rotate at different speeds. When a robot with a single motor turns, the outside wheel must travel a larger arc than the inside wheel. The differential lets the wheels spin at their own natural speeds, preventing tire slip, scrubbing, and mechanical strain on the drivetrain.

1.3 Constraints and Mobility

The presence of the wheels in contact with the ground introduces a series of *geometric* and *kinematic* constraints. We will also see the relationship between these two types of constraints and the implications in mobility.

1.3.1 Geometric Constraints

We have seen how the configuration $\mathbf{q}$ is in general a point of a manifold $\mathcal{C}$, that is locally homeomorphic to $\mathbb{R}^n$, with n being the dimensionality of the configuration space $\mathcal{C}$. We will focus on equality time-invariant constraints. All the geometric constraints are expressed in the following form:

$$h(\mathbf{q}) = 0 \tag{1.13}$$

and they depend only on the configuration. Usually we have $k < n$ geometric constraints, and they are reflected in the degrees of freedom of a robot:

$$h_1(\mathbf{q}) = 0 \tag{1.14}$$

$$\vdots \tag{1.15}$$

$$h_k(\mathbf{q}) = 0. \tag{1.16}$$

If the configuration space is a 2-manifold, one single geometric constraint will reduce the degrees of freedom of the robot, constraining it to move on a 1-dimensional manifold (a curve).

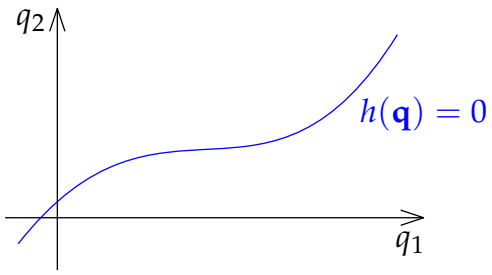


Figure 1.6: If the configuration space is $\mathbb{R}^2$, one constraint will let the configurations lie on a curve in the plane.

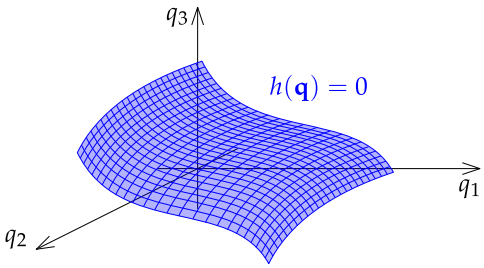


Figure 1.7: If the configuration space is $\mathbb{R}^3$, one constraint will let the configurations lie on a surface in the space.

In general, a robot with a configuration space that is an n-manifold with k independent geometric constraints, will move on an $(n-k)$-manifold. Instead of using the vector $\mathbf{q}$, we can reduce the number of coordinates in many ways, like for one constraint:

$$h(\mathbf{q}) = 0 \tag{1.17}$$

we can solve the equation for a coordinate q_i:

$$q_i = \tilde{h}(q_1, \ldots, q_{i-1}, q_{i+1}, \ldots, q_n). \tag{1.18}$$

For example, consider a 2 degrees of freedom robot with the following constraint:

$$q_1^2 + q_2^2 = 1 \tag{1.19}$$

the configuration space is limited to the unit circle in $\mathbb{R}^2$ (1-manifold), we solve the constraint for q_2:

$$q_2 = \pm\sqrt{1 - q_1^2}. \tag{1.20}$$

Actually this is not a good way to choose a coordinate, since the $\pm$ term prevents us from assigning a single value of q_2 to each value of q_1, requiring a local resolution. For this example of the circle it is better to use the arc length $s \in S^1$ as the single coordinate.

Observation 1.3.1 : Sub-Manifold

If the configuration space $\mathcal{C}$ is an n-manifold, the k independent constraints define a $(n-k)$-manifold embedded in $\mathcal{C}$, that we also call sub-manifold.

1.3.2 Kinematic Constraints

A kinematic constraint is local, it prevents the configuration from moving towards some directions. A general constraint of this kind is in the form:

$$a(\mathbf{q}, \dot{\mathbf{q}}) = 0 \tag{1.21}$$

it depends on the state of the robot. In every given configuration $\mathbf{q}^*$, we can immediately infer from $a(\mathbf{q}^*, \dot{\mathbf{q}}) = 0$ which velocities are allowed. All the relevant kinematic constraints in the robotic domain are in **Pfaffian form**:

$$\mathbf{a}^\top(\mathbf{q})\dot{\mathbf{q}} = 0 \tag{1.22}$$

with $\mathbf{a}$ being an n dimensional vector that depends on the configuration. The Pfaffian form expresses a condition of orthogonality: in a given configuration the velocity must be orthogonal to the vector $\mathbf{a}$. More kinematic constraints can be arranged in matrix form:

$$\begin{pmatrix} \mathbf{a}_1^\top(\mathbf{q}) \\ \mathbf{a}_2^\top(\mathbf{q}) \\ \vdots \\ \mathbf{a}_k^\top(\mathbf{q}) \end{pmatrix} \dot{\mathbf{q}} = \mathbf{0} \implies A(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0} \iff \dot{\mathbf{q}} \in \mathcal{N}(A). \tag{1.23}$$

The admissible velocities are the ones that lie in the null space of A[1], hence they lie in a linear space.

Observation 1.3.2 : Relationship Between Geometric and Kinematic Constraints

Any geometric constraint implies a kinematic constraint.

Consider a geometric constraint

$$h(\mathbf{q}) = 0 \tag{1.24}$$

this constraint must be satisfied as the time goes on, hence its derivative should also satisfy the following relation:

$$\dot{h}(\mathbf{q}) = 0 \tag{1.25}$$

by applying the chain rule:

$$\dot{h}(\mathbf{q}) = \frac{\partial h}{\partial \mathbf{q}}\dot{\mathbf{q}} = 0 \tag{1.26}$$

$\frac{\partial h}{\partial \mathbf{q}}$ is a row vector, the transpose of the gradient of h:

$$(\nabla_{\mathbf{q}} h)^\top \dot{\mathbf{q}} = 0 \tag{1.27}$$

this is exactly a kinematic constraint in Pfaffian form. The gradient of h evaluated on a point is the normal vector, i.e. orthogonal to the manifold on which the configuration is constrained to move, hence the resulting kinematic constraint is preventing the robot from going out of the sub-manifold defined by the geometric constraints.

[1] Sometimes, we leave the dependencies of a term implicit.

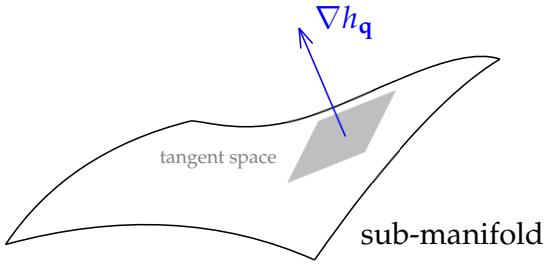


Figure 1.8: The admissible velocities are the ones that lie on the tangent space of the sub-manifold in the given configuration.

We have seen how a geometric constraint always implies a kinematic one. The opposite is not generally true, consider a generic kinematic constraint:

$$\mathbf{a}^\top(\mathbf{q})\dot{\mathbf{q}} = 0 \tag{1.28}$$

this implies a geometric constraint if there exists a function $h(\mathbf{q})$ such that

$$\nabla_{\mathbf{q}} h = \mathbf{a} \tag{1.29}$$

so that $\mathbf{a}^\top(\mathbf{q})\dot{\mathbf{q}} = \dot{h} = 0$ can be integrated to $h(\mathbf{q}) = c$; hence the constraint defined by $\mathbf{a}(\mathbf{q})$ should be *integrable*.

Definition 1.3.1 : Holonomic Constraint

An integrable kinematic constraint is called **holonomic**, and it defines a geometric constraint.

What conditions determine if a constraint is integrable or not? In general, the conditions are stated in the *Frobenius Theorem*, we will not discuss this result, which belongs to the domain of differential geometry. We will determine if a given constraint is holonomic or not in a different way.

Example: Consider the following kinematic constraint for a 2-dimensional robot:

$$q_1\dot{q}_1 + q_2\dot{q}_2 = 0 \tag{1.30}$$

is this integrable? The answer is yes. The resulting geometric constraint is

$$q_1^2 + q_2^2 = c, \quad c > 0. \tag{1.31}$$

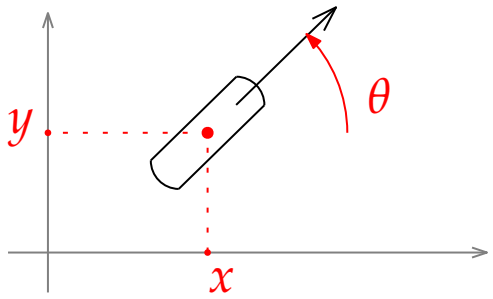
In general, in Equation (1.29) we can add a scalar integrating factor $\gamma(\mathbf{q}) \neq 0$ that depends on the configuration (the constraint is integrable if the Pfaffian form $\mathbf{a}^\top(\mathbf{q})\dot{\mathbf{q}}$ becomes an exact differential after the multiplication by γ):

$$\frac{\partial h}{\partial \mathbf{q}} = \gamma(\mathbf{q})\mathbf{a}^\top(\mathbf{q}) \tag{1.32}$$

this is allowed because the constraint $\mathbf{a}^\top\dot{\mathbf{q}} = 0$ is equal to zero, hence multiplying it by a non-zero factor γ does not change it. In the previous example, $\frac{\partial h}{\partial \mathbf{q}} = 2\begin{pmatrix} q_1 & q_2 \end{pmatrix} = \gamma\,\mathbf{a}^\top$ with $\gamma = 2$.

1.3.3 Non-Holonomic Constraints

A non-holonomic constraint defines a local limitation, it stops the robot from moving towards a certain direction when it is in that specific configuration, it does not prevent the robot from reaching a configuration by following an allowed path. The simplest example of a system with a non-holonomic constraint is a rolling disk.



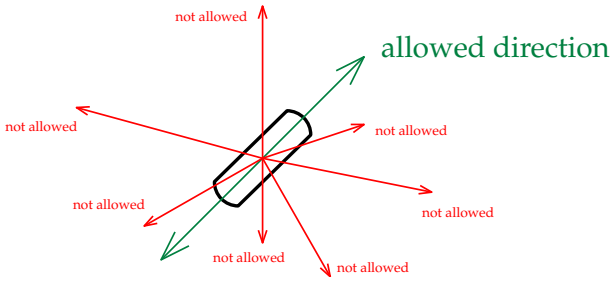
The configuration is given by the vector $\begin{pmatrix} x & y & \theta \end{pmatrix}^\top$, θ is its orientation with respect to the horizontal axis. It is clear that the disk in pure rolling (no slippage) can move *only* in the direction it is pointing (by rolling), hence can't move laterally. This constraint is expressed as follows:

$$\sin\theta\dot{x} - \cos\theta\dot{y} = 0 \tag{1.33}$$

or in vector form:

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 \end{pmatrix} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = 0 \tag{1.34}$$

$\mathbf{n}^\top = \begin{pmatrix} \sin\theta & -\cos\theta \end{pmatrix}$ is the vector *normal* to the direction. Any velocity that has a component on the line spanned by $\mathbf{n}$ is not allowed.



This constraint does not imply any global limitation, any configuration is still reachable, hence it is a non-holonomic constraint. To prove this, instead of using the Frobenius theorem, we give a non-formal proof by showing that, for any pair of configurations $\mathbf{q}_s$, $\mathbf{q}_g$, there exists a trajectory in the configuration space that connects these two configurations without violating the kinematic constraint. Let

$$\mathbf{q}_s = \begin{pmatrix} x_s \\ y_s \\ \theta_s \end{pmatrix}, \quad \mathbf{q}_g = \begin{pmatrix} x_g \\ y_g \\ \theta_g \end{pmatrix}. \tag{1.35}$$

1. $\dot{\theta}$ is not constrained, first of all we move the orientation from θ_s to θ^*, where θ^* is defined as the angle such that
$$\begin{pmatrix} \sin\theta^* & -\cos\theta^* \end{pmatrix} \begin{pmatrix} x_g - x_s \\ y_g - y_s \end{pmatrix} = 0. \tag{1.36}$$
2. Then, we can move on the direction $\begin{pmatrix} x_g - x_s & y_g - y_s \end{pmatrix}^\top$, in this way we can reach the desired position $\begin{pmatrix} x_g & y_g \end{pmatrix}^\top$.
3. At the end, we move the orientation from θ^* to θ_g.

This is called **universal maneuver**.

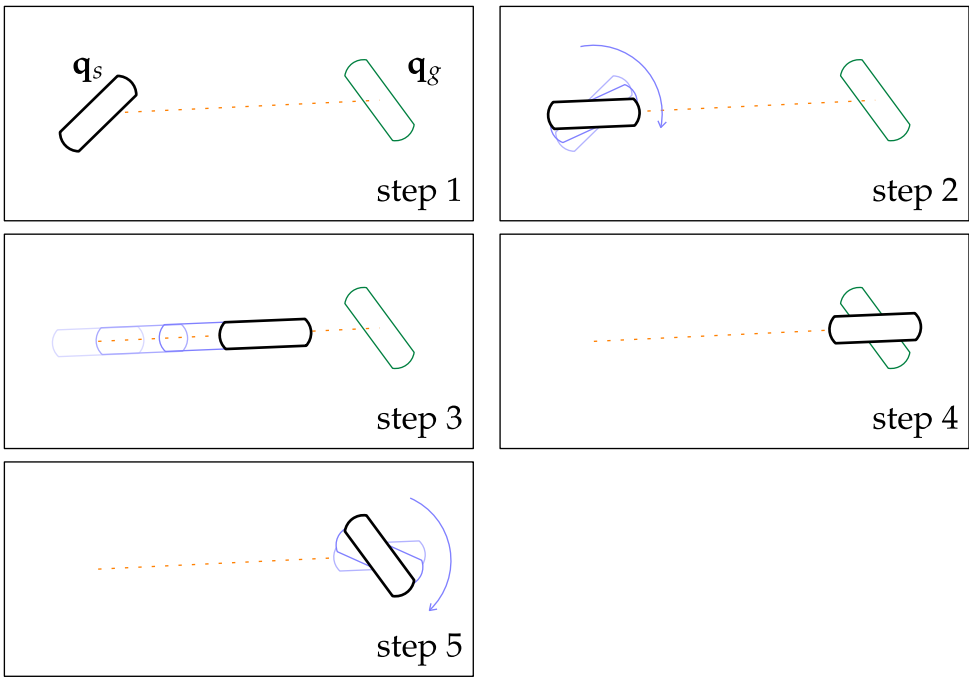
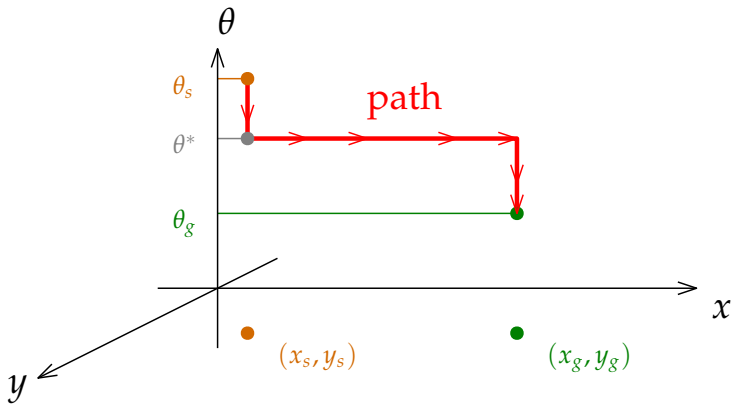


Figure 1.9: The universal maneuver in five steps.

There are no global mobility limitations. We can also observe this trajectory in the configuration space:



Note how for simplicity we have drawn the configuration space as Euclidean. Of course it is not Euclidean because $\theta \in S^1$.

1.4 Kinematic Models

We have seen how the kinematic constraints are defined in Pfaffian form as $A(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$, with A being a $k \times n$ matrix. We are interested in finding a basis for the null space of $A(\mathbf{q})$:

$$\mathcal{N}(A(\mathbf{q})) = \text{span}\{\mathbf{g}_1(\mathbf{q}), \mathbf{g}_2(\mathbf{q}), \ldots, \mathbf{g}_m(\mathbf{q})\} \tag{1.37}$$

$\mathcal{N}(A(\mathbf{q}))$ has dimension $m := n - k$ (assuming that the k constraints are linearly independent, i.e., $A(\mathbf{q})$ has full row rank). Each vector $\mathbf{g}_i(\mathbf{q})$ of the basis is a vector field on $\mathcal{C}$ with values in $\mathbb{R}^n$, any admissible velocity is a linear combination of the vectors in the basis:

$$\dot{\mathbf{q}} = \sum_{i=1}^{m} u_i \mathbf{g}_i(\mathbf{q}) \tag{1.38}$$

we can write any admissible velocity in compact form by using an m dimensional vector $\mathbf{u}$:

$$\dot{\mathbf{q}} = \underbrace{\begin{pmatrix} | & | & & | \\ \mathbf{g}_1(\mathbf{q}) & \mathbf{g}_2(\mathbf{q}) & \cdots & \mathbf{g}_m(\mathbf{q}) \\ | & | & & | \end{pmatrix}}_{G(\mathbf{q})} \underbrace{\begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix}}_{\mathbf{u}} \tag{1.39}$$

the **kinematic model** is:

$$\dot{\mathbf{q}} = G(\mathbf{q})\mathbf{u} \tag{1.40}$$

with $\mathbf{u}$ being a velocity-level quantity. We can see the model as a dynamical system, with $\mathbf{q}$ being the n-dimensional state vector and $\mathbf{u}$ the m-dimensional input vector, of course it is a nonlinear system due to $G(\mathbf{q})$ that depends (in general) nonlinearly on $\mathbf{q}$. This system satisfies the **driftless property**:

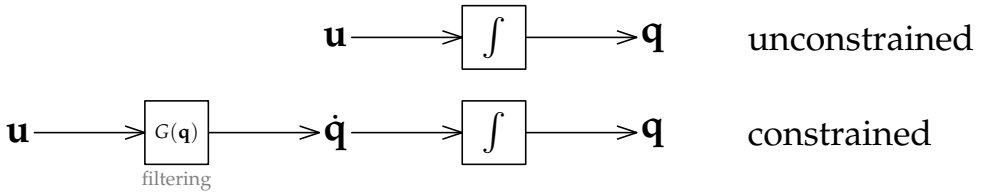
$$\mathbf{u} = \mathbf{0} \implies \dot{\mathbf{q}} = \mathbf{0} \tag{1.41}$$

it does not have a free evolution. The kinematic model is not unique, since we can select a different choice for the basis vectors $\mathbf{g}_1, \ldots, \mathbf{g}_m$, all the bases are equivalent for the description of the system, but the interpretation of the vector $\mathbf{u}$ depends on the choice of the basis. We can interpret $\mathbf{u}$ as the "virtual" velocities in the kinematically constrained configuration space.

Now we want to observe the differences between a kinematically constrained robot and an unconstrained robot. Consider an open-chain manipulator with revolute joints, with $\mathbf{q}$ being the joint variables (DH coordinates). This choice of coordinates makes the manipulator free from any constraint, the configuration can "fly" in the configuration space, any joint velocity $\dot{\mathbf{q}}$ is always admissible, hence its kinematic model is

$$\dot{\mathbf{q}} = \mathbf{u}. \tag{1.42}$$

This is a simple integrator, we can impose any joint velocity. For a robot with non-holonomic constraint this does not hold. The kinematic model is not a simple integrator since it has a *filtering* component given by the matrix $G(\mathbf{q})$, the vector $\mathbf{u}$ does not directly represent the generalized velocity $\dot{\mathbf{q}}$, and it is called **pseudo-velocity**.



Let's discuss the controllability of the constrained model. The controllability of a system can be proved formally through nonlinear control theory, or constructively, by the definition of a universal maneuver (as we did in the previous section).

Definition 1.4.1 : Controllability

The kinematic model of a robot is controllable if any configuration can be reached from any initial configuration in finite time.

A holonomic constraint defines a geometric constraint that prevents the robot from accessing a specific region in the configuration space, hence a robot with geometric constraints is never controllable. By this reasoning, there is a 1-to-1 correspondence between controllability and the *absence* of holonomic constraints.

Observation 1.4.1

The kinematic model of a robot is controllable **if and only if** the set of kinematic constraints is (completely) non-holonomic, i.e., no integrable constraint can be derived from it.

1.4.1 The Unicycle Model

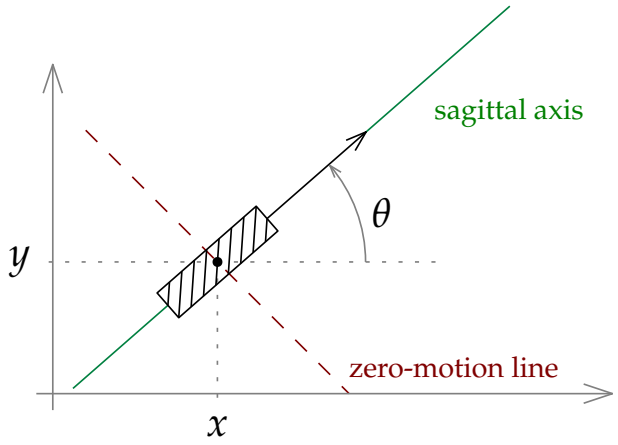
Let's define the kinematic model of a unicycle, that is equivalent to the rolling disk seen in Section 1.3.3. It has one kinematic constraint, the pure rolling condition imposes that:

$$\sin\theta\dot{x} - \cos\theta\dot{y} = 0. \tag{1.43}$$

It says that the velocity of the contact point must be aligned with the *sagittal axis* defined by θ, i.e. it can't have components on the zero-motion line. In matrix form the constraint is:

$$\underbrace{\begin{pmatrix}\sin\theta & -\cos\theta & 0\end{pmatrix}}_{A(\mathbf{q})}\begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{pmatrix} = 0. \tag{1.44}$$

The dimension of $\mathcal{N}(A)$ is 2, hence we will have a 2-dimensional vector of the pseudo-velocity.



A basis for the null space of A is the following:

$$\mathbf{g}_1(\mathbf{q}) = \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix}, \quad \mathbf{g}_2(\mathbf{q}) = \begin{pmatrix}0\\ 0\\ 1\end{pmatrix} \tag{1.45}$$

the kinematic model is

$$\dot{\mathbf{q}} = G(\mathbf{q})\mathbf{u} = \begin{pmatrix}\cos\theta & 0\\ \sin\theta & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}u_1\\ u_2\end{pmatrix} \implies \begin{cases}\dot{x} = \cos\theta u_1\\ \dot{y} = \sin\theta u_1\\ \dot{\theta} = u_2\end{cases}. \tag{1.46}$$

u_1 is the scalar velocity along the sagittal axis

$$u_1 = \pm\sqrt{\dot{x}^2 + \dot{y}^2} \tag{1.47}$$

while u_2 is exactly $\dot{\theta}$. We rename these quantities:

$$v = u_1 \tag{1.48}$$
$$\omega = u_2 \tag{1.49}$$

v is related to the rolling velocity ω_{roll} and the radius of the wheel:

$$v = \omega_{\text{roll}} \cdot r. \tag{1.50}$$

We can rewrite the kinematic model by splitting the Cartesian part from the angular one:

$$\dot{\mathbf{q}} = \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix} v + \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\omega. \tag{1.51}$$

Is this model controllable? Yes, we already proved for the rolling disk that a universal maneuver exists.

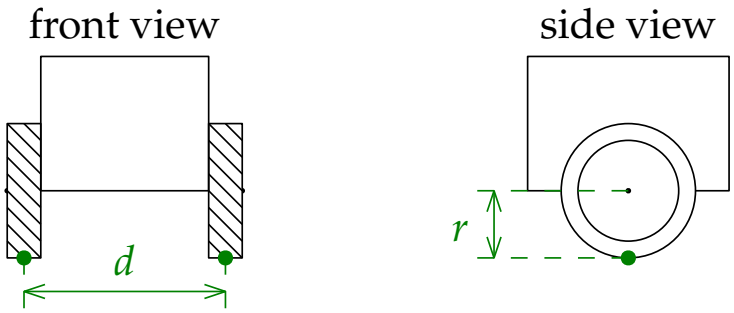
What are the real world robots that are kinematically and mechanically equivalent to the unicycle? First of all the differential drive, since it can rotate about its vertical axis, and move in the Cartesian plane only towards its sagittal axis (it can't slide laterally). It has two actuated wheels, of radius r and rolling velocities ω_R, ω_L. Its Cartesian velocity v is given by the average of the rolling velocities of the two wheels, multiplied by the radius:

$$v = r\frac{\omega_R + \omega_L}{2} \tag{1.52}$$

while its angular velocity is given by the difference:

$$\omega = (\omega_R - \omega_L)\frac{r}{d} \tag{1.53}$$

where d is the distance between the contact points of the wheels.



The other robot that is equivalent to the unicycle is the synchro-drive, it has two motors equipped with a pulley and belt mechanism, one controls the orientation of all the wheels, the other the rolling speed, hence these motors directly control v and ω, where v is the velocity of the geometric center of the three contact points.

1.4.2 The Bicycle Model

This kinematic model has two wheels, one fixed at the back (called rear wheel) and one steerable in the front.

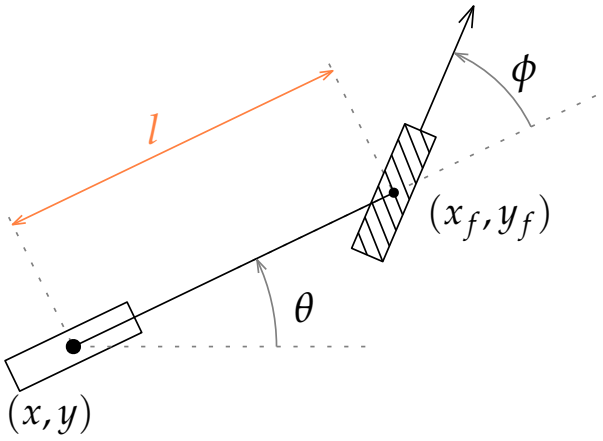


Figure 1.10: The bicycle model.

As described in Figure 1.10, the configuration space of the bicycle is $\mathcal{C} = \mathbb{R}^2 \times SO(2) \times SO(2)$, the dimension is $n = 4$. l is the distance between the two contact points of the two wheels, the first identified by (x, y), the second by (x_f, y_f), clearly:

$$x_f = x + l\cos\theta \tag{1.54}$$

$$y_f = y + l\sin\theta \tag{1.55}$$

the configuration vector is

$$\mathbf{q} = \begin{pmatrix} x & y & \theta & \phi \end{pmatrix}^\top. \tag{1.56}$$

Since there are two wheels, we will have two constraints, that impose the pure rolling condition on both wheels. For the rear wheel and the front wheel, respectively:

$$\sin\theta\dot{x} - \cos\theta\dot{y} = 0 \tag{1.57}$$
$$\sin(\theta+\phi)\dot{x}_f - \cos(\theta+\phi)\dot{y}_f = 0 \tag{1.58}$$

in vector form:

$$\underbrace{\begin{pmatrix} \sin\theta & -\cos\theta \end{pmatrix}}_{\mathbf{a}_1^\top} \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = 0 \tag{1.59}$$

$$\underbrace{\begin{pmatrix} \sin(\theta+\phi) & -\cos(\theta+\phi) \end{pmatrix}}_{\mathbf{a}_2^\top} \begin{pmatrix} \dot{x}_f \\ \dot{y}_f \end{pmatrix} = 0 \tag{1.60}$$

$\mathbf{a}_1$ and $\mathbf{a}_2$ identify the (forbidden) sliding direction of the two wheels. The two lines meet at the point known as **instantaneous center of rotation (ICR)**, as shown in Figure 1.11.

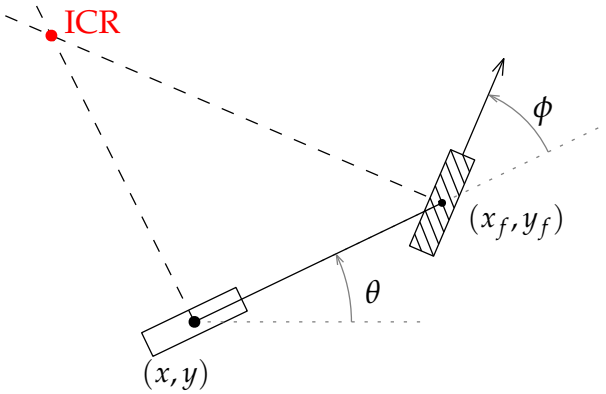


Figure 1.11: The instantaneous center of rotation.

The ICR represents the geometric point around which all the parts of the robot are rotating at a given instant. Now we want to express $(\dot{x}_f, \dot{y}_f)$ as a function of $\mathbf{q}$ and $\dot{\mathbf{q}}$, in order to write the constraints only as a function of $\mathbf{q}$ and $\dot{\mathbf{q}}$.

$$\begin{cases} x_f = x + l\cos\theta \\ y_f = y + l\sin\theta \end{cases} \implies \tag{1.61}$$

$$\begin{cases} \dot{x}_f = \dot{x} - \dot{\theta} l\sin\theta \\ \dot{y}_f = \dot{y} + \dot{\theta} l\cos\theta \end{cases} \tag{1.62}$$

we rewrite the constraint in Equation (1.58):

$$\sin(\theta+\phi)\dot{x}_f - \cos(\theta+\phi)\dot{y}_f = 0 \implies \tag{1.63}$$
$$\sin(\theta+\phi)(\dot{x} - \dot{\theta} l\sin\theta) - \cos(\theta+\phi)(\dot{y} + \dot{\theta} l\cos\theta) = 0 \implies \tag{1.64}$$
$$\sin(\theta+\phi)\dot{x} - \cos(\theta+\phi)\dot{y} - l\dot{\theta}\cos\phi = 0. \tag{1.65}$$

The constraints in matrix form are:

$$\underbrace{\begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 \\ \sin(\theta+\phi) & -\cos(\theta+\phi) & -l\cos\phi & 0 \end{pmatrix}}_{A(\mathbf{q})} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \mathbf{0} \tag{1.66}$$

with $\dim \mathcal{N}(A) = 2$, we should find two vectors for the basis of $\mathcal{N}(A)$. Since the last column of A is zero, one of the two vectors is:

$$\mathbf{g}_2(\mathbf{q}) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.67}$$

We want to find $\mathbf{g}_1(\mathbf{q})$, a vector similar to the one used for the unicycle is

$$\mathbf{g}_1(\mathbf{q}) = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \alpha \\ 0 \end{pmatrix} \tag{1.68}$$

with α a term that has yet to be chosen. We see that:

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 \\ \sin(\theta+\phi) & -\cos(\theta+\phi) & -l\cos\phi & 0 \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \\ \alpha \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \sin(\theta+\phi)\cos\theta - \cos(\theta+\phi)\sin\theta - l\cos\phi\alpha \end{pmatrix} \tag{1.69}$$

now we choose α in order to make the second component zero:

$$\sin(\theta+\phi)\cos\theta - \cos(\theta+\phi)\sin\theta - l\cos\phi\alpha = 0 \implies \tag{1.70}$$

$$\sin\phi - l\cos\phi\alpha = 0 \implies \tag{1.71}$$

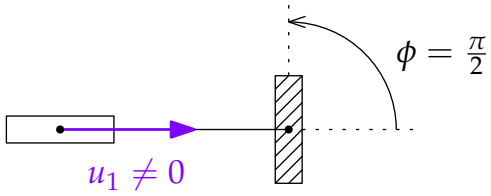
$$\alpha = \frac{1}{l}\tan\phi \tag{1.72}$$

we have found our basis, now we can write out the kinematic model:

$$\dot{\mathbf{q}} = \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{l}\tan\phi \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2 \tag{1.73}$$

$u_1 = \pm\sqrt{\dot{x}^2 + \dot{y}^2}$ is the magnitude of the velocity of the rear wheel contact point. If the robot is driven by the front wheel this model isn't appropriate, since u_1 is the velocity of the rear wheel.

This model has a singularity, since $\tan\phi$ isn't defined at $\phi = \pm\frac{\pi}{2}$, hence $\mathbf{g}_1$ can't be evaluated, this is due to the fact that the model assumes only pure rolling for the wheels, and if $\phi = \frac{\pi}{2}$ any non zero u_1 will make the front wheel slide.



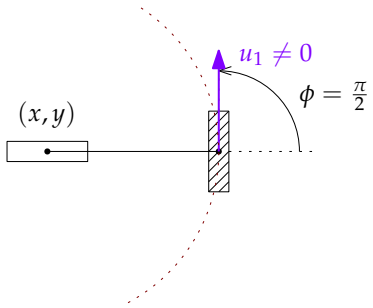
This problem is easily avoided by considering limits on the joint values. Now let's derive a model where u_1 is the velocity of the front wheel, to do so, we have to modify the first vector of the basis:

$$\widehat{\mathbf{g}}_1 = \mathbf{g}_1 \cos\phi \tag{1.74}$$

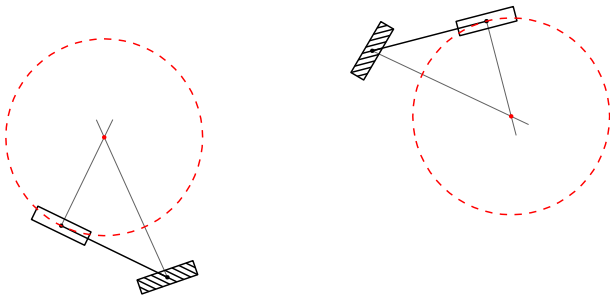
then

$$\dot{\mathbf{q}} = \underbrace{\begin{pmatrix} \cos\theta\cos\phi \\ \sin\theta\cos\phi \\ \frac{1}{l}\sin\phi \\ 0 \end{pmatrix}}_{\widehat{\mathbf{g}}_1} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2 \tag{1.75}$$

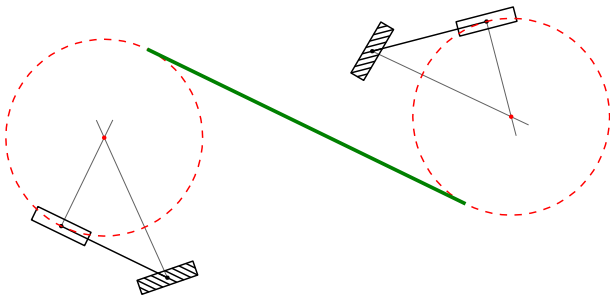
now u_1 is $\pm\sqrt{\dot{x}_f^2 + \dot{y}_f^2}$. Now the singularity on ϕ disappears, here a non zero u_1 when $\phi = \frac{\pi}{2}$ represents a motion around (x, y).



The bicycle constraints are non-holonomic and the model is controllable, we prove this fact by showing (in a non formal way) a universal maneuver. Consider two generic configurations $\mathbf{q}_s, \mathbf{q}_g$, we can draw the circles centered in the ICR that represent the Cartesian trajectory that the robot will perform by actuating just the rear wheel.



The robot in $\mathbf{q}_s$ (left) can start tracking the circular trajectory until it is on the *common tangent* between the two circles.



There, it will steer by making $\phi = 0$ and it will track the common tangent in order to reach the circular path of $\mathbf{q}_g$. There, it will steer again and reach the goal configuration.

Which robots are mechanically equivalent to a bicycle? One example is the tricycle. A tricycle with two rear wheels and one steerable front wheel can be modeled as a bicycle by defining the midpoint between the two rear wheels as (x, y). A car-like robot can also be represented by the bicycle model, but it presents a kinematic challenge: because it has two steerable front wheels, their respective zero-motion lines might intersect the zero-motion line of the rear axle at different points. When this happens, a single Instantaneous Center of Rotation (ICR) cannot be identified. This situation is illustrated in Figure 1.12.

To overcome this problem, we use a mechanism called **Ackermann steering**, which, during a circular path, automatically orients the front wheels so that their zero-motion lines intersect at a single ICR.

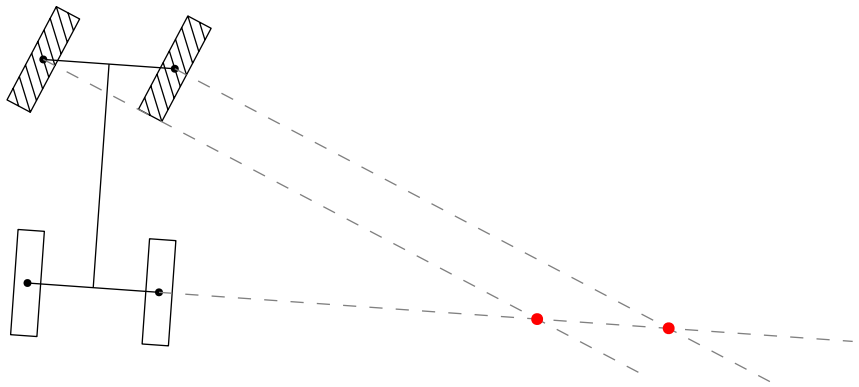


Figure 1.12: The intersections identify two different ICRs.

CHAPTER

2

CONTROLLABILITY OF MOBILE ROBOTS

2.1 Nonlinear Controllability

Let's discuss the controllability of a nonlinear *driftless* system:

$$\dot{\mathbf{x}} = \sum_{i=1}^{m} \mathbf{g}_i(\mathbf{x}) u_i \tag{2.1}$$

with $\mathbf{g}_i : \mathbb{R}^n \to \mathbb{R}^n$ being a vector field and $u_1, \ldots, u_m$ the inputs. In this general discussion we use the symbol $\mathbf{x}$ to denote the state of our system, although we usually use $\mathbf{q}$ (in the robotics domain).

Definition 2.1.1 : Controllability of a Nonlinear System

The system in Equation (2.1) is **controllable** if, for any pair of states $(\mathbf{x}_i, \mathbf{x}_f)$, with $\mathbf{x}(0) = \mathbf{x}_i$, there exists a profile $\mathbf{u}(t)$ for the inputs in $t \in [0, T]$ such that the system will reach at the end the state $\mathbf{x}(T) = \mathbf{x}_f$.

In linear systems, the controllability is guaranteed when the controllability matrix of the system has full rank, we will see how also for the nonlinear case, the controllability test will result in the derivation of the rank of a matrix.

Definition 2.1.2 : Lie Brackets

Let $\mathbf{g}_1, \mathbf{g}_2$ be two vector fields on $\mathbb{R}^n$. The **Lie Bracket** of these vector fields is the new vector field $[\mathbf{g}_1, \mathbf{g}_2]$ defined as follows:

$$[\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}) = \frac{\partial \mathbf{g}_2}{\partial \mathbf{x}} \mathbf{g}_1(\mathbf{x}) - \frac{\partial \mathbf{g}_1}{\partial \mathbf{x}} \mathbf{g}_2(\mathbf{x}). \tag{2.2}$$

Let's see an example in $\mathbb{R}^3$:

$$\mathbf{g}_1 = \begin{pmatrix} \cos x_3 \\ \sin x_3 \\ 0 \end{pmatrix}, \quad \mathbf{g}_2 = \begin{pmatrix} 0 \\ 0 \\ x_1 \end{pmatrix} \tag{2.3}$$

the Jacobians are:

$$\frac{\partial \mathbf{g}_1}{\partial \mathbf{x}} = \begin{pmatrix} 0 & 0 & -\sin x_3 \\ 0 & 0 & \cos x_3 \\ 0 & 0 & 0 \end{pmatrix}, \quad \frac{\partial \mathbf{g}_2}{\partial \mathbf{x}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{2.4}$$

hence:

$$[\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}) = \begin{pmatrix} x_1 \sin x_3 \\ -x_1 \cos x_3 \\ \cos x_3 \end{pmatrix} \tag{2.5}$$

notice how in this case $[\mathbf{g}_1, \mathbf{g}_2]$ is linearly independent from $\mathbf{g}_1, \mathbf{g}_2$.

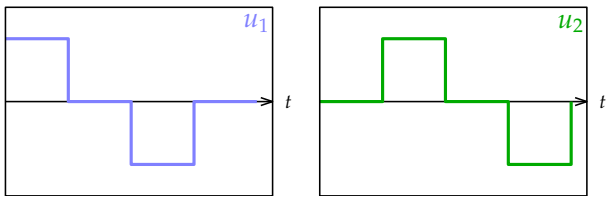
2.1.1 Lie Bracket Motion

Now we want to give an *interpretation* of the Lie bracket. Consider the following 2-inputs system:

$$\dot{\mathbf{x}} = \mathbf{g}_1(\mathbf{x})u_1 + \mathbf{g}_2(\mathbf{x})u_2 \tag{2.6}$$

the interpretation of the Lie brackets can be defined by considering the following time profile for u_1, u_2 in $t \in [0, 4\varepsilon]$:

$$(u_1, u_2) = (1, 0) \quad t \in [0, \varepsilon] \tag{2.7}$$
$$(u_1, u_2) = (0, 1) \quad t \in [\varepsilon, 2\varepsilon] \tag{2.8}$$
$$(u_1, u_2) = (-1, 0) \quad t \in [2\varepsilon, 3\varepsilon] \tag{2.9}$$
$$(u_1, u_2) = (0, -1) \quad t \in [3\varepsilon, 4\varepsilon]. \tag{2.10}$$



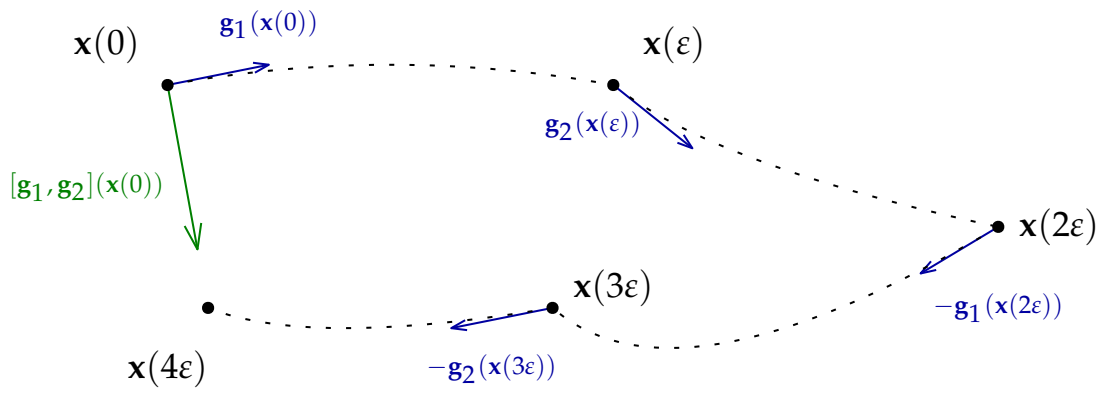
This input profile is called **Lie bracket motion**, and after applying it the system will be in the state:

$$\mathbf{x}(4\varepsilon) = \mathbf{x}(0) + \underbrace{\varepsilon^2[\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}(0)) + O(\varepsilon^3)}_{\text{displacement}}. \tag{2.11}$$

This result affirms that the displacement will be in the direction of the Lie bracket, with an error given by the term $O(\varepsilon^3)$: if $\varepsilon \to 0$, the terms in $O(\varepsilon^3)$ will vanish faster, and the displacement will be defined mainly by $\varepsilon^2[\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}(0))$.

Observation 2.1.1

The Lie bracket gives the direction of the displacement of that particular input sequence.



Definition 2.1.3 : Distribution

Let $\mathcal{C}$ be a smooth manifold. A *Distribution* Δ is a map that assigns to each point $\mathbf{x} \in \mathcal{C}$ a subspace of the tangent space $T_{\mathbf{x}}\mathcal{C}$:

$$\Delta : \mathbf{x} \mapsto \Delta(\mathbf{x}) \subset T_{\mathbf{x}}\mathcal{C}. \tag{2.12}$$

The tangent space in each point of a n-manifold has dimension n, hence the dimension of the subspace $\Delta(\mathbf{x})$ can be at most n (depending on the point on where we are evaluating Δ). For a system in the form (2.1), the *input distribution* is defined as follows:

$$\Delta(\mathbf{x}) = \text{span}\{\mathbf{g}_1(\mathbf{x}), \mathbf{g}_2(\mathbf{x}), \dots, \mathbf{g}_m(\mathbf{x})\}. \tag{2.13}$$

The controllability problem for nonlinear systems is strictly correlated to a particular distribution, called the **accessibility distribution** $\Delta_{\mathcal{A}}$, defined as follows:

$$\Delta_{\mathcal{A}}(\mathbf{x}) = \text{span}\{\mathbf{g}_1(\mathbf{x}), \dots, \mathbf{g}_m(\mathbf{x}), \dots, [\mathbf{g}_i, \mathbf{g}_j](\mathbf{x}), \dots, [\mathbf{g}_i, [\mathbf{g}_j, \mathbf{g}_k]](\mathbf{x}), \dots\}. \tag{2.14}$$

Let's break it down:

- With $[\mathbf{g}_i, \mathbf{g}_j]$ we denote all Lie brackets given by all the possible pairs of vector fields for $1 \leq i, j \leq m$. These are also called first order Lie brackets.
- Each pair $\mathbf{g}_i, \mathbf{g}_j$ (with $i \neq j$) appears only one time, we don't have to add also $[\mathbf{g}_j, \mathbf{g}_i]$ since it is equal to $-[\mathbf{g}_i, \mathbf{g}_j]$ (they are linearly dependent, the span needs only one of them).
- $[\mathbf{g}_i, [\mathbf{g}_j, \mathbf{g}_k]]$ for $1 \leq i, j, k \leq m$ are the second order Lie brackets.
- This nested definition can continue, for example the third order Lie brackets are in the form $[\mathbf{g}_i, [\mathbf{g}_j, [\mathbf{g}_k, \mathbf{g}_h]]]$.
- The definition of $\Delta_{\mathcal{A}}$ consider all the possible combination of Lie brackets, for each possible order, up to infinity.

Of course $\Delta_{\mathcal{A}}(\mathbf{x})$ can have different dimension depending on $\mathbf{x}$, for a nonlinear system, to assert controllability, it is *mandatory* that $\Delta_{\mathcal{A}}(\mathbf{x})$ has dimension n at every single $\mathbf{x} \in \mathcal{C}$.

Definition 2.1.4 : Dimension of a Distribution

The distribution $\Delta_{\mathcal{A}}$ has dimension k if $\dim \Delta_{\mathcal{A}}(\mathbf{x}) = k$ for every $\mathbf{x} \in \mathcal{C}$. In that case we write $\dim \Delta_{\mathcal{A}} = k$.

The nonlinear controllability problem is fully described by the following result.

Theorem 2.1.1 : Accessibility Criterion (Chow)

A nonlinear system of the form (2.1) with an n-dimensional state vector is controllable *if and only if* its accessibility distribution has dimension n:

$$\dim \Delta_{\mathcal{A}} = n. \tag{2.15}$$

So in practice, we have to build the matrix M whose columns are the vector fields $\mathbf{g}_i$ and all their Lie brackets, and check if the rank is full.

$$M(\mathbf{x}) = \begin{pmatrix} | & & | & & | & & | \\ \mathbf{g}_1(\mathbf{x}) & \dots & \mathbf{g}_m(\mathbf{x}) & \dots & [\mathbf{g}_i, \mathbf{g}_j](\mathbf{x}) & \dots & [\mathbf{g}_i, [\mathbf{g}_j, \mathbf{g}_k]](\mathbf{x}) \dots \\ | & & | & & | & & | \end{pmatrix} \tag{2.16}$$

It's impossible to write out all the columns, we can check each iteratively for the vector fields $\mathbf{g}_1, \dots, \mathbf{g}_m$, then for the first order Lie brackets, then for the higher order, as soon as the rank of the matrix reaches size n, the criterion is satisfied ($\dim \Delta_{\mathcal{A}} = n$) and the system is controllable. All remaining infinite

columns can be ignored. For systems described by analytic functions, the chain of Lie brackets stabilizes after a finite number of steps. If at a certain point the new Lie brackets no longer increase the rank and it remains lower than n, the process stops and the system is not controllable.

Proposition 2.1.1

If the system is analytic, the accessibility distribution stabilizes. The chain of Lie brackets cannot generate distributions of infinitely increasing size in a space of dimension n. Consequently, in the worst case, it is not necessary to go beyond the Lie brackets of order $n-1$ (or in any case limited according to the size of the state).

In computational practice (for example using symbolic calculation software such as MATLAB or Python), the brackets are calculated in ascending order of degree and those that do not increase the rank are immediately discarded, blocking the algorithm as soon as the rank is equal to n.

Proposition 2.1.2

Consider the vector fields $\mathcal{B} = \{\mathbf{g}_1, \dots, \mathbf{g}_m\}$. If all the Lie brackets of first order are linearly dependent on the vector fields in $\mathcal{B}$

$$[\mathbf{g}_i, \mathbf{g}_j] \in \text{span}\mathcal{B}, \quad 1 \leq i, j \leq m \tag{2.17}$$

then also the higher order Lie brackets are linearly dependent on the vector fields in $\mathcal{B}$.

Proposition 2.1.2 provides a criterion to stop computing Lie brackets for a system if all the first order Lie brackets are linearly dependent on the vector fields in $\mathcal{B}$.

2.1.2 The Unicycle Case

The unicycle model is:

$$\dot{\mathbf{q}} = \underbrace{\begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}}_{\mathbf{g}_1} v + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{g}_2} \omega \tag{2.18}$$

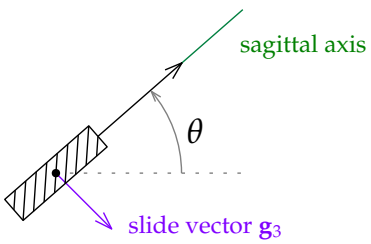
we compute the first order Lie bracket:

$$\mathbf{g}_3(\mathbf{q}) := [\mathbf{g}_1, \mathbf{g}_2](\mathbf{q}) = -\begin{pmatrix} 0 & 0 & -\sin\theta \\ 0 & 0 & \cos\theta \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \sin\theta \\ -\cos\theta \\ 0 \end{pmatrix} \tag{2.19}$$

we denote $\mathbf{g}_3$ the Lie bracket, notice how it is the unit vector orthogonal to the sagittal axis, called the **slide vector field**. Let's check the matrix given by $\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3$:

$$\text{rank} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ \sin\theta & 0 & -\cos\theta \\ 0 & 1 & 0 \end{pmatrix} = \text{rank} \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & -\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = 3 \tag{2.20}$$

The rank is full, we don't have to check the higher order Lie brackets, hence the system is controllable.



In this case the Lie bracket defines a velocity that is not instantaneously realizable, but specifies a direction achievable through a sequence of motions.

2.1.3 The Bicycle Case

The model of the bicycle is:

$$\dot{\mathbf{q}} = \underbrace{\begin{pmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{l}\tan\phi \\ 0 \end{pmatrix}}_{\mathbf{g}_1} u_1 + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{g}_2} u_2 \tag{2.21}$$

the first Lie bracket (we don't show the computation) is:

$$\mathbf{g}_3(\mathbf{q}) := [\mathbf{g}_1, \mathbf{g}_2](\mathbf{q}) = \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{l\cos^2\phi} \\ 0 \end{pmatrix} \tag{2.22}$$

this vector is called **wriggle** and is clearly linearly independent from the others, hence the rank is at least 3. Let's compute the second order Lie bracket:

$$\mathbf{g}_4 := [\mathbf{g}_1, \mathbf{g}_3](\mathbf{q}) = \begin{pmatrix} -\frac{\sin\theta}{l\cos^2\phi} \\ \frac{\cos\theta}{l\cos^2\phi} \\ 0 \\ 0 \end{pmatrix} \tag{2.23}$$

$\mathbf{g}_4$ is parallel to the slide vector field. It can be verified that:

$$\operatorname{rank} \begin{pmatrix} \cos\theta & 0 & 0 & -\frac{\sin\theta}{l\cos^2\phi} \\ \sin\theta & 0 & 0 & \frac{\cos\theta}{l\cos^2\phi} \\ \frac{1}{l}\tan\phi & 0 & -\frac{1}{l\cos^2\phi} & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = 4 \tag{2.24}$$

hence the model is controllable.

2.1.4 Chained-Form

Definition 2.1.5 : Chained-Form Systems

Consider a nonlinear system with an n-dimensional state vector $\mathbf{z} \in \mathbb{R}^n$ and 2 inputs $\mathbf{w} \in \mathbb{R}^2$. The system is in chained-form if it has the following structure:

$$\dot{z}_1 = w_1 \tag{2.25}$$
$$\dot{z}_2 = w_2 \tag{2.26}$$
$$\dot{z}_3 = z_2 w_1 \tag{2.27}$$
$$\dot{z}_4 = z_3 w_1 \tag{2.28}$$
$$\vdots \tag{2.29}$$
$$\dot{z}_n = z_{n-1} w_1. \tag{2.30}$$

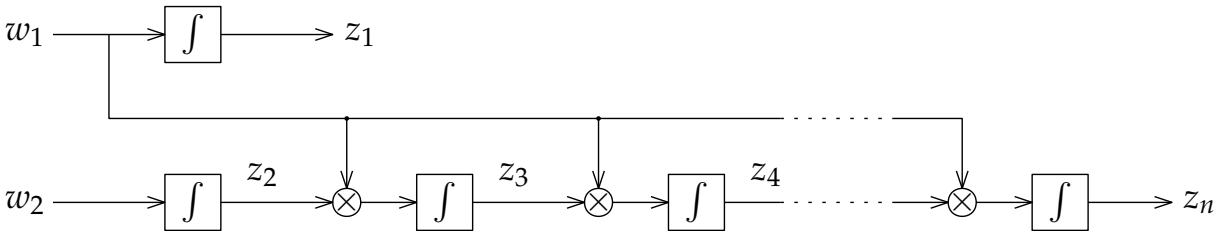


Figure 2.1: Block diagram of a chained-form system.

We can observe how, if the input w_1 is constant, the system becomes linear in $\mathbf{z}$ (with w_2 as the only input), hence in digital control, where the inputs are piece-wise constant, the system is piece-wise linear.

Proposition 2.1.3

Any chained-form system is controllable.

Many controllable driftless systems with two inputs (in particular, the kinematic models of the mobile robots of our interest) can be rewritten in chained-form under a coordinate and input transformation:

$$\mathbf{z} = \alpha(\mathbf{q}) \tag{2.31}$$
$$\mathbf{w} = \beta(\mathbf{q})\mathbf{u}. \tag{2.32}$$

Now we will show how to transform the unicycle model in chained-form. The original model is:

$$\dot{x} = v\cos\theta \tag{2.33}$$
$$\dot{y} = v\sin\theta \tag{2.34}$$
$$\dot{\theta} = \omega \tag{2.35}$$

we define the following transformation for the coordinates:

$$\mathbf{z} = \alpha(\mathbf{q}) = \begin{cases} z_1 = \theta \\ z_2 = x\cos\theta + y\sin\theta \\ z_3 = x\sin\theta - y\cos\theta \end{cases} \tag{2.36}$$

and the following transformation for the inputs:

$$\mathbf{w} = \beta(\mathbf{q})\begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{cases} w_1 = \omega \\ w_2 = v - z_3\omega \end{cases} . \tag{2.37}$$

Now we rewrite the model and see if it is in chained-form. From Equation (2.36) we get:

$$\dot{z}_1 = \dot{\theta} = \omega = w_1 \tag{2.38}$$

then:

$$\dot{z}_2 = \underbrace{\dot{x}\cos\theta + \dot{y}\sin\theta}_{v} - x\sin\theta\dot{\theta} + y\cos\theta\dot{\theta} \tag{2.39}$$
$$= v - \underbrace{\dot{\theta}}_{\omega}\underbrace{(x\sin\theta - y\cos\theta)}_{z_3} = \underbrace{v - \omega z_3 = w_2}_{\text{input transformation}} \tag{2.40}$$

ultimately:

$$\dot{z}_3 = \dot{x}\sin\theta - \dot{y}\cos\theta + x\cos\theta\dot{\theta} + y\sin\theta\dot{\theta} \tag{2.41}$$
$$= \underbrace{(\dot{x}\sin\theta - \dot{y}\cos\theta)}_{\text{0 due to the constraint}} + \dot{\theta}(x\cos\theta + y\sin\theta) \tag{2.42}$$
$$= \underbrace{\dot{\theta}}_{\omega = w_1}\underbrace{(x\cos\theta + y\sin\theta)}_{z_2} = z_2 w_1 \tag{2.43}$$

so the model after the transformations is:

$$\dot{z}_1 = w_1 \tag{2.44}$$
$$\dot{z}_2 = w_2 \tag{2.45}$$
$$\dot{z}_3 = z_2 w_1 \tag{2.46}$$

and it is in chained-form. z_2 and z_3 have a clear interpretation: z_2 is the x coordinate of the unicycle in the reference frame rotated by θ with respect to the base frame (i.e., with its horizontal axis aligned with the sagittal axis), while z_3 is the opposite of the y coordinate in the same frame, as shown in Figure 2.2.

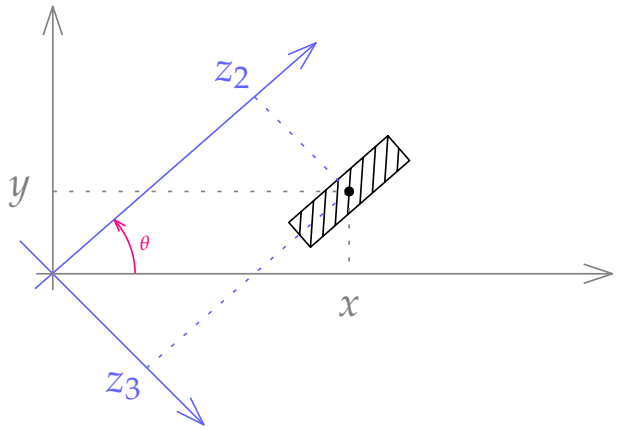


Figure 2.2: Interpretation of z_2 and z_3.

Setting aside the unicycle model: an interesting property of **trailed vehicles** emerges when considering a car-like robot: if the trailer is hitched exactly at the midpoint between the two rear wheels (denoted by the point (x, y)), the resulting kinematic model can still be converted into chained-form. This property holds recursively if the trailer itself has two rear wheels and an additional trailer is hitched at their midpoint, the overall kinematic model can still be expressed in chained-form.

2.2 Path and Trajectory Planning

It is in our interest to define a method to *plan* a motion that transfers a mobile robot from an initial configuration $\mathbf{q}_i$ to a final configuration $\mathbf{q}_f$, hence, we should define a trajectory:

$$\mathbf{q}(s), \quad s \in [s_i, s_f] \tag{2.47}$$

such that $\mathbf{q}(s_i) = \mathbf{q}_i$ and $\mathbf{q}(s_f) = \mathbf{q}_f$, without violating the kinematic constraints on $\dot{\mathbf{q}}$. In general, as we have already seen in previous robotics courses, we separate the geometric path from the timing law:

$$\mathbf{q}(s), \quad s \in [s_i, s_f] \tag{2.48}$$
$$s(t), \quad t \in [t_i, t_f]. \tag{2.49}$$

The path parameter s is usually chosen as the arc length.

- By **path planning** we mean the plan of a geometric path $\mathbf{q}(s)$ in the configuration space.
- By **trajectory planning** instead, we also include the time, i.e. we mean the plan of a time evolution of the configuration: $\mathbf{q}(t)$. This is usually done indirectly: first plan a path, then add a timing law.

Let us compute the velocity along a trajectory $\mathbf{q}(s(t))$, which must satisfy the kinematic constraints:

$$\dot{\mathbf{q}} = \frac{d\mathbf{q}}{dt} = \frac{d\mathbf{q}}{ds}\dot{s} = \mathbf{q}'\dot{s} \tag{2.50}$$

we denote $\mathbf{q}'$ the derivative of $\mathbf{q}$ with respect to s, this represents the *tangent vector* of the geometric path in the configuration space. If s is the arc length, then $\|\mathbf{q}'\| = 1$.

Example: Consider a robot with $\mathbf{q} = (x, y)$, we want to go from $(1, 0)$ to $(0, 1)$, one possible choice for the geometric path is:

$$\mathbf{q}(s) = \begin{cases} \cos s \\ \sin s \end{cases} \tag{2.51}$$

with $s \in \left[0, \frac{\pi}{2}\right]$. This is exactly the arc length, indeed:

$$\mathbf{q}'(s) = \begin{cases} -\sin s \\ \cos s \end{cases} \implies \|\mathbf{q}'(s)\| = \sqrt{(-\sin s)^2 + \cos^2 s} = 1. \tag{2.52}$$

Another choice of parametrization is

$$\mathbf{q}(s) = \begin{cases} \sqrt{1 - s^2} \\ s \end{cases} \tag{2.53}$$

with $s \in [0, 1]$, here it's not the arc length:

$$\mathbf{q}'(s) = \begin{cases} -\frac{s}{\sqrt{1-s^2}} \\ 1 \end{cases} \implies \|\mathbf{q}'(s)\| = \frac{1}{\sqrt{1 - s^2}}. \tag{2.54}$$

Now we take into account the kinematic constraints, we know that:

$$A(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0} \tag{2.55}$$

$\dot{\mathbf{q}}$ is the time derivative of the planned trajectory $\mathbf{q}(s(t))$:

$$A(\mathbf{q})\dot{\mathbf{q}} = A(\mathbf{q})\mathbf{q}'\dot{s} = \mathbf{0} \implies A(\mathbf{q})\mathbf{q}' = \mathbf{0} \tag{2.56}$$

we have assumed $\dot{s} \neq 0$, the result is that the tangent vector $\mathbf{q}'$ must lie in the null space of A:

$$\mathbf{q}' \in \mathcal{N}(A) \tag{2.57}$$

hence $\mathbf{q}'$ can be expressed as a linear combination of the basis vectors $\mathbf{g}_1, \ldots, \mathbf{g}_m$, exactly as the kinematic model, hence we can write the so-called **geometric model**:

$$\mathbf{q}' = G(\mathbf{q})\tilde{\mathbf{u}} \tag{2.58}$$

this model describes explicitly the admissible tangent vectors. $\tilde{\mathbf{u}}$ is the vector of the **geometric inputs**, by multiplying both sides by $\dot{s}$ we can conclude that:

$$\underbrace{\mathbf{u}(t)}_{\text{pseudo velocity}} = \tilde{\mathbf{u}}(s)\dot{s}(t). \tag{2.59}$$

After we have decided a profile for the geometric inputs $\tilde{\mathbf{u}}(s)$, we have determined a geometric path in the configuration space, only after having defined also $s(t)$ this will identify a particular trajectory.

Example: Let's apply this reasoning to the unicycle model, the pure rolling constraint for the tangent vector is:

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 \end{pmatrix} \mathbf{q}' = 0 \tag{2.60}$$

this means that the tangent vector $\mathbf{q}'$ must always be aligned with the sagittal axis of the unicycle: paths such as the one in Figure 2.3 are not admissible.

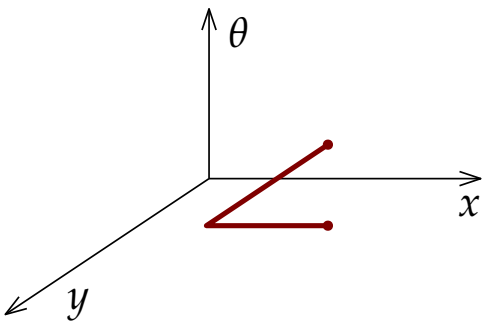


Figure 2.3: Geometric path with tangent vector non-aligned with the sagittal axis.

2.2.1 Flat Outputs

Definition 2.2.1 : Differential Flatness

Consider a general nonlinear system

$$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}. \tag{2.61}$$

The system is **differentially flat** if there exists a vector $\mathbf{y} = \Psi(\mathbf{x})$, called the **flat outputs**, such that the state and the control inputs can be expressed algebraically as a function of $\mathbf{y}$ and its time derivatives up to a finite order:

$$\mathbf{x} = \Phi_x(\mathbf{y}, \dot{\mathbf{y}}, \ddot{\mathbf{y}}, \dots, \mathbf{y}^{(r_x)}) \tag{2.62}$$

$$\mathbf{u} = \Phi_u(\mathbf{y}, \dot{\mathbf{y}}, \ddot{\mathbf{y}}, \dots, \mathbf{y}^{(r_u)}) \tag{2.63}$$

In Definition 2.2.1 we talk about time derivatives (since that is the domain of dynamical systems), but of course it also holds if all the derivatives are taken with respect to the parameter s (the case of our geometric model). The flatness property is useful in control, if we define a trajectory for $\mathbf{y}$ that is differentiable up to the $\max(r_x, r_u)$-th order, we can reconstruct the state evolution $\mathbf{x}$ and the required inputs $\mathbf{u}$ through the reconstruction maps Φ_x, Φ_u.

We immediately show this property with the unicycle model.

$$x' = \tilde{v}\cos\theta \tag{2.64}$$

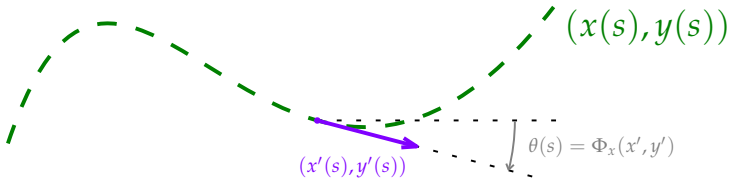
$$y' = \tilde{v}\sin\theta \tag{2.65}$$

$$\theta' = \tilde{\omega} \tag{2.66}$$

The intuition behind flatness is the following: If I know perfectly the path of the Cartesian position of the unicycle (x, y), I can infer what the path of the orientation θ is, together with the input profile that executes that path. For example, by knowing $(x(s), y(s))$, I can conclude that, at a given $s = s^*$, the orientation must be aligned with the direction of motion, i.e. the vector $(x'(s^*), y'(s^*))$.

$$\theta(s) = \Phi_x(x', y') = \text{atan2}(y'(s), x'(s)) + k\pi. \tag{2.67}$$

$k \in \{0, 1\}$ is a constant that defines the direction of motion.



Also the input $\tilde{v}$ can be reconstructed from the path of x, y, since it is exactly the norm of the Cartesian velocity:

$$\begin{cases} x' = \tilde{v}\cos\theta \\ y' = \tilde{v}\sin\theta \end{cases} \implies \tilde{v} = \pm\sqrt{(x')^2 + (y')^2} \tag{2.68}$$

the same can be done for the angular velocity (we skip the proof):

$$\tilde{\omega} = \frac{y''x' - x''y'}{(x')^2 + (y')^2}. \tag{2.69}$$

Equations (2.68),(2.69) are the reconstruction map for the input $\tilde{\mathbf{u}} = \Phi_u(x', y', x'', y'')$. These reconstructions have a problem that arises if $x' = y' = 0$: atan2 is not defined and $\tilde{\omega}$ would have a zero in the denominator. This singularity has an intuitive meaning: if the robot stops and its Cartesian velocity is zero, we cannot know whether it is steering in place or not, hence we cannot infer the evolution of $\theta(s)$ when it is not moving.

Proposition 2.2.1

A driftless system with two inputs is differentially flat if and only if it can be transformed into chained-form.

Proposition 2.2.2

A general dynamic system is differentially flat if and only if it can be linearized via dynamic feedback.

The wheeled robots in our interest can be put in chained-form, hence, they admit flat outputs, this is a great simplification from a control point of view, since we can directly execute a desired trajectory through the reconstruction maps. Let's see the flat outputs for the bicycle, the model is:

$$x' = \tilde{v}\cos\theta \tag{2.70}$$
$$y' = \tilde{v}\sin\theta \tag{2.71}$$
$$\theta' = \frac{\tilde{v}}{l}\tan\phi \tag{2.72}$$
$$\phi' = \tilde{\omega} \tag{2.73}$$

x, y, θ can be reconstructed in the same way as for the unicycle case, to reconstruct ϕ we use θ':

$$\theta(s) = \text{atan2}(y', x') + k\pi \tag{2.74}$$
$$\implies \theta'(s) = \frac{x'y'' - y'x''}{(x')^2 + (y')^2} \tag{2.75}$$

hence

$$\theta' = \frac{\tilde{v}}{l}\tan\phi \tag{2.76}$$
$$\implies \phi(s) = \arctan\left(\frac{l\theta'}{\tilde{v}}\right) \tag{2.77}$$
$$= \arctan\left(\frac{l\frac{x'y''-y'x''}{(x')^2+(y')^2}}{\pm\sqrt{(x')^2+(y')^2}}\right) \tag{2.78}$$
$$= \arctan\left(\pm\frac{l(x'y''-y'x'')}{((x')^2+(y')^2)^{3/2}}\right). \tag{2.79}$$

The input $\tilde{\omega} = \phi'$ is obtained by differentiating the last expression, hence it depends on the derivatives of x, y up to the third order (we omit the explicit expression).

Now we use Proposition 2.2.1 and show how a 2-input/3-state chained-form system admits flat outputs. The system is the following:

$$z_1' = \tilde{w}_1 \tag{2.80}$$
$$z_2' = \tilde{w}_2 \tag{2.81}$$
$$z_3' = z_2\tilde{w}_1 \tag{2.82}$$

for a chained-form system the flat outputs **are always the first and the last variables**:

$$\mathbf{y} = \begin{pmatrix} z_1 \\ z_3 \end{pmatrix} \tag{2.83}$$

the reconstruction maps are the following:

$$\Phi_x(z_1', z_3') = \frac{z_3'}{z_1'} \tag{2.84}$$

$$\Phi_u(z_1', z_1'', z_3', z_3'') = \begin{cases} \tilde{w}_1 = z_1' \\ \tilde{w}_2 = \dfrac{z_1' z_3'' - z_1'' z_3'}{(z_1')^2} \end{cases} . \tag{2.85}$$

2.2.2 Path Planning for the Unicycle

Any flat-output profile is automatically mapped to an admissible path. An algorithm for path planning exploiting differential flatness is the following:

1. Compute the initial and final values $\mathbf{y}_i, \mathbf{y}_f$ for the flat outputs from $\mathbf{q}_i, \mathbf{q}_f$.
2. Define $\mathbf{y}(s)$ by interpolating $\mathbf{y}_i, \mathbf{y}_f$ with the appropriate boundary conditions for $\mathbf{q}$.
3. Use the reconstruction formulas to find $\mathbf{q}(s)$ and $\tilde{\mathbf{u}}(s)$.

The same algorithm is valid for trajectory planning by interpolating over t instead of s. Let's apply this algorithm to the unicycle. We use a parametrization with $s \in [0, 1]$, from

$$\mathbf{q}_i = \begin{pmatrix} x_i \\ y_i \\ \theta_i \end{pmatrix}, \quad \mathbf{q}_f = \begin{pmatrix} x_f \\ y_f \\ \theta_f \end{pmatrix} \tag{2.86}$$

we extract

$$\mathbf{y}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}, \quad \mathbf{y}_f = \begin{pmatrix} x_f \\ y_f \end{pmatrix} \tag{2.87}$$

and perform an interpolation between these two points with a cubic polynomial function:

$$x(s) = s^3 x_f - (s-1)^3 x_i + \alpha_x s^2 (s-1) + \beta_x s (s-1)^2 \tag{2.88}$$

$$y(s) = s^3 y_f - (s-1)^3 y_i + \alpha_y s^2 (s-1) + \beta_y s (s-1)^2 \tag{2.89}$$

this polynomial already satisfies the boundary conditions:

$$x(0) = x_i, \quad x(1) = x_f \tag{2.90}$$

$$y(0) = y_i, \quad y(1) = y_f \tag{2.91}$$

then $\alpha_x, \beta_x, \alpha_y, \beta_y$ must be chosen in order to satisfy two additional boundary conditions that come from the kinematic constraints, (x', y') must be aligned with θ, and θ_i, θ_f are given, hence there are the following conditions on the orientation:

$$\text{atan2}(y'(0), x'(0)) = \theta_i \tag{2.92}$$

$$\text{atan2}(y'(1), x'(1)) = \theta_f. \tag{2.93}$$

The initial and final Cartesian velocities must be:

$$x'(0) = k_i \cos\theta_i, \quad x'(1) = k_f \cos\theta_f \tag{2.94}$$

$$y'(0) = k_i \sin\theta_i, \quad y'(1) = k_f \sin\theta_f \tag{2.95}$$

with k_i, k_f being two arbitrary positive constants, for simplicity we set $k_i = k_f = k$. To impose these conditions, we can solve algebraically for $\alpha_x, \beta_x, \alpha_y, \beta_y$ and obtain:

$$\alpha_x = k\cos\theta_f - 3x_f, \quad \beta_x = k\cos\theta_i + 3x_i \tag{2.96}$$

$$\alpha_y = k\sin\theta_f - 3y_f, \quad \beta_y = k\sin\theta_i + 3y_i. \tag{2.97}$$

It will be $\tilde{v}(0) = \tilde{v}(1) = k$, hence k can be chosen to shape the trajectory, by affecting the length of the first and last "linear" segments of the path, as shown in Figure 2.4.

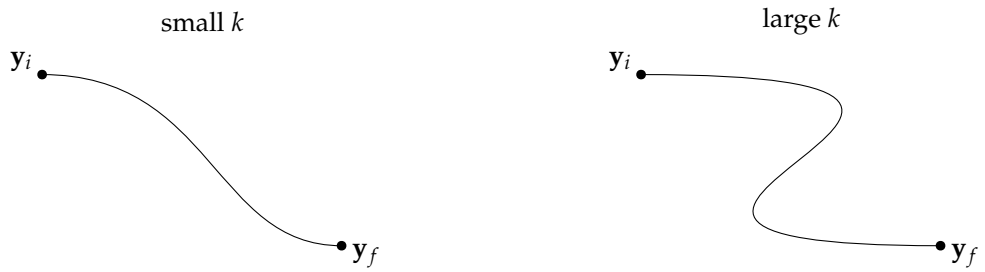


Figure 2.4: Difference between two paths with the same interpolating points but different choice of k.

2.2.3 Boundary Conditions for the General Case

In the unicycle example the boundary conditions on the flat outputs were obtained by inspection. Let's now formalize how they must be chosen for a generic differentially flat system

$$\dot{\mathbf{q}} = \mathbf{G}(\mathbf{q})\mathbf{u}, \qquad \mathbf{q} \in \mathbb{R}^n,\ \mathbf{u} \in \mathbb{R}^m, \tag{2.98}$$

with flat output $\mathbf{y} \in \mathbb{R}^m$. By definition of flatness, the state and the input can be reconstructed from $\mathbf{y}$ and a finite number of its derivatives:

$$\mathbf{q} = \Phi_x\big(\mathbf{y}, \mathbf{y}', \ldots, \mathbf{y}^{(r)}\big) \tag{2.99}$$

$$\tilde{\mathbf{u}} = \Phi_u\big(\mathbf{y}, \mathbf{y}', \ldots, \mathbf{y}^{(r+1)}\big) \tag{2.100}$$

To simplify the notation we collect the flat output and its derivatives up to order r in the vector

$$\bar{\mathbf{y}}(s) = \begin{pmatrix} \mathbf{y}(s) \\ \mathbf{y}'(s) \\ \vdots \\ \mathbf{y}^{(r)}(s) \end{pmatrix} \in \mathbb{R}^{m(r+1)} \tag{2.101}$$

so that (2.99) becomes $\mathbf{q} = \Phi_x(\bar{\mathbf{y}})$.

Conditions on the boundary values: Given $\mathbf{q}_i, \mathbf{q}_f$, the profile $\mathbf{y}(s)$ must produce a path that starts in $\mathbf{q}_i$ and ends in $\mathbf{q}_f$. Because of (2.99), this is not a condition on $\mathbf{y}(0), \mathbf{y}(1)$ alone, but on the whole vectors $\bar{\mathbf{y}}(0), \bar{\mathbf{y}}(1)$:

$$\Phi_x\big(\bar{\mathbf{y}}(0)\big) = \mathbf{q}_i, \qquad \Phi_x\big(\bar{\mathbf{y}}(1)\big) = \mathbf{q}_f. \tag{2.102}$$

Each of the two conditions in (2.102) is a set of n scalar equations in the $m(r+1)$ unknowns $\bar{\mathbf{y}}_i = \bar{\mathbf{y}}(0)$, $\bar{\mathbf{y}}_f = \bar{\mathbf{y}}(1)$ respectively. Since Φ_x is determined by the system alone, the two boundaries are decoupled, and can be treated separately. Let $\mathbf{h}(\mathbf{q})$ be the map that extracts the flat output from the state, as in the first step of the algorithm. Then the first m equations of (2.102) are simply

$$\mathbf{y}(0) = \mathbf{y}_i = \mathbf{h}(\mathbf{q}_i), \qquad \mathbf{y}(1) = \mathbf{y}_f = \mathbf{h}(\mathbf{q}_f) \tag{2.103}$$

while the remaining $n - m$ equations are the constraints that the kinematics (or dynamics) of the system impose on the derivatives $\mathbf{y}^{(k)}(0), \mathbf{y}^{(k)}(1)$ for $k = 1, \ldots, r$. These are the equations that, in the unicycle case ($n = 3$, $m = 2$, $r = 1$), reduce to the alignment of (x', y') with θ at the two ends.

The number of scalar unknowns at each boundary exceeds the number of equations by

$$d = m(r+1) - n \ \geq 0. \tag{2.104}$$

Hence, at each boundary, n components of $\bar{\mathbf{y}}$ are fixed by $\mathbf{q}_i$ (resp. $\mathbf{q}_f$), while d components are free and can be used to shape the path. In general, the solution set of (2.102) can be parametrized as

$$\bar{\mathbf{y}}_i = \sigma(\mathbf{q}_i, \boldsymbol{\kappa}_i), \qquad \bar{\mathbf{y}}_f = \sigma(\mathbf{q}_f, \boldsymbol{\kappa}_f), \qquad \boldsymbol{\kappa}_i, \boldsymbol{\kappa}_f \in \mathbb{R}^d \tag{2.105}$$

where σ is a right inverse of Φ_x, i.e. $\Phi_x(\sigma(\mathbf{q}, \boldsymbol{\kappa})) = \mathbf{q}$. For the unicycle $d = 2 \cdot 2 - 3 = 1$, and the free parameters κ_i, κ_f are exactly the constants k_i, k_f that we introduced, with $\sigma(\mathbf{q}, k) = (x,\ y,\ k\cos\theta,\ k\sin\theta)$.

The free parameters cannot be chosen arbitrarily, they must be such that:

1. $\bar{\mathbf{y}}_i, \bar{\mathbf{y}}_f$ belong to the domain where Φ_x and Φ_u are well defined and regular, i.e. the reconstruction formulas do not hit a singularity at the boundaries. For the unicycle the singularity is at $(x', y') = (0, 0)$, where θ is undefined, hence it must be $k_i, k_f \neq 0$ (in particular, positive for a forward motion);
2. the path has the required regularity: $\tilde{\mathbf{u}}$ depends on $\mathbf{y}^{(r+1)}$, hence $\mathbf{y}(s)$ must be at least C^{r+1} on $[0, 1]$.

Moreover, if the initial and final inputs are also prescribed, $\tilde{\mathbf{u}}(0) = \tilde{\mathbf{u}}_i$ and $\tilde{\mathbf{u}}(1) = \tilde{\mathbf{u}}_f$, then from (2.100) the derivative of order $r + 1$ must be constrained as well: $\Phi_u(\bar{\mathbf{y}}_i, \mathbf{y}^{(r+1)}(0)) = \tilde{\mathbf{u}}_i$, and similarly at $s = 1$. This adds m equations for each boundary, which are solved for $\mathbf{y}^{(r+1)}(0), \mathbf{y}^{(r+1)}(1)$ and consume m additional degrees of freedom in the interpolation.

Once $\bar{\mathbf{y}}_i, \bar{\mathbf{y}}_f$ are fixed, $\mathbf{y}(s)$ is any sufficiently smooth function such that

$$\mathbf{y}^{(k)}(0) = \mathbf{y}_i^{(k)}, \qquad \mathbf{y}^{(k)}(1) = \mathbf{y}_f^{(k)}, \qquad k = 0, \dots, r. \tag{2.106}$$

This corresponds to $2(r + 1)$ conditions for each component of $\mathbf{y}$, hence the lowest-degree polynomial that satisfies them is the Hermite polynomial of degree $2r + 1$, which is unique. The cubic interpolation used for the unicycle is the case $r = 1$. Higher-order derivatives of $\mathbf{y}$ at the boundaries (if needed to impose $\tilde{\mathbf{u}}_i, \tilde{\mathbf{u}}_f$) require increasing the degree of the polynomial accordingly.

2.2.4 Path Planning for a Chained-Form System

Consider a 2-input/3-state chained-form system, we apply the path planning algorithm to this case. We recall that the flat outputs are z_1, z_3, hence we have to interpolate (z_{1i}, z_{3i}) with (z_{1f}, z_{3f}). We interpolate linearly z_1 and with a cubic polynomial z_3:

$$z_1(s) = z_{1i} + s(z_{1f} - z_{1i}) \tag{2.107}$$

$$z_3(s) = s^3 z_{3f} - (s - 1)^3 z_{3i} + \alpha_3 s^2 (s - 1) + \beta_3 s (s - 1)^2 \tag{2.108}$$

that already satisfies the boundary conditions on z_1, z_3. We must impose the additional boundary conditions for z_2:

$$z_{2i} = \frac{z_3'(0)}{z_1'(0)}, \quad z_{2f} = \frac{z_3'(1)}{z_1'(1)} \tag{2.109}$$

we solve for α_3, β_3:

$$\alpha_3 = z_{2f}(z_{1f} - z_{1i}) - 3z_{3f} \tag{2.110}$$

$$\beta_3 = z_{2i}(z_{1f} - z_{1i}) + 3z_{3i} \tag{2.111}$$

After having $z_1(s), z_3(s)$ we can reconstruct $z_2(s), \tilde{w}_1(s), \tilde{w}_2(s)$ through the reconstruction maps, then $\mathbf{q}(s), \tilde{v}(s), \tilde{\omega}(s)$ are computed by inverting the chained form coordinate and input transformation.

By recalling that $z_1 = \theta$: one problem arises when the initial and final orientations are identical $z_{1i} = z_{1f}$, since we would have:

$$z_1'(s) = z_{1f} - z_{1i} = 0 \tag{2.112}$$

and then z_{2i}, z_{2f} can't be inferred since we would have a division by zero, in that case it is sufficient to set:

$$z_{1f} = z_{1i} + 2\pi \tag{2.113}$$

or add a mid-point $\mathbf{q}_m = \begin{pmatrix} x_m & y_m & \theta_m \end{pmatrix}^\top$ in the trajectory with $\theta_m \neq \theta_i$.

2.2.5 Path Planning via Parametrized Inputs

A different approach, not directly based on flatness, consists of writing the inputs in parametrized form and then imposing that the robot goes from $\mathbf{q}_i$ to $\mathbf{q}_f$. We assume that the n-state robot model can be rewritten in chained form:

$$\begin{cases} z_1' = \tilde{w}_1 \\ z_2' = \tilde{w}_2 \\ z_3' = z_2 \tilde{w}_1 \\ \vdots \\ z_n' = z_{n-1} \tilde{w}_1 \end{cases} \tag{2.114}$$

which is linear in $\mathbf{z}$ when $\tilde{w}_1$ is constant. Let $\mathbf{z}_i, \mathbf{z}_f$ be the initial and final configurations in the chained-form system coordinates, we define Δ as the difference between the final and initial values for z_1:

$$\Delta = z_{1f} - z_{1i} \tag{2.115}$$

and we choose the following profiles for the geometric inputs:

$$\begin{cases} \tilde{w}_1 = \text{sign}(\Delta) \\ \tilde{w}_2 = \sum_{j=0}^{n-2} c_j s^j \end{cases} \qquad s \in [0, |\Delta|] \tag{2.116}$$

by doing so, we can integrate $z_2' = \tilde{w}_2$ analytically to get the profiles (as functions of s) of $z_2, z_3, \ldots, z_n$, and impose $z_j(|\Delta|) = z_{jf}$ for each j. This leads to a linear system of the following form:

$$D(\Delta) \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-2} \end{pmatrix} = \mathbf{d}(\mathbf{z}_i, \mathbf{z}_f, \Delta) \tag{2.117}$$

if $D(\Delta)$ is invertible we can find the values for the coefficients. For example, with a system of 2 inputs and 3 states:

$$D = \begin{pmatrix} |\Delta| & \Delta^2/2 \\ \text{sign}(\Delta)\Delta^2/2 & \Delta^3/6 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} z_{2f} - z_{2i} \\ z_{3f} - z_{3i} - z_{2i}\Delta \end{pmatrix} \tag{2.118}$$

Here $\det D = -\text{sign}(\Delta)\frac{\Delta^4}{12}$, hence D is invertible as long as $\Delta \neq 0$, i.e., when the initial and final values of $z_1 = \theta$ are different (compare with the previous subsection).

2.2.6 Choice of the Timing Law

We have separated the timing law $s(t)$ from the geometric path $\mathbf{q}(s)$, now we have to assign a timing law that interpolates s_i to s_f:

$$s(t_i) = s_i, \quad s(t_f) = s_f. \tag{2.119}$$

We will select a timing law that is monotonic increasing, since it does not have any meaning for the trajectory to go backwards (in that case, this has to be defined geometrically), hence:

$$\dot{s} > 0. \tag{2.120}$$

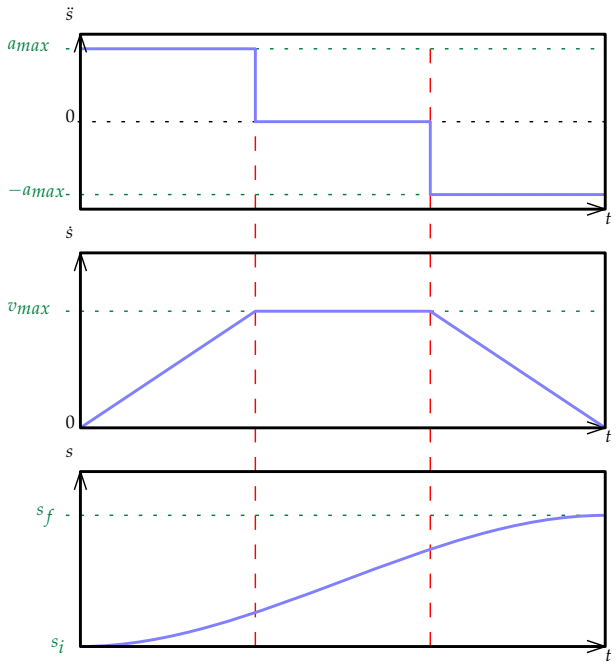
The simplest timing law is given by linear interpolation:

$$s(t) = s_i + \frac{s_f - s_i}{t_f - t_i}(t - t_i) \tag{2.121}$$

usually we use $t \in [0, T]$:

$$s(t) = s_i + (s_f - s_i)\frac{t}{T}. \tag{2.122}$$

One of the most used timing laws is the bang-coast-bang acceleration profile (which gives a trapezoidal velocity profile, hence the name **trapezoidal timing law**), since it starts and ends with zero velocity.



$$\ddot{s}(t) = \begin{cases} a_{max} & \text{if } t \in [t_i, t_i + t_m] \\ 0 & \text{if } t \in (t_i + t_m, t_f - t_m] \\ -a_{max} & \text{if } t \in (t_f - t_m, t_f] \end{cases}. \tag{2.123}$$

Here t_m is the duration of the acceleration phase (which is equal to the duration of the deceleration phase). Now let's take into account the bounds on velocity, for example, consider a unicycle subject to the following bounds for the linear and angular velocities:

$$|v| \leq v_{max}, \quad |\omega| \leq \omega_{max} \tag{2.124}$$

assume that a path $\mathbf{q}(s)$ has already been generated, with geometric inputs $\tilde{v}(s)$ and $\tilde{\omega}(s)$, we choose a linear timing law parametrized by a positive real number k:

$$s(t) = s_i + \frac{t}{k}, \quad t \in [0, k(s_f - s_i)]. \tag{2.125}$$

We assume that $\mathbf{q}(s)$ is defined such that s is the arc length. The timing law derivative is $\dot{s} = \frac{1}{k}$, with

$$v(t) = \frac{\tilde{v}(s(t))}{k}, \quad \omega(t) = \frac{\tilde{\omega}(s(t))}{k} \tag{2.126}$$

in order to respect the bounds we have to choose k as follows:

$$k = \max \left\{ \frac{\tilde{v}_{peak}}{v_{max}}, \frac{\tilde{\omega}_{peak}}{\omega_{max}} \right\} \tag{2.127}$$

where

$$\tilde{v}_{peak} = \max_{s \in [s_i, s_f]} |\tilde{v}(s)| \tag{2.128}$$

$$\tilde{\omega}_{peak} = \max_{s \in [s_i, s_f]} |\tilde{\omega}(s)|. \tag{2.129}$$

CHAPTER

3

MOTION CONTROL

In order to follow a desired trajectory, or to regulate the robot into a desired configuration, we can't simply proceed by applying the necessary inputs computed from the flat outputs, in order to reject the disturbances and the errors we must add a feedback action. In previous robotics courses, we have considered the dynamic model of the manipulators, controlled by the torques on the joints, for wheeled mobile robotics we can consider only the kinematic model, since the dynamic model is simpler, can be canceled via feedback and the wheels are always equipped with low-level control loops that can regulate velocities in a very reliable way.

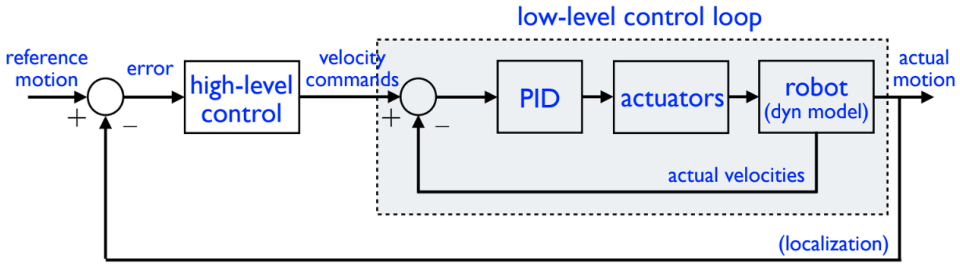


Figure 3.1: Actual control scheme.

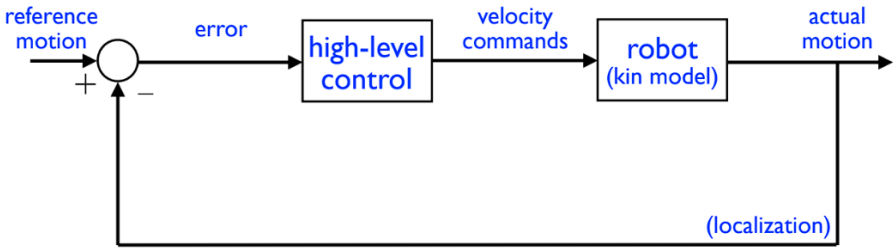


Figure 3.2: Simplified control scheme (for design).

We cover two types of control tasks, *trajectory tracking* and *regulation*, we will provide a full description of this problem for the unicycle model. In **trajectory tracking** the robot must converge to a desired Cartesian trajectory $(x_d(t), y_d(t))$, while in **posture regulation** the robot must converge to a desired con-

figuration $\mathbf{q}_d$. Trajectory tracking is more relevant in practice, since moving along planned trajectories reduces the risk of collision, however, posture regulation does not require any planning.

3.1 Trajectory Tracking

Let $(x_d(t), y_d(t))$ be an admissible desired Cartesian trajectory for the unicycle, there exist control inputs v_d, ω_d such that:

$$\begin{cases} \dot{x}_d = v_d \cos\theta_d \\ \dot{y}_d = v_d \sin\theta_d \\ \dot{\theta}_d = \omega_d \end{cases} . \tag{3.1}$$

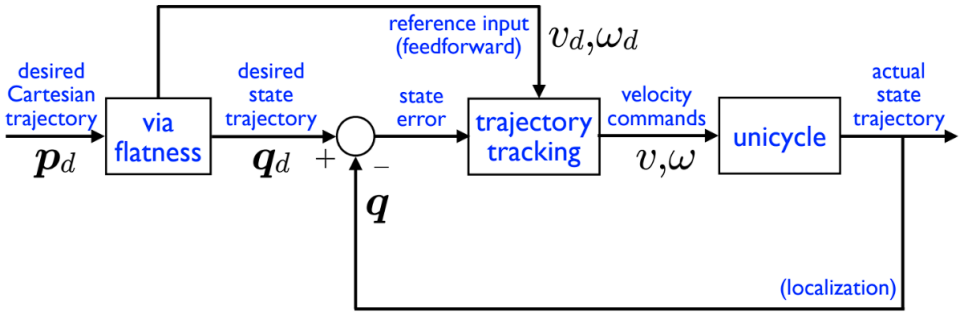
Thanks to the flatness property, from the desired trajectory $(x_d(t), y_d(t))$ we can compute the evolution of the full state and the required inputs:

$$\theta_d(t) = \text{atan2}(\dot{y}_d(t), \dot{x}_d(t)) + k\pi \quad k \in \{0, 1\} \tag{3.2}$$

$$v_d(t) = \pm\sqrt{\dot{x}_d^2(t) + \dot{y}_d^2(t)} \tag{3.3}$$

$$\omega_d(t) = \frac{\ddot{y}_d(t)\dot{x}_d(t) - \ddot{x}_d(t)\dot{y}_d(t)}{\dot{x}_d^2(t) + \dot{y}_d^2(t)}. \tag{3.4}$$

We recall that an admissible trajectory requires $(\dot{x}_d(t), \dot{y}_d(t))$ to be continuous over time, the desired state trajectory $\mathbf{q}_d(t)$ that comes from the flat outputs will be used to compute the state error, then used in the controller. The resulting block diagram will be:



The objective is to drive the state error to zero, we will express the error in a coordinate frame rotated by θ, with its x-axis parallel to the sagittal axis of the robot:

$$\mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_d - x \\ y_d - y \\ \theta_d - \theta \end{pmatrix} . \tag{3.5}$$

(e_1, e_2) is the Cartesian error $\mathbf{e}_p = (x_d - x,\ y_d - y)^\top$ expressed in a frame rotated by θ, shown in Figure 3.3. Differentiating $\mathbf{e}$ and using the unicycle kinematic model together with the desired one, we obtain the **error dynamics**:

$$\dot{e}_1 = v_d \cos e_3 - v + e_2\,\omega \tag{3.6}$$

$$\dot{e}_2 = v_d \sin e_3 - e_1\,\omega \tag{3.7}$$

$$\dot{e}_3 = \omega_d - \omega \tag{3.8}$$

which is **nonlinear** and **time-varying**, since it depends on the reference inputs $v_d(t), \omega_d(t)$.

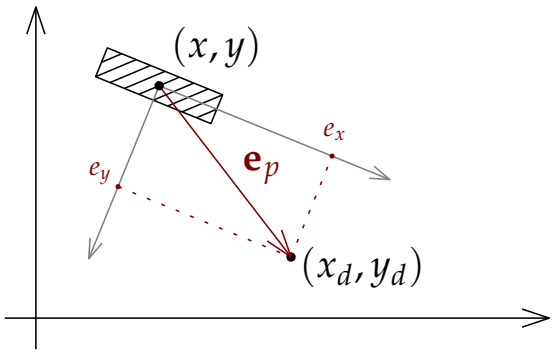


Figure 3.3: Error expressed in the rotated frame.

3.1.1 Approximate Linearization

Before applying it to the unicycle, let us recall the idea of stabilization via approximate linearization: to stabilize a nonlinear system at an equilibrium, we stabilize its approximate linearization around it. Consider a generic nonlinear system

$$\dot{\mathbf{x}} = \boldsymbol{\eta}(\mathbf{x}, \mathbf{u}) = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u} \qquad \mathbf{x} \in \mathbb{R}^n,\ \mathbf{u} \in \mathbb{R}^m \tag{3.9}$$

and assume, without loss of generality, that the origin is the desired **unforced equilibrium**, i.e. $\boldsymbol{\eta}(\mathbf{0}, \mathbf{0}) = \mathbf{0}$, or equivalently $f(\mathbf{0}) = \mathbf{0}$. The first-order Taylor expansion of $\boldsymbol{\eta}$ around $\mathbf{x} = \mathbf{0}, \mathbf{u} = \mathbf{0}$ gives

$$\boldsymbol{\eta}(\mathbf{x}, \mathbf{u}) \approx \underbrace{\boldsymbol{\eta}(\mathbf{0}, \mathbf{0})}_{=\mathbf{0}} + \left.\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{x}}\right|_{\substack{\mathbf{x}=\mathbf{0} \\ \mathbf{u}=\mathbf{0}}} \mathbf{x} + \left.\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{u}}\right|_{\substack{\mathbf{x}=\mathbf{0} \\ \mathbf{u}=\mathbf{0}}} \mathbf{u} \tag{3.10}$$

and the **approximate linearization** of the system at the origin is

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \tag{3.11}$$

where A and B are the two Jacobian matrices evaluated at the origin. Let now $\mathbf{u} = K\mathbf{x}$, so that

$$\dot{\mathbf{x}} = A\mathbf{x} + BK\mathbf{x} = (A + BK)\mathbf{x}. \tag{3.12}$$

If K is chosen in such a way that $A + BK$ is **Hurwitz**[1] (certainly possible if (A, B) is controllable), then the approximate linearization is asymptotically stable.

Theorem 3.1.1 : Lyapunov Indirect Method

If the approximate linearization of a nonlinear system at an equilibrium is asymptotically stable, then the equilibrium is **locally** asymptotically stable for the nonlinear system.

Let us apply this approach to the error dynamics of the unicycle. In the form we derived, the origin is **not** an unforced equilibrium: for $\mathbf{e} = \mathbf{0}$ the terms $v_d - v$ and $\omega_d - \omega$ must vanish to have $\dot{\mathbf{e}} = \mathbf{0}$. This is easily rectified by the following (invertible) input transformation:

$$w_1 = v_d \cos e_3 - v \tag{3.13}$$

$$w_2 = \omega_d - \omega \tag{3.14}$$

and, by substituting $\omega = \omega_d - w_2$, the error dynamics $\dot{\mathbf{e}} = \boldsymbol{\eta}(\mathbf{e}, \mathbf{w})$ takes the form

$$\dot{\mathbf{e}} = \underbrace{\begin{pmatrix} \omega_d e_2 \\ -\omega_d e_1 + v_d \sin e_3 \\ 0 \end{pmatrix}}_{f(\mathbf{e})} + \underbrace{\begin{pmatrix} 1 & -e_2 \\ 0 & e_1 \\ 0 & 1 \end{pmatrix}}_{G(\mathbf{e})} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \tag{3.15}$$

[1] Its eigenvalues have strictly negative real part.

which is at rest when $\mathbf{e} = \mathbf{0}, \mathbf{w} = \mathbf{0}$, i.e. $\boldsymbol{\eta}(\mathbf{0}, \mathbf{0}) = \mathbf{0}$. Both the drift and the input term are nonlinear, and in general the drift is also time-varying. The Jacobian matrices are

$$\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{e}} = \begin{pmatrix} 0 & \omega_d - w_2 & 0 \\ -\omega_d + w_2 & 0 & v_d \cos e_3 \\ 0 & 0 & 0 \end{pmatrix} \qquad \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{w}} = \begin{pmatrix} 1 & -e_2 \\ 0 & e_1 \\ 0 & 1 \end{pmatrix} \tag{3.16}$$

and by letting $\mathbf{e} = \mathbf{0}, \mathbf{w} = \mathbf{0}$ we obtain

$$A(t) = \begin{pmatrix} 0 & \omega_d & 0 \\ -\omega_d & 0 & v_d \\ 0 & 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3.17}$$

This system is controllable if and only if the matrix

$$\begin{pmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{pmatrix} \tag{3.18}$$

have full rank, this can be seen by building just $\begin{pmatrix} B & AB \end{pmatrix}$:

$$\operatorname{rank} \begin{pmatrix} B & AB \end{pmatrix} = \operatorname{rank} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -\omega_d & v_d \\ 0 & 1 & 0 & 0 \end{pmatrix} = 3. \tag{3.19}$$

This holds if ω_d or v_d is non-zero, so the system loses controllability when the reference is at rest ($v_d = \omega_d = 0$). Note that A is actually $A(t)$, due to the dependence of the reference inputs v_d and ω_d on time, hence the linear approximation is still a **time-varying** system. Let us define the linear feedback

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \underbrace{\begin{pmatrix} -k_1 & 0 & 0 \\ 0 & -k_2 & -k_3 \end{pmatrix}}_{K} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = K\mathbf{e} \tag{3.20}$$

the closed-loop error dynamics is

$$\dot{\mathbf{e}} = (A(t) + BK)\mathbf{e} = A_{cl}(t)\mathbf{e} = \begin{pmatrix} -k_1 & \omega_d & 0 \\ -\omega_d & 0 & v_d \\ 0 & -k_2 & -k_3 \end{pmatrix} \mathbf{e} \tag{3.21}$$

which is still time-varying. Letting

$$k_1 = k_3 = 2\zeta a \qquad k_2 = \frac{a^2 - \omega_d^2}{v_d} \tag{3.22}$$

with $a > 0$ and $\zeta \in (0, 1)$, the characteristic polynomial of $A_{cl}(t)$ becomes time-invariant and Hurwitz:

$$p(\lambda) = (\lambda + 2\zeta a)(\lambda^2 + 2\zeta a\lambda + a^2) \tag{3.23}$$

it has one real negative eigenvalue and two complex eigenvalues with damping ζ and natural frequency a.

Observation 3.1.1 : Caveats

Having the eigenvalues of $A_{cl}(t)$ fixed at every instant **does not guarantee** asymptotic stability of a time-varying system, unless v_d and ω_d are constant (rectilinear and circular trajectories). Even in this case, the asymptotic stability is only **local** (not global), since it comes from the indirect Lyapunov method. Moreover, $k_2 \to \infty$ as $v_d \to 0$, hence this controller can only be used with **persistent** Cartesian trajectories (stops are not allowed).

The **actual** velocity inputs v, ω are obtained by plugging the feedback controls w_1, w_2 in the input transformation:

$$v = v_d \cos e_3 - w_1 = v_d \cos e_3 + k_1 e_1 \tag{3.24}$$

$$\omega = \omega_d - w_2 = \omega_d + k_2 e_2 + k_3 e_3. \tag{3.25}$$

Note that $(v, \omega) \to (v_d, \omega_d)$ as $\mathbf{e} \to \mathbf{0}$, i.e. the control action reduces to the pure feedforward term. The controller is **static** and based on the state error, it needs v_d, ω_d (computed via flatness) and it needs θ both for the error rotation and for the input transformation. **Global stability** can be guaranteed by a nonlinear version of the feedback:

$$w_1 = -k_1(v_d, \omega_d)\, e_1 \tag{3.26}$$

$$w_2 = -k_2\, v_d \frac{\sin e_3}{e_3} e_2 - k_3(v_d, \omega_d)\, e_3 \tag{3.27}$$

with $k_2 > 0$ constant and k_1, k_3 bounded, positive and with bounded derivatives, provided that v_d, ω_d are bounded with bounded derivatives and do not both converge to zero. The actual inputs are then recovered from the same input transformation.

3.1.2 Output Error Feedback

Another approach is to develop the feedback action from the **output (Cartesian) error** only, without computing a desired state trajectory $\mathbf{q}_d(t)$, using as **feedforward term** the velocity $\dot{\mathbf{p}}_d$ along the reference trajectory, where $\mathbf{p} = (x, y)^\top$ is the Cartesian position of the unicycle and $\mathbf{p}_d$ the desired one. We will design this controller through the technique of **exact input-output linearization**.

Exact Input-Output Linearization

Consider a driftless nonlinear system with the same number of inputs and outputs:

$$\dot{\mathbf{x}} = G(\mathbf{x})\mathbf{u} \qquad \mathbf{x} \in \mathbb{R}^n,\ \mathbf{u} \in \mathbb{R}^m,\ \mathbf{y} \in \mathbb{R}^m \tag{3.28}$$

$$\mathbf{y} = h(\mathbf{x}) \tag{3.29}$$

differentiating the output we obtain

$$\dot{\mathbf{y}} = \frac{\partial h}{\partial \mathbf{x}}\dot{\mathbf{x}} = \underbrace{\frac{\partial h}{\partial \mathbf{x}} G(\mathbf{x})}_{T(\mathbf{x})}\, \mathbf{u} = T(\mathbf{x})\mathbf{u} \tag{3.30}$$

where T is an $m \times m$ matrix called the **decoupling matrix**. If T is invertible, we can set the **linearizing feedback**

$$\mathbf{u} = T^{-1}(\mathbf{x})\mathbf{w} \implies \dot{\mathbf{y}} = T(\mathbf{x})T^{-1}(\mathbf{x})\mathbf{w} = \mathbf{w} \tag{3.31}$$

i.e. an **exactly linear** map between the new linearizing inputs $\mathbf{w}$ and (the time derivative of) the outputs: under the action of the linearizing feedback the system behaves as a simple (vector) integrator from $\mathbf{w}$ to $\mathbf{y}$. Given a reference output $\mathbf{y}_d(t)$, the dynamics of the output error $\mathbf{e} = \mathbf{y}_d - \mathbf{y}$ is

$$\dot{\mathbf{e}} = \dot{\mathbf{y}}_d - \dot{\mathbf{y}} = \dot{\mathbf{y}}_d - \mathbf{w}. \tag{3.32}$$

Let $\mathbf{w} = \dot{\mathbf{y}}_d + K\mathbf{e}$ (feedforward plus proportional feedback), to obtain

$$\dot{\mathbf{e}} = -K\mathbf{e} \tag{3.33}$$

which is **globally exponentially stable** provided that the eigenvalues of K lie in the right half-plane. The final control law is

$$\mathbf{u} = T^{-1}(\mathbf{x})(\dot{\mathbf{y}}_d + K\mathbf{e}). \tag{3.34}$$

3.1.3 Static Feedback

Let us apply the exact input-output linearization to the unicycle, with output $\mathbf{p} = (x, y)^\top$. However, the decoupling matrix associated to the Cartesian position turns out to be **singular**:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} \tag{3.35}$$

hence exact input-output linearization is **not possible** for the output (x, y). The solution is to change slightly the output so that the new input-output map is invertible: we displace the output from the contact point of the wheel to a point B along the sagittal axis, at distance b:

$$\begin{pmatrix} x_B \\ y_B \end{pmatrix} = \begin{pmatrix} x + b\cos\theta \\ y + b\sin\theta \end{pmatrix}. \tag{3.36}$$

Differentiating with respect to time:

$$\begin{pmatrix} \dot{x}_B \\ \dot{y}_B \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & -b\sin\theta \\ \sin\theta & b\cos\theta \end{pmatrix}}_{T(\theta),\ \det T = b} \begin{pmatrix} v \\ \omega \end{pmatrix} = T(\theta) \begin{pmatrix} v \\ \omega \end{pmatrix}. \tag{3.37}$$

If $b \neq 0$ we may set

$$\begin{pmatrix} v \\ \omega \end{pmatrix} = T^{-1}(\theta) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\dfrac{\sin\theta}{b} & \dfrac{\cos\theta}{b} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \tag{3.38}$$

obtaining an input-output linearized system:

$$\dot{x}_B = w_1 \tag{3.39}$$
$$\dot{y}_B = w_2 \tag{3.40}$$
$$\dot{\theta} = \frac{w_2\cos\theta - w_1\sin\theta}{b}. \tag{3.41}$$

We achieve **global exponential convergence** of x_B, y_B to the desired trajectory by letting

$$w_1 = \dot{x}_d + k_x(x_d - x_B) \tag{3.42}$$
$$w_2 = \dot{y}_d + k_y(y_d - y_B) \tag{3.43}$$

with $k_x, k_y > 0$. The orientation θ is **not** controlled by this scheme, which is based on output error feedback: there is a **zero dynamics** (the last equation above). Still, θ must evolve as dictated by flatness. An advantage of this approach is that the desired trajectory for B can be **arbitrary**, in particular square corners may be included. The controller is **static**, based on the output error, it needs $\dot{\mathbf{p}}_d$, and it needs x, y, θ for the output reconstruction and θ also for the input transformation.

3.1.4 Dynamic Feedback

Rather than displacing the controlled output to B, we can keep the (flat) output (x, y) and achieve exact linearization by using a **dynamic compensator**. To do this, we perform a **dynamic extension** on the driving velocity channel (we "slow down" the input), by considering the velocity v as a state of the system and its acceleration a as a new input:

$$\dot{v} = a. \tag{3.44}$$

We can now differentiate further the output:

$$\ddot{x} = a\cos\theta - v\sin\theta\,\omega \tag{3.45}$$
$$\ddot{y} = a\sin\theta + v\cos\theta\,\omega \tag{3.46}$$

the new decoupling matrix is

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\theta & -v\sin\theta \\ \sin\theta & v\cos\theta \end{pmatrix}}_{T(v,\theta),\ \det T = v} \begin{pmatrix} a \\ \omega \end{pmatrix} = T(v,\theta) \begin{pmatrix} a \\ \omega \end{pmatrix} \tag{3.47}$$

which is nonsingular if $v \neq 0$. Under this assumption we may set

$$\begin{pmatrix} a \\ \omega \end{pmatrix} = T^{-1}(v,\theta) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\dfrac{\sin\theta}{v} & \dfrac{\cos\theta}{v} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \tag{3.48}$$

obtaining a fully linearized system $\ddot{x} = w_1$, $\ddot{y} = w_2$. We achieve **global exponential convergence** of x, y to the desired trajectory by letting

$$w_1 = \ddot{x}_d + k_{p,x}(x_d - x) + k_{d,x}(\dot{x}_d - \dot{x}) \tag{3.49}$$

$$w_2 = \ddot{y}_d + k_{p,y}(y_d - y) + k_{d,y}(\dot{y}_d - \dot{y}) \tag{3.50}$$

with $k_{p,x}, k_{p,y}, k_{d,x}, k_{d,y} > 0$. Since we are controlling the original flat output, there is **no zero dynamics** with this approach. The desired trajectory must be twice differentiable and **persistent**, i.e. it must be $v_d \neq 0$ always (this singularity is structural in nonholonomic systems). The controller is **dynamic** (the velocity v is the state of the compensator, obtained by integrating a), based on the output error, it needs $\dot{\mathbf{p}}_d$ and $\ddot{\mathbf{p}}_d$, and it needs x, y for the error computation and θ for the input transformation.

3.1.5 Comparison of the Controllers

The three controllers have been compared in simulation while tracking a circle, a square and a figure 8, with the following outcomes.

- **Approximate linearization**: only **local** stability is guaranteed (and only for constant v_d, ω_d). When tracking a circle it converges to zero error; when tracking a square a new transient appears at each corner; for the figure 8 local stability is not guaranteed, but the observed performance is good.
- **Static input-output linearization**: the steady-state error norm is equal to b, since it is the point B that converges to the desired trajectory, while the contact point of the wheel stays at distance b from it. The displaced output provides a **lookahead** behavior when tracking a square. Reducing b reduces the steady-state error, but the steering velocity increases (the term $1/b$ appears in T^{-1}).
- **Dynamic input-output linearization**: zero steady-state error, reasonable velocities and faster transients (e.g. at the corners of the square).

3.2 Regulation

Let's address the problem of posture regulation: we want to bring a mobile robot from a starting configuration $\mathbf{q}_i$ to a configuration $\mathbf{q}_d$ with a feedback control law. One may ask, why can't we just use trajectory tracking with a trajectory that ends in the desired configuration $\mathbf{q}_d$? None of the methods described in the previous section works for regulation: the controllers based on approximate linearization and input-output linearization via dynamic feedback require a persistent trajectory. For example, the decoupling matrix $T(v, \theta)$ is not invertible if the desired velocity v is zero. Furthermore, the input-output linearization via static feedback would lead the point $\begin{pmatrix} x_B & y_B \end{pmatrix}^\top$ to the desired destination, and not the wheel contact point.

There is a *general result* that shows that nonholonomic systems, like wheeled mobile robots, do not admit universal controllers, i.e., controllers that can stabilize arbitrary trajectories, whether persistent or not.

3.2.1 Recall on Lyapunov Theory

Let's recap some of the basic results of Lyapunov theory. Consider a general control system:

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) \tag{3.51}$$

let $\mathbf{u} = g(\mathbf{x})$ be our control input, the closed loop system is:

$$\dot{\mathbf{x}} = f(\mathbf{x}, g(\mathbf{x})) = f'(\mathbf{x}) \tag{3.52}$$

let $\mathbf{x}_e$ be an equilibrium point, without loss of generality we can assume that $\mathbf{x}_e = \mathbf{0}$:

$$f'(\mathbf{x}_e) = f'(\mathbf{0}) = \mathbf{0}. \tag{3.53}$$

Theorem 3.2.1 : Lyapunov Stability

Consider the closed loop control system of Equation (3.52). Let $V(\mathbf{x})$ be a positive definite (continuously differentiable) function, and let $\dot{V}$ be its time derivative along the trajectories of the closed loop system. Then:

- If $\dot{V}$ is negative definite, then the equilibrium $\mathbf{x}_e = \mathbf{0}$ is asymptotically stable.
- If $\dot{V}$ is negative semi-definite, then the equilibrium $\mathbf{x}_e = \mathbf{0}$ is stable (but not necessarily asymptotically stable).

Lemma 3.2.1 : Barbalat

Consider the closed loop control system of Equation (3.52), and let V be a function of the state such that

- V is positive semi-definite
- $\dot{V}$ is negative semi-definite

if $\ddot{V}$ is bounded (hence $\dot{V}$ is uniformly continuous), then $\lim_{t\to\infty} \dot{V}(t) = 0$. Moreover, since V is non-increasing and bounded from below, it converges to a limit:

$$\lim_{t\to\infty} V(t) = V_l \in [0, +\infty). \tag{3.54}$$

3.2.2 Regulation of the Cartesian Position

We build our discussion focusing on the unicycle model. Before defining a regulation control law for the whole configuration $\mathbf{q}$, we define a **partial controller** that stabilizes a desired Cartesian position, regardless of the orientation θ.

Without loss of generality, we can assume that the desired Cartesian position is $(x \quad y)^\top = (0 \quad 0)^\top$, this can be done through an appropriate coordinate transformation. Hence we define the Cartesian error as

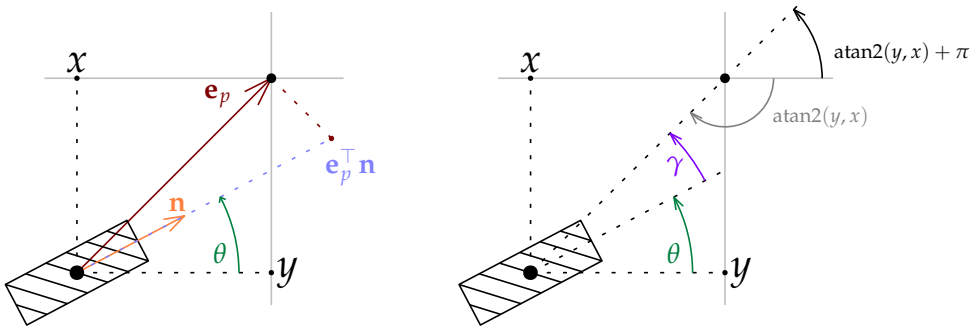
$$\mathbf{e}_p = \begin{pmatrix} x_d \\ y_d \end{pmatrix} - \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \tag{3.55}$$

Let $\mathbf{n}$ be the unit vector aligned with the sagittal axis; we will find useful the projection of $\mathbf{e}_p$ on the sagittal axis:

$$\mathbf{e}_p^\top \mathbf{n} = \begin{pmatrix} -x & -y \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = -x\cos\theta - y\sin\theta. \tag{3.56}$$

We also define the **pointing error** γ (the angle between the sagittal axis and the Cartesian error vector, normalized in $(-\pi, \pi]$):

$$\gamma = \text{atan2}(y, x) + \pi - \theta. \tag{3.57}$$



In order to converge to the desired Cartesian position, we use the feedback control law described in Definition 3.2.1.

Definition 3.2.1 : Unicycle Regulation Control Law (Cartesian Position)

Let $k_1, k_2 > 0$.

$$v = -k_1(x\cos\theta + y\sin\theta) = k_1\mathbf{e}_p^\top\mathbf{n} \tag{3.58}$$
$$\omega = k_2(\text{atan2}(y,x) + \pi - \theta) = k_2\gamma \tag{3.59}$$

The geometrical interpretation is clear:

- v is proportional to the orthogonal projection of the Cartesian error on the sagittal axis.
- ω is proportional to the pointing error.

In order to prove that this control law stabilizes the system on the desired position, we define the following Lyapunov function:

$$V = \frac{1}{2}(x^2 + y^2) \geq 0 \tag{3.60}$$

It's important to outline that $V(\mathbf{q})$ is **not positive definite**, but positive semi-definite, since it is zero when $x = y = 0$, but depends also on θ (the full configuration), hence there are infinitely many non-zero vectors that make V zero:

$$x = y = 0 \implies V(x,y,\theta) = 0 \quad \forall\theta. \tag{3.61}$$

Now we compute the derivative:

$$\dot{V} = x\dot{x} + y\dot{y} \tag{3.62}$$

here we can use the kinematic model of the unicycle to substitute $\dot{x}, \dot{y}$:

$$\dot{V} = xv\cos\theta + yv\sin\theta = v(x\cos\theta + y\sin\theta) \tag{3.63}$$

due to the fact that we are considering the closed loop system, we can substitute v with the control law:

$$\dot{V} = -k_1(x\cos\theta + y\sin\theta)^2 = -k_1(\mathbf{e}_p^\top\mathbf{n})^2 \leq 0. \tag{3.64}$$

For the same reason as for V, $\dot{V}$ is negative semi-definite, zero on all the configurations with $x = y = 0$ (and, in general, whenever $\mathbf{e}_p^\top\mathbf{n} = 0$). To apply Barbalat's lemma, we need to show that $\ddot{V}$ is bounded.

Proposition 3.2.1

$\ddot{V}$ is bounded, i.e. $\exists c \in \mathbb{R}^+$ such that $|\ddot{V}| \leq c$.

Proof:

$$\ddot{V} = \frac{d}{dt}\left[-k_1(x\cos\theta + y\sin\theta)^2\right] \tag{3.65}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\frac{d}{dt}(x\cos\theta + y\sin\theta) \tag{3.66}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\left(\dot{x}\cos\theta - x\dot{\theta}\sin\theta + \dot{y}\sin\theta + y\dot{\theta}\cos\theta\right) \tag{3.67}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\left[(\dot{x} + y\dot{\theta})\cos\theta + (\dot{y} - x\dot{\theta})\sin\theta\right] \tag{3.68}$$

we use the unicycle model and the definition of the control law:

$$\ddot{V} = -2k_1(x\cos\theta + y\sin\theta)\left[(\dot{x} + y\dot{\theta})\cos\theta + (\dot{y} - x\dot{\theta})\sin\theta\right] \tag{3.69}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\left[(v\cos\theta + y\omega)\cos\theta + (v\sin\theta - x\omega)\sin\theta\right] \tag{3.70}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\left[v(\cos^2\theta + \sin^2\theta) + \omega(y\cos\theta - x\sin\theta)\right] \tag{3.71}$$
$$= -2k_1(x\cos\theta + y\sin\theta)\left[v + \omega(y\cos\theta - x\sin\theta)\right] \tag{3.72}$$
$$= 2k_1(\mathbf{e}_p^\top\mathbf{n})\left[k_1\mathbf{e}_p^\top\mathbf{n} - k_2\gamma(\mathbf{e}_p^\top\mathbf{n}^\perp)\right] \tag{3.73}$$
$$= 2k_1^2(\mathbf{e}_p^\top\mathbf{n})^2 - 2k_1k_2\gamma(\mathbf{e}_p^\top\mathbf{n})(\mathbf{e}_p^\top\mathbf{n}^\perp) \tag{3.74}$$

where $\mathbf{n}^{\perp} = (-\sin\theta \quad \cos\theta)^{\top}$. Due to the definition of the Lyapunov function, we can state that $\mathbf{e}_p$ is bounded, since V is proportional to the squared norm of $\mathbf{e}_p$, and V never increases, hence $\|\mathbf{e}_p(t)\| \leq \|\mathbf{e}_p(0)\|, \forall t$. Since all the quantities in Equation (3.74) are bounded (γ is bounded since it is an angle in $(-\pi, \pi]$), $\ddot{V}$ is bounded. ■

Having proved that $\ddot{V}$ is bounded, by Barbalat's lemma we can say that $\dot{V}$ will converge to zero at steady state:

$$\dot{V} = -k_1(\mathbf{e}_p^{\top}\mathbf{n})^2 = 0 \iff \mathbf{e}_p^{\top}\mathbf{n} = 0 \tag{3.75}$$

so we can assert that at steady state $\mathbf{e}_p^{\top}\mathbf{n} = 0$, this is true only in two possible cases:

- $\mathbf{e}_p$ is zero, hence we have correctly regulated the desired Cartesian position.
- $\mathbf{e}_p$ is orthogonal to $\mathbf{n}$.

Observation 3.2.1

At steady state, it is impossible to have a non-zero $\mathbf{e}_p$ orthogonal to $\mathbf{n}$.

This is pretty easy to show, since the control law for ω is

$$\omega = k_2\gamma$$

where γ is the pointing error, with a non-zero $\mathbf{e}_p$ orthogonal to $\mathbf{n}$ we would have $\gamma = \pm\frac{\pi}{2}$, hence the angular velocity would be non-zero: $\dot{\theta} = \omega = \pm k_2\frac{\pi}{2}$, and by definition this is not steady state: the only possible steady state condition is $\mathbf{e}_p = \mathbf{0}$.

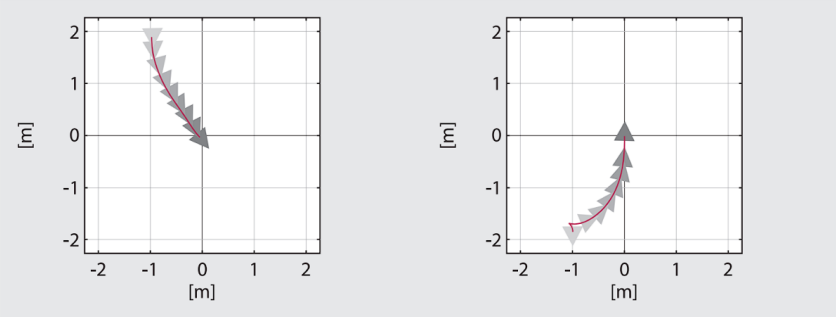


Figure 3.4: Simulations of the unicycle under the regulation control law for the Cartesian position.

3.2.3 Regulation of the Whole Configuration

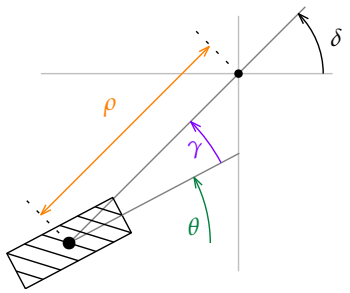
Now we define a control law to achieve any desired posture $\mathbf{q}_d$, including the orientation θ. We will work in polar coordinates, hence we define the following coordinate transformation:

$$\rho = \sqrt{x^2 + y^2} \tag{3.76}$$

$$\gamma = \text{atan2}(y, x) + \pi - \theta \tag{3.77}$$

$$\delta = \gamma + \theta. \tag{3.78}$$

ρ is the Cartesian distance from the origin, δ the angle between the Cartesian error vector and the horizontal axis, γ the angle between the sagittal axis and the Cartesian error vector.



Let's derive the kinematic model, starting from the time evolution of ρ:

$$\dot{\rho} = \frac{1}{\sqrt{x^2+y^2}}(x\dot{x}+y\dot{y}) \tag{3.79}$$
$$= \frac{v}{\rho}(x\cos\theta + y\sin\theta) \tag{3.80}$$

we denote by β the angle between the horizontal axis and the line from the origin to the wheel contact point:

$$\beta = \text{atan2}(y,x) \implies \begin{cases} x = \rho\cos\beta \\ y = \rho\sin\beta \end{cases} \tag{3.81}$$

hence

$$\dot{\rho} = \frac{v}{\rho}(\rho\cos\beta\cos\theta + \rho\sin\beta\sin\theta) \tag{3.82}$$
$$= v\cos(\beta-\theta) \tag{3.83}$$
$$= -v\cos(\beta-\theta+\pi) \tag{3.84}$$
$$= -v\cos(\text{atan2}(y,x)-\theta+\pi) = -v\cos\gamma. \tag{3.85}$$

We proceed with the computation of $\dot{\gamma}$:

$$\dot{\gamma} = \frac{d}{dt}\left(\text{atan2}(y,x)+\pi-\theta\right) \tag{3.86}$$
$$= \frac{x\dot{y}-y\dot{x}}{x^2+y^2} - \dot{\theta} \tag{3.87}$$
$$= \frac{x(v\sin\theta)-y(v\cos\theta)}{\rho^2} - \omega \tag{3.88}$$
$$= \frac{v}{\rho^2}(x\sin\theta - y\cos\theta) - \omega \tag{3.89}$$
$$= \frac{v}{\rho^2}\left[-\rho\cos(\gamma+\theta)\sin\theta + \rho\sin(\gamma+\theta)\cos\theta\right] - \omega \tag{3.90}$$
$$= \frac{v}{\rho}\left[\sin(\gamma+\theta)\cos\theta - \cos(\gamma+\theta)\sin\theta\right] - \omega \tag{3.91}$$
$$= \frac{v}{\rho}\sin\left((\gamma+\theta)-\theta\right) - \omega \tag{3.92}$$
$$= \frac{\sin\gamma}{\rho}v - \omega \tag{3.93}$$

for $\dot{\delta}$ the computation is straightforward:

$$\dot{\delta} = \frac{d}{dt}(\gamma+\theta) \tag{3.94}$$
$$= \dot{\gamma}+\dot{\theta} \tag{3.95}$$
$$= \left(\frac{\sin\gamma}{\rho}v - \omega\right) + \omega \tag{3.96}$$
$$= \frac{\sin\gamma}{\rho}v. \tag{3.97}$$

The kinematic model of the unicycle in polar coordinates is:

$$\dot{\rho} = -v\cos\gamma \tag{3.98}$$

$$\dot{\gamma} = \frac{\sin\gamma}{\rho}v - \omega \tag{3.99}$$

$$\dot{\delta} = \frac{\sin\gamma}{\rho}v. \tag{3.100}$$

How to define a control law to stabilize $(\rho \quad \gamma \quad \delta)^\top$ to zero? In the previous subsection, we defined a control law and then proved its correctness through Lyapunov analysis, now we first define a Lyapunov candidate, and then design a law for v and ω in order to make $\dot{V}$ negative definite. Consider:

$$V = \frac{1}{2}(\rho^2 + \gamma^2 + \delta^2) = \frac{1}{2}\left\|(\rho \quad \gamma \quad \delta)^\top\right\|^2 \geq 0 \tag{3.101}$$

differently from the previous case, this V is *positive definite*, zero only at $\rho = \gamma = \delta = 0$. Its derivative is

$$\dot{V} = \rho\dot{\rho} + \gamma\dot{\gamma} + \delta\dot{\delta} \tag{3.102}$$

$$= -v\cos\gamma\rho + \left(\frac{\sin\gamma}{\rho}v - \omega\right)\gamma + \frac{\sin\gamma}{\rho}v\delta \tag{3.103}$$

v and ω appear in $\dot{V}$, hence by an appropriate choice we can modify $\dot{V}$ in order to obtain the desired property. We rewrite $\dot{V}$ as follows:

$$\dot{V} = -\rho v\cos\gamma + v\frac{\sin\gamma}{\rho}(\gamma + \delta) - \gamma\omega \tag{3.104}$$

the first choice that we make is $v = k_1\rho\cos\gamma$, in this way:

$$\dot{V} = -\rho v\cos\gamma + v\frac{\sin\gamma}{\rho}(\gamma + \delta) - \gamma\omega \tag{3.105}$$

$$\dot{V} = -\rho(k_1\rho\cos\gamma)\cos\gamma + (k_1\rho\cos\gamma)\frac{\sin\gamma}{\rho}(\gamma + \delta) - \gamma\omega \tag{3.106}$$

$$= \underbrace{-k_1\rho^2\cos^2\gamma}_{\leq 0} + k_1\sin\gamma\cos\gamma(\gamma + \delta) - \gamma\omega \tag{3.107}$$

now we can proceed with an appropriate choice of ω, let $\omega = k_2\gamma + \alpha$, with α yet to be chosen:

$$\dot{V} = -k_1\rho^2\cos^2\gamma + k_1\sin\gamma\cos\gamma(\gamma + \delta) - \gamma\omega \tag{3.108}$$

$$= -k_1\rho^2\cos^2\gamma + k_1\sin\gamma\cos\gamma(\gamma + \delta) - \gamma(k_2\gamma + \alpha) \tag{3.109}$$

$$= \underbrace{-k_1\rho^2\cos^2\gamma}_{\leq 0} + k_1\sin\gamma\cos\gamma(\gamma + \delta) \underbrace{-k_2\gamma^2}_{\leq 0} - \gamma\alpha \tag{3.110}$$

at this point we use α to cancel the remaining terms that are not guaranteed to be non-positive:

$$\text{we want: } -\gamma\alpha + k_1\sin\gamma\cos\gamma(\gamma + \delta) = 0 \tag{3.111}$$

$$\text{we set: } \alpha = \frac{1}{\gamma}k_1\sin\gamma\cos\gamma(\gamma + \delta) \tag{3.112}$$

hence:

$$\dot{V} = -k_1\rho^2\cos^2\gamma + k_1\sin\gamma\cos\gamma(\gamma + \delta) - k_2\gamma^2 - \gamma\alpha \tag{3.113}$$

$$= -k_1\rho^2\cos^2\gamma + k_1\sin\gamma\cos\gamma(\gamma + \delta) - k_2\gamma^2 - \gamma\frac{1}{\gamma}k_1\sin\gamma\cos\gamma(\gamma + \delta) \tag{3.114}$$

$$= -k_1\rho^2\cos^2\gamma - k_2\gamma^2 \leq 0. \tag{3.115}$$

$\dot{V}$ is negative semi-definite, having its zeros on the line $\gamma = \rho = 0$, for any value of δ.

Definition 3.2.2 : Unicycle Regulation Control Law (Full Configuration)

The control law found through Lyapunov analysis is:

$$v = k_1 \rho \cos\gamma \tag{3.116}$$

$$\omega = k_2 \gamma + k_1 \frac{\sin\gamma \cos\gamma}{\gamma}(\gamma + \delta). \tag{3.117}$$

Notice that $\frac{\sin\gamma}{\gamma}$ is extended by continuity to 1 for $\gamma = 0$ (the apparent singularity in α is removable). It can be shown (proof omitted) that $\ddot{V}$ is bounded, hence by Barbalat's lemma $\dot{V}$ will converge to zero. At steady state we have $\gamma = \rho = 0$, and this (from the kinematic model in Equation (3.100)) implies that at steady state $\dot{\delta} = 0$ (indeed, by substituting $v = k_1 \rho \cos\gamma$ in Equation (3.100) we get $\dot{\delta} = k_1 \sin\gamma \cos\gamma$, which vanishes for $\gamma = 0$, avoiding the indeterminate form at $\rho = 0$), hence also δ will converge to a limit, and this will be exactly zero (proof omitted).

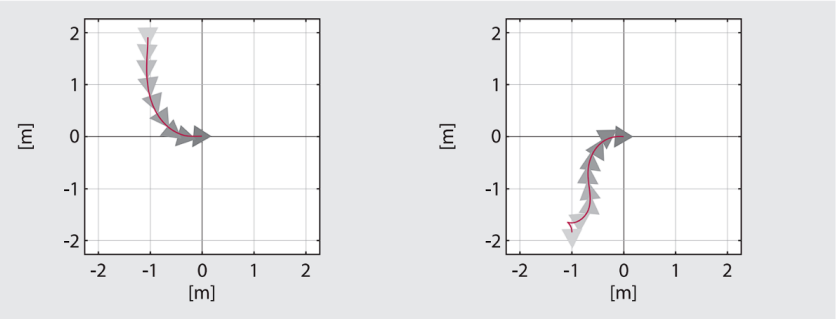


Figure 3.5: Simulations of the unicycle under the regulation control law for the whole configuration.

CHAPTER

4

MOBILE MANIPULATORS

In this short chapter we discuss the kinematics of mobile manipulators, i.e., manipulators mounted on a moving base. Let $\mathbf{q}_m \in \mathbb{R}^{n_m}$ be the configuration (joint coordinates) of a manipulator placed on a mobile robot with configuration $\mathbf{q}_b \in \mathbb{R}^{n_b}$. The configuration of the whole system is $\mathbf{q} = (\mathbf{q}_b \quad \mathbf{q}_m)^\top$. We have seen that, for a manipulator, the kinematic model is trivial (since it is unconstrained):

$$\dot{\mathbf{q}}_m = \mathbf{u}_m \tag{4.1}$$

hence, the kinematic model of the whole system is

$$\dot{\mathbf{q}} = \begin{pmatrix} \dot{\mathbf{q}}_b \\ \dot{\mathbf{q}}_m \end{pmatrix} = \begin{pmatrix} G_b(\mathbf{q}_b)\mathbf{u}_b \\ \mathbf{u}_m \end{pmatrix} = \begin{pmatrix} G_b(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \begin{pmatrix} \mathbf{u}_b \\ \mathbf{u}_m \end{pmatrix} = G(\mathbf{q})\mathbf{u} \tag{4.2}$$

where $\mathcal{O}$ is a matrix with all zero entries, while $G_b(\mathbf{q}_b)$ is the matrix of the mobile robot kinematic model that we have seen in the previous chapters. For example, if the mobile base is a unicycle we have:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{q}_{m,1} \\ \vdots \\ \dot{q}_{m,n_m} \end{pmatrix} = \left(\begin{array}{cc|ccc} \cos\theta & 0 & 0 & \dots & 0 \\ \sin\theta & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \hline 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{array}\right) \begin{pmatrix} v \\ \omega \\ u_{m,1} \\ \vdots \\ u_{m,n_m} \end{pmatrix}. \tag{4.3}$$

Motion planning for mobile manipulators relies fundamentally on kinematic or geometric modeling. When planning takes place directly in the configuration space (such as finding a collision-free path between a start configuration and a goal configuration), the problem conveniently decouples. This separation allows us to plan paths for the mobile platform and the robotic arm independently, using well-established standalone algorithms for each subsystem.

In practice, however, mobile manipulators typically perform manipulation tasks where the end-effector must follow a strictly prescribed trajectory in the workspace. Introducing task-space constraints couples the two subsystems and requires solving the *inverse differential kinematics problem*. To do this, we define a unified differential kinematic map that transforms the desired end-effector velocities directly into coordinated velocity commands for both the base and the arm joints.

4.1 Differential Kinematics

Consider the end-effector pose:

$$\mathbf{y}_e = k_e(\mathbf{q}) = k_e(\mathbf{q}_b, \mathbf{q}_m) \tag{4.4}$$

$\mathbf{y}_e$ is composed of the Cartesian position $\mathbf{p}_e$ and a representation of its orientation (Euler angles, rotation matrix, or others). The corresponding differential kinematics is

$$\dot{\mathbf{y}}_e = \underbrace{\frac{\partial k_e}{\partial \mathbf{q}_b}}_{J_b} \dot{\mathbf{q}}_b + \underbrace{\frac{\partial k_e}{\partial \mathbf{q}_m}}_{J_m} \dot{\mathbf{q}}_m \tag{4.5}$$

we substitute $\dot{\mathbf{q}}_b$ with the mobile base kinematic model

$$\dot{\mathbf{y}}_e = J_b(\mathbf{q}) G_b(\mathbf{q}_b) \mathbf{u}_b + J_m(\mathbf{q}) \mathbf{u}_m \tag{4.6}$$

in matrix form:

$$\dot{\mathbf{y}}_e = \begin{pmatrix} J_b(\mathbf{q}) G_b(\mathbf{q}_b) & J_m(\mathbf{q}) \end{pmatrix} \begin{pmatrix} \mathbf{u}_b \\ \mathbf{u}_m \end{pmatrix} = J_{mm}(\mathbf{q}) \mathbf{u}. \tag{4.7}$$

The $r \times (m_b + n_m)$ matrix $J_{mm}(\mathbf{q})$, where r is the dimension of the task vector ($r = 6$ for a full spatial pose) and $m_b = \dim \mathbf{u}_b$ is the number of pseudo-velocities of the mobile base, is called the **Jacobian of the mobile manipulator**, although it is not a Jacobian in the strict mathematical sense (it is not the matrix of the partial derivatives of a function, since it contains the mobile base kinematic model $G_b(\mathbf{q}_b)$). Typically, $m_b + n_m > r$, hence mobile manipulators are often kinematically redundant. The matrix J_{mm} is used for singularity analysis and to solve the inverse differential kinematics problem.

4.1.1 Example: 2R Planar Manipulator on a Unicycle

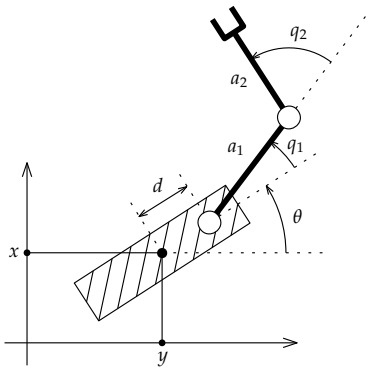
We analyze the model of a 2R planar robot mounted on a unicycle, the arm is mounted on the sagittal axis, at distance d from the center of the wheel axle. Its configuration is

$$\mathbf{q} = \begin{pmatrix} x & y & \theta & q_1 & q_2 \end{pmatrix}^\top \tag{4.8}$$

and the velocity inputs are

$$\mathbf{u} = \begin{pmatrix} v & \omega & \dot{q}_1 & \dot{q}_2 \end{pmatrix}^\top. \tag{4.9}$$

We denote by a_1, a_2 the lengths of the links of the arm.



The position of the end-effector (our task vector) is:

$$\mathbf{y}_e = \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} x + d\cos\theta + a_1\cos(\theta + q_1) + a_2\cos(\theta + q_1 + q_2) \\ y + d\sin\theta + a_1\sin(\theta + q_1) + a_2\sin(\theta + q_1 + q_2) \end{pmatrix} \tag{4.10}$$

to simplify the notation we write:

$$c_\theta = \cos\theta \qquad s_\theta = \sin\theta \tag{4.11}$$

$$c_{\theta 1} = \cos(\theta + q_1) \qquad s_{\theta 1} = \sin(\theta + q_1) \tag{4.12}$$

$$c_{\theta 12} = \cos(\theta + q_1 + q_2) \qquad s_{\theta 12} = \sin(\theta + q_1 + q_2) \tag{4.13}$$

so

$$\begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} x + dc_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} \\ y + ds_\theta + a_1 s_{\theta 1} + a_2 s_{\theta 12} \end{pmatrix} \tag{4.14}$$

now we compute the two Jacobian matrices:

$$J_b = \begin{pmatrix} 1 & 0 & -ds_\theta - a_1 s_{\theta 1} - a_2 s_{\theta 12} \\ 0 & 1 & dc_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} \end{pmatrix}, \quad J_m = \begin{pmatrix} -a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_2 s_{\theta 12} \\ a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_2 c_{\theta 12} \end{pmatrix} \tag{4.15}$$

the unicycle kinematic model matrix is:

$$G_b(\mathbf{q}_b) = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{pmatrix} \tag{4.16}$$

so

$$J_b(\mathbf{q}) G_b(\mathbf{q}_b) = \begin{pmatrix} c_\theta & -d\, s_\theta - a_1 s_{\theta 1} - a_2 s_{\theta 12} \\ s_\theta & d\, c_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} \end{pmatrix} \tag{4.17}$$

finally, the mobile manipulator Jacobian is:

$$J_{mm}(\mathbf{q}) = \begin{pmatrix} c_\theta & -d\, s_\theta - a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_2 s_{\theta 12} \\ s_\theta & d\, c_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_2 c_{\theta 12} \end{pmatrix} \tag{4.18}$$

If $d > 0$, J_{mm} is never singular (it always has rank 2), hence any end-effector velocity can be realized at any configuration (adding the degrees of freedom of the base allows us to remove the singularities of the 2R arm alone).

4.2 Inverse Differential Kinematics

The methods for solving the IDK in presence of kinematic redundancy that we have seen in previous robotics courses apply to mobile manipulators with minimal adaptation. We use the pseudo-inverse matrix $J_{mm}^{\#}$: if $\dot{\mathbf{y}}_d$ is the desired task velocity, we select as input velocities

$$\mathbf{u} = J_{mm}^{\#} \dot{\mathbf{y}}_d \tag{4.19}$$

this realizes the desired velocity (provided that J_{mm} has full row rank):

$$\dot{\mathbf{y}}_e = J_{mm} J_{mm}^{\#} \dot{\mathbf{y}}_d = \dot{\mathbf{y}}_d \tag{4.20}$$

in particular, $J_{mm}^{\#}$ provides the minimum norm solution. We recall that $J_{mm}^{\#} = J_{mm}^{\top}(J_{mm} J_{mm}^{\top})^{-1}$ can be computed analytically if J_{mm} has full row rank, otherwise it can be computed via numerical methods (e.g., the SVD). In order to exploit the additional degrees of freedom given by the kinematic redundancy, we add to the solution in Equation (4.19) a term that is in the null space of J_{mm}, in this way we can move the configuration along a desired direction of the configuration space without affecting the task velocity. This is done in the following way:

- Let $\mathbf{u}_N$ be an arbitrary vector that represents a desired direction in the configuration space.
- We project $\mathbf{u}_N$ onto the null space of J_{mm} through the projector:

$$\underbrace{(I - J_{mm}^{\#} J_{mm})}_{\Pi} \mathbf{u}_N. \tag{4.21}$$

- By doing so we have:

$$\Pi \mathbf{u}_N \in \mathcal{N}(J_{mm}) \implies J_{mm}(\Pi \mathbf{u}_N) = \mathbf{0}. \tag{4.22}$$

The solution to the IDK is then completed with such a null-space projection:

$$\mathbf{u} = J_{mm}^{\#}\dot{\mathbf{y}}_d + (I - J_{mm}^{\#}J_{mm})\mathbf{u}_N. \tag{4.23}$$

How do we choose $\mathbf{u}_N$? Usually it is a direction that tends to maximize (or minimize) a performance (or cost) criterion. Let $H(\mathbf{q})$ be a scalar function that assigns a value of "goodness" to a given configuration ($H(\mathbf{q})$ might be, e.g., the distance from a singular configuration). Since we want to maximize (or minimize) $H(\mathbf{q})$, a preferred direction is its weighted gradient $\pm\alpha\nabla H(\mathbf{q})$ (the sign establishes whether H should be maximized or minimized).

However, we can't simply set $\mathbf{u}_N = \pm\alpha\nabla H(\mathbf{q})$, since the first m_b components of the vector $\mathbf{u}$ are pseudo-velocities, rather than generalized velocities, and the nonholonomic constraints might prevent the motion defined by the first m_b components of $\mathbf{u}_N$. In order to identify a suitable velocity command $\mathbf{u}_N$ that does not violate the nonholonomic constraints of the mobile base, we have to look at the time derivative of $H(\mathbf{q})$ along the system motion:

$$\dot{H}(\mathbf{q}) = \frac{\partial H}{\partial \mathbf{q}}\dot{\mathbf{q}} = (\nabla H(\mathbf{q}))^\top G(\mathbf{q})\mathbf{u} \tag{4.24}$$

so, to obtain the maximum local improvement of $H(\mathbf{q})$, we have to set:

$$\mathbf{u}_N = \pm\alpha G^\top(\mathbf{q})\nabla H(\mathbf{q}) = \pm\alpha \begin{pmatrix} G_b^\top(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \nabla H(\mathbf{q}). \tag{4.25}$$

To improve robustness, instead of using $\dot{\mathbf{y}}_d$ directly as the desired velocity, we can close the loop with a feedback action on the error $\mathbf{y}_d - \mathbf{y}_e$:

$$\dot{\mathbf{y}}_d + K(\mathbf{y}_d - \mathbf{y}_e) \tag{4.26}$$

with K being a positive definite matrix. Ultimately, to follow a task vector trajectory $\mathbf{y}_d(t)$ over time we select (at each time step) the following control law:

$$\mathbf{u} = J_{mm}^{\#}\left(\dot{\mathbf{y}}_d + K(\mathbf{y}_d - \mathbf{y}_e)\right) \pm \alpha(I - J_{mm}^{\#}J_{mm}) \begin{pmatrix} G_b^\top(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \nabla H(\mathbf{q}). \tag{4.27}$$

However, keep in mind that kinematic control of mobile manipulators only results in acceptable performance if the required velocities are small, so that the inertial couplings between the coordinates of the base and those of the manipulator remain negligible. In general, as for manipulators, robust feedback control laws for mobile manipulators should be designed on the basis of their dynamic model.

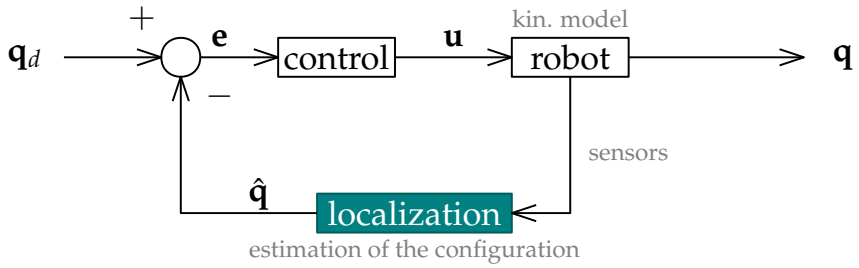
CHAPTER

5 LOCALIZATION

5.1 Odometric localization

All the discussion about planning and control in the previous chapters assumes that, in the control loop, the information about the current configuration $\mathbf{q}$ is known. Clearly, this is not true a priori: we need to use sensors to measure the internal and external state of our robot and, from them, reconstruct the configuration $\mathbf{q}$.

The encoders of mobile robots are incremental and measure only the rotation of the wheels (i.e., their angular displacement over time); **localization** is the procedure of estimating the robot configuration $\mathbf{q}$ in real time.



We will explore how to build the localization block in this scheme. We will use the equations of motion of the robot and, by integrating them, we will track the configuration evolution. This kind of state reconstruction uses only the internal sensors of the robot, which measure only the internal state (wheel angular speed) and not the environment.

We need to use a *digital control scheme* for the low-level control of the robot; this assumes that the control inputs are constant during the sampling interval. We use the unicycle model as an example. Let T_s be the sampling interval; the time window where the inputs v_k, ω_k are constant is:

$$[t_k, t_{k+1}] = [t_k, t_k + T_s] \tag{5.1}$$

where k is the time step. Given the unicycle model, it is easy to see how a constant set of inputs v_k, ω_k results in an arc of a circle in the Cartesian plane.

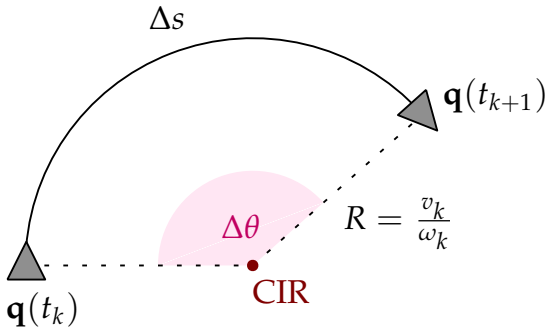


Figure 5.1: Arc of a circle traced during the sampling interval.

In Figure 5.1, $\Delta\theta$ is the variation of the orientation angle and Δs is the length of the path traveled in the sampling interval; the circle has radius $R = \frac{v_k}{\omega_k}$ (for $\omega_k \neq 0$).

5.1.1 Euler Integration

Now let us see how to integrate the equations numerically. The kinematic model is nonlinear:

$$\dot{\mathbf{q}} = f(\mathbf{q}, \mathbf{u}) \tag{5.2}$$

Consider the time interval $t \in [t_k, t_{k+1}]$; starting from $\mathbf{q}(t_k)$ we want to compute the displacement $\Delta\mathbf{q} = \mathbf{q}(t_{k+1}) - \mathbf{q}(t_k)$ in order to obtain the next configuration $\mathbf{q}(t_{k+1})$:

$$\mathbf{q}(t_{k+1}) - \mathbf{q}(t_k) = \int_{t_k}^{t_{k+1}} \dot{\mathbf{q}}\, dt = \int_{t_k}^{t_{k+1}} f(\mathbf{q}(t), \mathbf{u}(t))\, dt \tag{5.3}$$

$$\implies \mathbf{q}_{k+1} = \mathbf{q}_k + \int_{t_k}^{t_{k+1}} f(\mathbf{q}(t), \mathbf{u}(t))\, dt \tag{5.4}$$

One simple method is **Euler Integration**, where we assume that $f(\mathbf{q}, \mathbf{u})$ is constant during $[t_k, t_{k+1}]$:

$$\mathbf{q}_{k+1} = \mathbf{q}_k + \int_{t_k}^{t_{k+1}} f(\mathbf{q}_k, \mathbf{u}_k) dt = \mathbf{q}_k + f(\mathbf{q}_k, \mathbf{u}_k) \int_{t_k}^{t_{k+1}} dt = \mathbf{q}_k + f(\mathbf{q}_k, \mathbf{u}_k) T_s \tag{5.5}$$

This assumption of course does not hold: $\mathbf{u}_k$ is constant due to the digital control, but $\mathbf{q}$ is time-varying, hence this integration method accumulates a small error at each time step. Geometrically, we are assuming that the robot moves on a straight line in a time interval (we have seen that it moves on an arc of a circle). Euler integration applied to the unicycle model gives the following:

$$x_{k+1} = x_k + v_k T_s \cos\theta_k \tag{5.6}$$

$$y_{k+1} = y_k + v_k T_s \sin\theta_k \tag{5.7}$$

$$\theta_{k+1} = \theta_k + \omega_k T_s \tag{5.8}$$

The update of θ_{k+1} is exact, while x_{k+1} and y_{k+1} are approximated.

5.1.2 Runge-Kutta Integration

The error accumulated over many steps by the Euler method is not negligible. A more precise method is the **Runge-Kutta Integration** (here, its second-order version), which uses the following approximation:

$$f(\mathbf{q}(t), \mathbf{u}(t)) \simeq f\left(\mathbf{q}_k + \frac{T_s}{2} f(\mathbf{q}_k, \mathbf{u}_k), \mathbf{u}_k\right) \tag{5.9}$$

in the time interval $[t_k, t_{k+1}]$. It uses the average orientation, that is $\theta_k + \frac{\omega_k T_s}{2}$, hence the update rule is

$$x_{k+1} = x_k + v_k T_s \cos\left(\theta_k + \frac{\omega_k T_s}{2}\right) \tag{5.10}$$

$$y_{k+1} = y_k + v_k T_s \sin\left(\theta_k + \frac{\omega_k T_s}{2}\right) \tag{5.11}$$

$$\theta_{k+1} = \theta_k + \omega_k T_s \tag{5.12}$$

Note that x_{k+1} and y_{k+1} are still approximate, but more accurate.

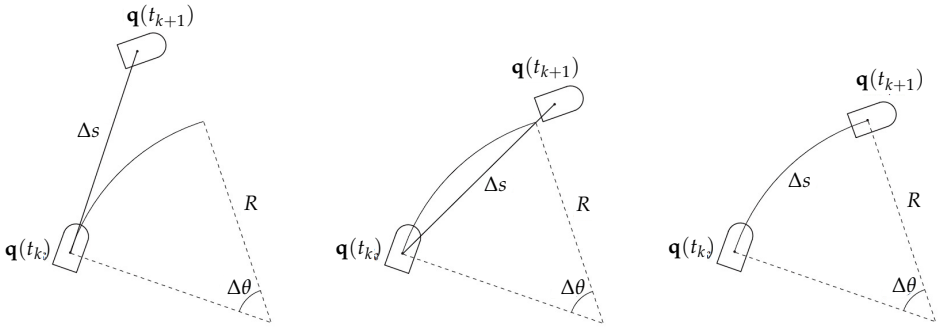
5.1.3 Exact Integration

The last method that we see simply computes the exact integral in the time interval $[t_k, t_{k+1}]$. This method requires the dynamics to have an analytical solution for the integral, hence it cannot be applied to all systems. The unicycle model has equations of motion that are easily integrable for constant v, ω:

$$\begin{cases} x_{k+1} = x_k + \frac{v_k}{\omega_k}(\sin\theta_{k+1} - \sin\theta_k) \\ y_{k+1} = y_k - \frac{v_k}{\omega_k}(\cos\theta_{k+1} - \cos\theta_k) \\ \theta_{k+1} = \theta_k + \omega_k T_s \end{cases} \tag{5.13}$$

$$\implies \begin{cases} x_{k+1} = x_k + \frac{v_k}{\omega_k}(\sin(\theta_k + \omega_k T_s) - \sin\theta_k) \\ y_{k+1} = y_k - \frac{v_k}{\omega_k}(\cos(\theta_k + \omega_k T_s) - \cos\theta_k) \\ \theta_{k+1} = \theta_k + \omega_k T_s \end{cases} . \tag{5.14}$$

For $\omega_k = 0$ the expressions above are undefined (division by zero), but their limit for $\omega_k \to 0$ exists and coincides with the solution given by Euler and Runge-Kutta, i.e., $x_{k+1} = x_k + v_k T_s \cos\theta_k$ and $y_{k+1} = y_k + v_k T_s \sin\theta_k$. When $\omega_k \simeq 0$, conditional instructions must be used in the implementation to avoid numerical problems; this is not desirable, and in real cases Euler integration is usually used, together with methods that correct the estimate using sensors.



The methods seen use the nominal inputs v_k, ω_k. Since no actuation system is ideal, we cannot assume that any requested input is immediately realized; hence, instead of using v_k and ω_k, we should compute their "real" values inferred from the encoder readings. First of all we use the following identities:

$$v_k T_s = \Delta s, \qquad \omega_k T_s = \Delta\theta \tag{5.15}$$

Δs and $\Delta\theta$ are reconstructed via proprioceptive sensors. For example, in a differential-drive robot we have access to the total rotations of the right and left wheels measured by the encoders in a time window:

$$\Delta\phi_R, \quad \Delta\phi_L \tag{5.16}$$

Δs and $\Delta\theta$ are then reconstructed as follows (where r is the wheel radius and d the distance between the wheels):

$$\Delta s = \frac{r}{2}(\Delta\phi_R + \Delta\phi_L), \qquad \Delta\theta = \frac{r}{d}(\Delta\phi_R - \Delta\phi_L). \tag{5.17}$$

Odometric localization maintains an estimate of a robot's configuration via iterative integration of its kinematic model. While fundamental to autonomous navigation, it inherently suffers from cumulative odometric drift, which grows without bound over extended trajectories. This divergence stems primarily from unmodeled physical perturbations such as wheel slippage, systematic uncertainties such as imperfect wheel radius calibration, and numerical integration errors. Consequently, practical and effective localization frameworks cannot rely solely on proprioceptive dead reckoning, necessitating the sensor fusion of proprioceptive measurements with exteroceptive sensing modalities to bound estimation errors over time.

5.2 Kalman Filter

In the previous section, we have seen how to integrate the equations of motion in order to track the configuration evolution over time. Odometric localization alone is subject to an error that diverges over time; effective localization methods use proprioceptive as well as exteroceptive sensors, exploiting the information about a known map in order to compare it with the actual sensor readings. Instead of maintaining a single hypothesis on the configuration, the **Kalman Filter** maintains a probability distribution over the space of all possible hypotheses.

Although in the robotics context we denote the configuration by $\mathbf{q}$, in this section we will present the filter for a generic dynamical system, denoting the state by $\mathbf{x}$.

5.2.1 Probability Recap

We recall three basic definitions of probability theory. Let $\mathbf{X}$ be a random variable over $\mathbb{R}^n$, with density function $f_X(\mathbf{x})$, its **expected value** is:

$$\mathbb{E}(\mathbf{X}) = \bar{\mathbf{X}} = \int_{\mathbb{R}^n} \mathbf{x} f_X(\mathbf{x}) d\mathbf{x}. \tag{5.18}$$

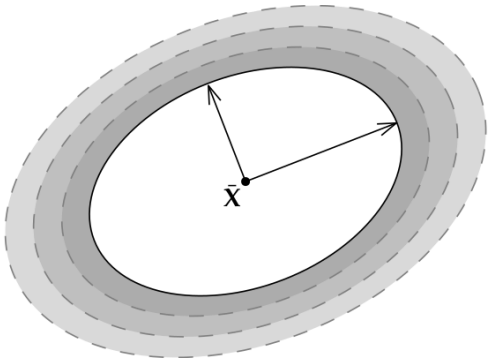
The **covariance matrix** is defined as follows:

$$P_X = \mathbb{E}\left((\mathbf{X} - \bar{\mathbf{X}})(\mathbf{X} - \bar{\mathbf{X}})^\top\right). \tag{5.19}$$

A random variable has a **multivariate Gaussian distribution** if its density function is:

$$f_X(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det P_X}} \exp\left(-\frac{1}{2}(\mathbf{x} - \bar{\mathbf{X}})^\top P_X^{-1}(\mathbf{x} - \bar{\mathbf{X}})\right). \tag{5.20}$$

In a Gaussian distribution the level sets of the density are ellipsoids described by the covariance matrix: the eigenvalues of P_X are proportional to the squared lengths of the principal semi-axes, and the directions of the axes are given by the eigenvectors.



5.2.2 Zero-Noise Kalman Filter

The Kalman Filter was initially developed for linear systems; we will describe its basic formulation and then extend it to nonlinear systems. Consider a linear discrete-time system:

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k \tag{5.21}$$

$$\mathbf{y}_k = C_k \mathbf{x}_k \tag{5.22}$$

$\mathbf{x}_k$ is the state and $\mathbf{y}_k$ the observations given by the sensors. We denote $\hat{\mathbf{x}}_k$ our state estimate at time step k. To estimate $\mathbf{x}_{k+1}$, we will use the equation of motion (5.21) to get a first estimate, and then use the observation $\mathbf{y}_{k+1}$ to correct such an estimate.

- **Prediction**: We generate an intermediate estimate $\widehat{\mathbf{x}}_{k+1|k}$ by propagating $\widehat{\mathbf{x}}$ using the equation of motion.
- **Correction**: We correct the prediction on the basis of the difference between the measured and the predicted output.

The prediction phase is just:

$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.23}$$

Now we have to use the measurement. To be consistent, we want our estimate to be mapped into the observation, hence $\widehat{\mathbf{x}}_{k+1}$ should lie in the affine subspace:

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n \ : \ C_{k+1}\mathbf{x} = \mathbf{y}_{k+1}\} \tag{5.24}$$

Hence our correction $\Delta\mathbf{x}$ should satisfy the following equation:

$$C_{k+1}(\widehat{\mathbf{x}}_{k+1|k} + \Delta\mathbf{x}) = \mathbf{y}_{k+1} \tag{5.25}$$

which can be rewritten as:

$$C_{k+1}\Delta\mathbf{x} = \mathbf{y}_{k+1} - C_{k+1}\widehat{\mathbf{x}}_{k+1|k}. \tag{5.26}$$

There are infinitely many corrections $\Delta\mathbf{x}$ that satisfy Equation (5.26), hence we choose the one with minimum norm, provided by the pseudo-inverse matrix:

$$\Delta\mathbf{x} = C^{\#}_{k+1}(\mathbf{y}_{k+1} - C_{k+1}\widehat{\mathbf{x}}_{k+1|k}) = C^{\#}_{k+1}\boldsymbol{\nu}_{k+1} \tag{5.27}$$

where we denoted by $\boldsymbol{\nu}_{k+1}$ the term called **innovation**. We have assumed that C_{k+1} has full row rank (so that the system in Equation (5.26) admits a solution).

Definition 5.2.1 : Two-step observer

$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.28}$$

$$\widehat{\mathbf{x}}_{k+1} = \widehat{\mathbf{x}}_{k+1|k} + C^{\#}_{k+1}\boldsymbol{\nu}_{k+1}. \tag{5.29}$$

5.2.3 Noise on the Dynamics

The full Kalman filter considers the dynamics and the observations to be stochastic, i.e., affected by noise. Let us proceed by including randomness only in the equation of motion. Let us assume that the dynamics includes a zero-mean Gaussian noise:

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k + \mathbf{v}_k \tag{5.30}$$

$$\mathbf{y}_k = C_k \mathbf{x}_k \tag{5.31}$$

where

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, V_k) \tag{5.32}$$

V_k is the covariance matrix of $\mathbf{v}_k$. It is a white noise: samples at different time steps are independent. Now the process is stochastic, hence we do not have to estimate a state, but a distribution of states, specified by a mean value $\widehat{\mathbf{x}}_k$ and a covariance matrix P_k (the distribution is Gaussian):

$$\mathcal{N}(\widehat{\mathbf{x}}_k, P_k). \tag{5.33}$$

We keep a prediction/correction structure. Suppose that at time step k we have our current estimate $\widehat{\mathbf{x}}_k$, P_k. We want to obtain a first estimate of the next state distribution through the system dynamics as before (with this prediction we update just the mean):

$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.34}$$

Normally we would also include the mean value of the noise, but it is zero, hence the prediction phase is unchanged. Then we use the observation to correct both the mean and the covariance. First, we

introduce an intermediate covariance, that is the covariance matrix of the prediction error with respect to our first estimate $\widehat{\mathbf{x}}_{k+1|k}$ of the next state:

$$P_{k+1|k} = \mathbb{E}\left((\mathbf{x}_{k+1} - \widehat{\mathbf{x}}_{k+1|k})(\mathbf{x}_{k+1} - \widehat{\mathbf{x}}_{k+1|k})^\top\right) \tag{5.35}$$

we use Equations (5.30) and (5.34):

$$\begin{aligned} P_{k+1|k} &= \mathbb{E}\left((A_k\mathbf{x}_k + B_k\mathbf{u}_k + \mathbf{v}_k - A_k\widehat{\mathbf{x}}_k - B_k\mathbf{u}_k)(A_k\mathbf{x}_k + B_k\mathbf{u}_k + \mathbf{v}_k - A_k\widehat{\mathbf{x}}_k - B_k\mathbf{u}_k)^\top\right) && (5.36)\\ &= \mathbb{E}\left((A_k\mathbf{x}_k + \mathbf{v}_k - A_k\widehat{\mathbf{x}}_k)(A_k\mathbf{x}_k + \mathbf{v}_k - A_k\widehat{\mathbf{x}}_k)^\top\right) && (5.37)\\ &= \mathbb{E}\left((A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k) + \mathbf{v}_k)(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k) + \mathbf{v}_k)^\top\right) && (5.38)\\ &= \mathbb{E}\left(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top + A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)\mathbf{v}_k^\top + \mathbf{v}_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top + \mathbf{v}_k\mathbf{v}_k^\top\right) && (5.39)\end{aligned}$$

the expected value is linear:

$$P_{k+1|k} = \mathbb{E}\left(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top\right) + \mathbb{E}\left(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)\mathbf{v}_k^\top + \mathbf{v}_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top\right) + \underbrace{\mathbb{E}\left(\mathbf{v}_k\mathbf{v}_k^\top\right)}_{V_k} \tag{5.40}$$

$$= \mathbb{E}\left(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top\right) + \mathbb{E}\left(A_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)\mathbf{v}_k^\top + \mathbf{v}_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top A_k^\top\right) + V_k \tag{5.41}$$

A_k is constant hence we exploit once again the linearity of the expected value:

$$P_{k+1|k} = A_k\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top\right)A_k^\top + A_k\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)\mathbf{v}_k^\top\right) + \mathbb{E}\left(\mathbf{v}_k(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top\right)A_k^\top + V_k \tag{5.42}$$

we use the fact that $\mathbb{E}(\mathbf{a}\mathbf{b}^\top) = \mathbb{E}(\mathbf{a})\mathbb{E}(\mathbf{b})^\top$ if $\mathbf{a}, \mathbf{b}$ are independent (here, $\mathbf{v}_k$ is white noise, hence independent of $\mathbf{x}_k - \widehat{\mathbf{x}}_k$):

$$P_{k+1|k} = A_k\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top\right)A_k^\top + A_k\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)\right)\mathbb{E}\left(\mathbf{v}_k\right)^\top + \mathbb{E}\left(\mathbf{v}_k\right)\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)\right)^\top A_k^\top + V_k \tag{5.43}$$

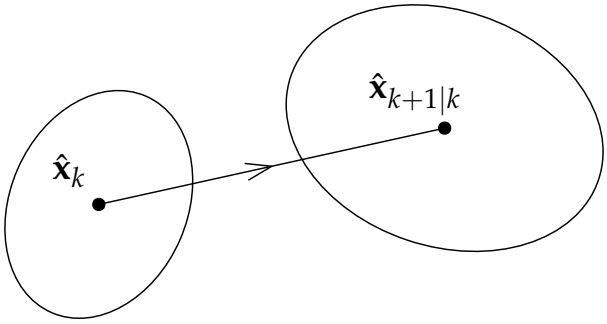
$\mathbb{E}(\mathbf{v}_k)$ is zero (hence also its transpose), so the two cross terms vanish:

$$P_{k+1|k} = A_k\underbrace{\mathbb{E}\left((\mathbf{x}_k - \widehat{\mathbf{x}}_k)(\mathbf{x}_k - \widehat{\mathbf{x}}_k)^\top\right)}_{P_k}A_k^\top + V_k \tag{5.44}$$

ultimately

$$\boxed{P_{k+1|k} = A_kP_kA_k^\top + V_k}. \tag{5.45}$$

$P_{k+1|k}$ is a measure of the uncertainty associated with the state estimate $\widehat{\mathbf{x}}_{k+1|k}$. Notice how at each step the previous covariance matrix P_k is stretched/compressed (and/or) rotated through the application of A_k, and is increased by the addition of V_k. Hence, at each step we are adding noise and the uncertainty tends to increase[1]; without corrections, it may grow without bound.



[1]The volume of the associated ellipsoid tends to grow.

Now that we have a first estimate $\mathcal{N}(\hat{\mathbf{x}}_{k+1|k}, P_{k+1|k})$, we can find the correction $\Delta\mathbf{x}$ by solving the system as in Equation (5.26), but instead of taking the minimum norm correction, we will consider the **maximum likelihood** correction. We want to select the value in Ω that maximizes the density of the Gaussian distribution $\mathcal{N}(\hat{\mathbf{x}}_{k+1|k}, P_{k+1|k})$, which is defined by:

$$p(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det P_{k+1|k}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k})^\top P_{k+1|k}^{-1}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k})\right) \tag{5.46}$$

Observation 5.2.1

Maximizing $p(\mathbf{x})$ is equivalent to minimizing $\frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k})^\top P_{k+1|k}^{-1}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k})$.

Formally, the state estimation that we want to find for step $k+1$ is given by the following minimization problem:

$$\arg\min_{\mathbf{x}\in\mathbb{R}^n} \frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k})^\top P_{k+1|k}^{-1}(\mathbf{x} - \hat{\mathbf{x}}_{k+1|k}) \tag{5.47}$$

$$\text{s.t. } C_{k+1}\mathbf{x} = \mathbf{y}_{k+1} \tag{5.48}$$

since $\mathbf{x} - \hat{\mathbf{x}}_{k+1|k}$ is exactly the correction $\Delta\mathbf{x}$, the problem can be rewritten in the variable $\Delta\mathbf{x}$:

$$\arg\min_{\Delta\mathbf{x}\in\mathbb{R}^n} \frac{1}{2}\Delta\mathbf{x}^\top P_{k+1|k}^{-1}\Delta\mathbf{x} \tag{5.49}$$

$$\text{s.t. } C_{k+1}\Delta\mathbf{x} = \underbrace{\mathbf{y}_{k+1} - C_{k+1}\hat{\mathbf{x}}_{k+1|k}}_{\boldsymbol{\nu}_{k+1}}. \tag{5.50}$$

We are trying to minimize a quadratic form on an affine subspace, hence the solution is given by the weighted pseudo-inverse of C_{k+1}, which uses the weight $M = P_{k+1|k}^{-1}$:

$$\boxed{\Delta\mathbf{x} = C_{k+1,M}^{\#}\,\boldsymbol{\nu}_{k+1} = P_{k+1|k}C_{k+1}^\top(C_{k+1}P_{k+1|k}C_{k+1}^\top)^{-1}\,\boldsymbol{\nu}_{k+1}} \tag{5.51}$$

so the next state estimate will be

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1|k} + C_{k+1,M}^{\#}\,\boldsymbol{\nu}_{k+1}. \tag{5.52}$$

Without giving the proof, the correction of the covariance matrix is the following:

$$P_{k+1} = P_{k+1|k} - C_{k+1,M}^{\#}C_{k+1}P_{k+1|k} \tag{5.53}$$

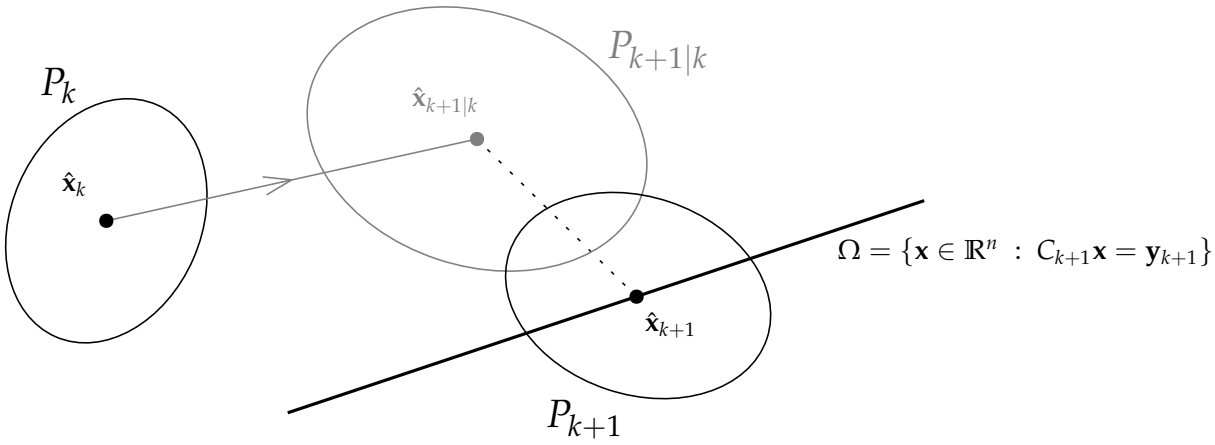


Figure 5.2: Geometric interpretation of the correction.

Definition 5.2.2 : Observer With Noise on the Dynamics

The Kalman filter for a system with noise on the dynamics is defined as follows:

- **prediction**:

$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.54}$$
$$P_{k+1|k} = A_k P_k A_k^\top + V_k \tag{5.55}$$

- **correction**:

$$\widehat{\mathbf{x}}_{k+1} = \widehat{\mathbf{x}}_{k+1|k} + C^{\#}_{k+1,M} \boldsymbol{\nu}_{k+1} \tag{5.56}$$
$$P_{k+1} = P_{k+1|k} - C^{\#}_{k+1,M} C_{k+1} P_{k+1|k}. \tag{5.57}$$

5.2.4 Complete Kalman Filter

Now the noise is considered also in the observations:

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k + \mathbf{v}_k \tag{5.58}$$
$$\mathbf{y}_k = C_k \mathbf{x}_k + \mathbf{w}_k \tag{5.59}$$

with $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, W_k)$. This noise usually comes from the sensors, the noise on the dynamics comes from non-modeled factors (for example, the slippage of the wheels, which the model does not include). The dynamics is unchanged and the prediction phase is identical to the one seen before:

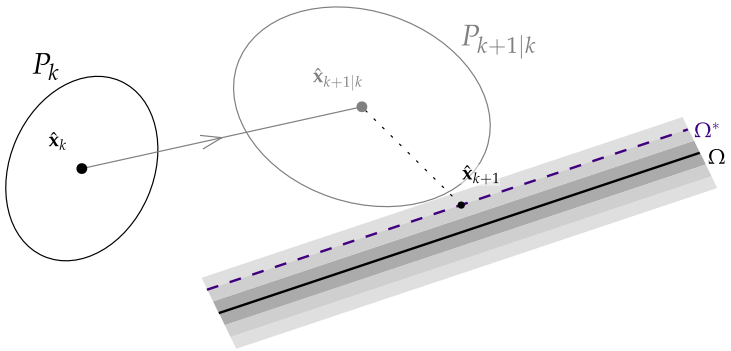
$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.60}$$
$$P_{k+1|k} = A_k P_k A_k^\top + V_k. \tag{5.61}$$

Due to the noise, the observation $\mathbf{y}_{k+1}$ is not exact: it is drawn from a Gaussian distribution with mean value $C_{k+1}\mathbf{x}_{k+1}$ and covariance W_{k+1}. We compute the **most likely** output value $\mathbf{y}^*_{k+1}$ associated to the prediction $\widehat{\mathbf{x}}_{k+1|k}$ and the measured output $\mathbf{y}_{k+1}$. Then the correction is identical, but instead of using

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n \ : \ C_{k+1}\mathbf{x} = \mathbf{y}_{k+1}\} \tag{5.62}$$

it uses

$$\Omega^* = \{\mathbf{x} \in \mathbb{R}^n \ : \ C_{k+1}\mathbf{x} = \mathbf{y}^*_{k+1}\}. \tag{5.63}$$



We will not dive into the details of the computation and give just the result: the correction uses the **Kalman Gain Matrix**:

$$\boxed{R_{k+1} = P_{k+1|k} C_{k+1}^\top (C_{k+1} P_{k+1|k} C_{k+1}^\top + W_{k+1})^{-1}}. \tag{5.64}$$

The matrix R_{k+1} weighs the accuracy of the prediction with respect to that of the measurements.

- R_{k+1} is large (in terms of eigenvalues) if the measurements are more reliable.
- R_{k+1} is small (in terms of eigenvalues) if the prediction is more reliable.

Definition 5.2.3 : Kalman Filter

The Kalman Filter is an observer defined in two steps:

- **prediction**:

$$\widehat{\mathbf{x}}_{k+1|k} = A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \tag{5.65}$$
$$P_{k+1|k} = A_k P_k A_k^\top + V_k \tag{5.66}$$

- **correction**:

$$\widehat{\mathbf{x}}_{k+1} = \widehat{\mathbf{x}}_{k+1|k} + R_{k+1} \boldsymbol{\nu}_{k+1} \tag{5.67}$$
$$P_{k+1} = P_{k+1|k} - R_{k+1} C_{k+1} P_{k+1|k}. \tag{5.68}$$

It can be proved that the Kalman filter provides an optimal estimate: at each time step k it minimizes the mean squared estimation error $\mathbb{E}\left(\|\mathbf{x}_{k+1} - \widehat{\mathbf{x}}_{k+1}\|^2\right)$. Notice how R_k does not depend on the measurements, hence the sequence $\{R_1, \dots, R_k, \dots\}$ can be precomputed a priori.

5.3 Extended Kalman Filter

Now we extend the Kalman filter to a general nonlinear system (such as the kinematic model of a mobile robot):

$$\begin{cases} \mathbf{x}_{k+1} = f_k(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k \\ \mathbf{y}_k = h_k(\mathbf{x}_k) + \mathbf{w}_k \end{cases} \tag{5.69}$$

with f_k, h_k being continuously differentiable for each k. What we do is apply approximate linearization around the current estimate and use the Kalman filter on that approximation.

Definition 5.3.1 : Extended Kalman Filter

For a system like the one in Equation (5.69), we consider the Jacobian matrices:

$$F_k = \left.\frac{\partial f_k}{\partial \mathbf{x}}\right|_{\mathbf{x}=\widehat{\mathbf{x}}_k, \mathbf{u}=\mathbf{u}_k}, \qquad H_{k+1} = \left.\frac{\partial h_{k+1}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\widehat{\mathbf{x}}_{k+1|k}}. \tag{5.70}$$

The extended Kalman filter is an observer defined in two steps:

- **prediction**:

$$\widehat{\mathbf{x}}_{k+1|k} = f_k(\widehat{\mathbf{x}}_k, \mathbf{u}_k) \tag{5.71}$$
$$P_{k+1|k} = F_k P_k F_k^\top + V_k \tag{5.72}$$

- **correction**:

$$\widehat{\mathbf{x}}_{k+1} = \widehat{\mathbf{x}}_{k+1|k} + R_{k+1} \boldsymbol{\nu}_{k+1} \tag{5.73}$$
$$P_{k+1} = P_{k+1|k} - R_{k+1} H_{k+1} P_{k+1|k}, \tag{5.74}$$

where the innovation is $\boldsymbol{\nu}_{k+1} = \mathbf{y}_{k+1} - h_{k+1}(\widehat{\mathbf{x}}_{k+1|k})$,

and the Kalman gain matrix is:

$$R_{k+1} = P_{k+1|k} H_{k+1}^\top (H_{k+1} P_{k+1|k} H_{k+1}^\top + W_{k+1})^{-1}. \tag{5.75}$$

The block scheme of the filter is shown in Figure 5.3.

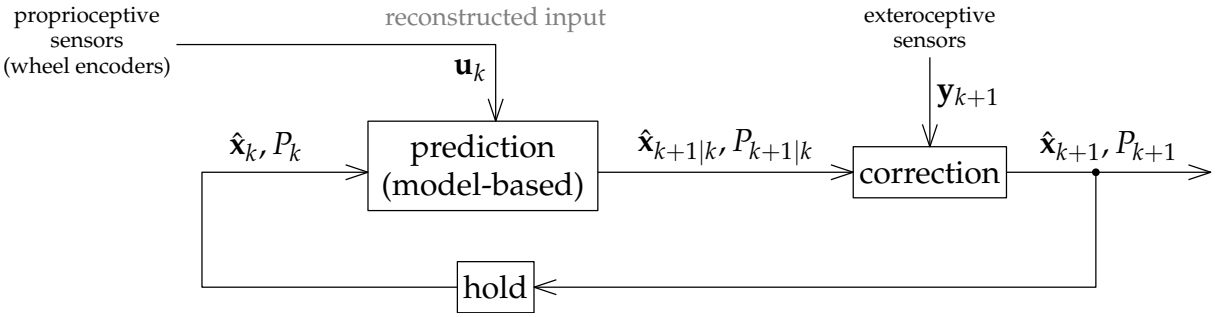
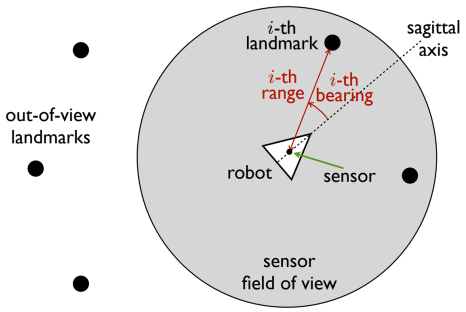


Figure 5.3: Block scheme of the extended Kalman filter.

5.4 Landmark-Based Localization

Now we apply the extended Kalman filter to a robotic system in order to localize the robot and obtain a proper estimate of its state. We base the discussion on the assumption that our mobile robot is equipped with a sensor that can measure the *relative distance (range) and orientation (bearing)* to a set of **landmarks**, which are point-like obstacles that may be artificial or natural. The positions of the landmarks are fixed and known.

There are various sensors that can provide this kind of measurement, such as laser rangefinders, depth cameras or RFID sensors.



Let us use the unicycle model, with the addition of a white Gaussian noise $\mathbf{v}_k$ with zero mean and covariance matrix V_k:

$$x_{k+1} = x_k + v_k T_s \cos\theta_k + v_{1k} \tag{5.76}$$
$$y_{k+1} = y_k + v_k T_s \sin\theta_k + v_{2k} \tag{5.77}$$
$$\theta_{k+1} = \theta_k + \omega_k T_s + v_{3k}. \tag{5.78}$$

About the Notation 5.4.1

With v_k, we denote the translational velocity input of the unicycle model at time step k, while with $\mathbf{v}_k = \begin{pmatrix} v_{1k} & v_{2k} & v_{3k} \end{pmatrix}^\top$ we denote the Gaussian noise on the dynamics at time step k.

Let L be the total number of landmarks, and let $L_k \leq L$ be the number of landmarks that the robot can sense at time step k. We identify the Cartesian positions of the landmarks as:

$$\mathbf{p}_{l\,i} = \begin{pmatrix} x_{l\,i} \\ y_{l\,i} \end{pmatrix}, \quad 1 \leq i \leq L. \tag{5.79}$$

The sensor does not directly measure the (known) landmark positions $\begin{pmatrix} x_{l\,i} & y_{l\,i} \end{pmatrix}^\top$, but only the relative range and bearing (the landmark positions are known a priori). We assume that the robot can identify and distinguish the landmarks from each other: each landmark is identified by an index from 1 to L.

At a given time step, there are L_k landmarks sensed, but the i-th landmark sensed is not necessarily the landmark i out of L. Hence, we use an association map a that assigns to each integer i from 1 to L_k (the i-th landmark sensed) the index j, between 1 and L, of the corresponding landmark:

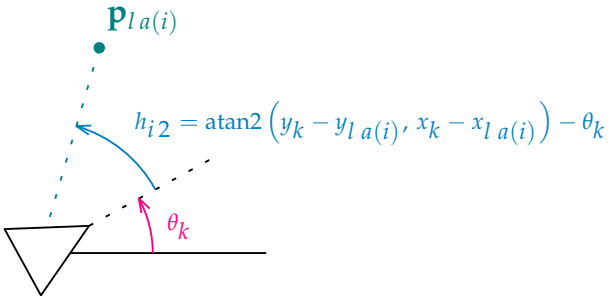
$$a : \{1, 2, \ldots, L_k\} \to \{1, 2, \ldots, L\}, \tag{5.80}$$

so we denote by $a(i)$ the index of the landmark observed at the i-th measurement. At each time step, we define the vector of the measurements $\mathbf{y}_k$; this is a vector of length $2L_k$ (the length of the vector changes over the time steps) and is defined as follows:

$$\mathbf{y}_k = \begin{pmatrix} \mathbf{h}_1(\mathbf{q}_k, a(1)) \\ \mathbf{h}_2(\mathbf{q}_k, a(2)) \\ \vdots \\ \mathbf{h}_{L_k}(\mathbf{q}_k, a(L_k)) \end{pmatrix} + \begin{pmatrix} \mathbf{w}_{1k} \\ \mathbf{w}_{2k} \\ \vdots \\ \mathbf{w}_{L_k k} \end{pmatrix}, \tag{5.81}$$

where $\mathbf{w}_k = \begin{pmatrix} \mathbf{w}_{1k}^\top & \ldots & \mathbf{w}_{L_k k}^\top \end{pmatrix}^\top \in \mathbb{R}^{2L_k}$ is a Gaussian white noise with covariance matrix W_k, and each $\mathbf{h}_i(\mathbf{q}_k, a(i))$ is a two-component vector with the range and bearing of the $a(i)$-th landmark. Of course, it depends on the current configuration $\mathbf{q}_k$:

$$\mathbf{h}_i(\mathbf{q}_k, a(i)) = \begin{pmatrix} \sqrt{(x_k - x_{l\,a(i)})^2 + (y_k - y_{l\,a(i)})^2} \\ \text{atan2}\left(y_{l\,a(i)} - y_k,\, x_{l\,a(i)} - x_k\right) - \theta_k \end{pmatrix}. \tag{5.82}$$



The bearing is the angle of the landmark in the robot frame; the angular component of the innovation must always be normalized in $(-\pi, \pi]$.

The kinematic model of the unicycle is nonlinear, so we need to linearize the equations of motion at a given configuration in order to apply the extended Kalman filter. We denote by F_k the Jacobian of the process dynamics at time k, which is defined as follows:

$$F_k = \begin{pmatrix} 1 & 0 & -v_k T_s \sin\widehat{\theta}_k \\ 0 & 1 & v_k T_s \cos\widehat{\theta}_k \\ 0 & 0 & 1 \end{pmatrix} \tag{5.83}$$

and is evaluated at the current configuration estimate $\widehat{\mathbf{q}}_k$. Now we have to differentiate the measurement equations $\mathbf{h}_i(\mathbf{q}_k, a(i))$ to obtain H_{k+1}:

$$H_{k+1} = \begin{pmatrix} \left.\dfrac{\partial \mathbf{h}_1}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} \\ \vdots \\ \left.\dfrac{\partial \mathbf{h}_{L_k}}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} \end{pmatrix} \in \mathbb{R}^{2L_k \times 3}, \tag{5.84}$$

where each $\frac{\partial \mathbf{h}_i}{\partial \mathbf{q}}$ is a matrix with two rows and three columns, defined as follows:

$$\left.\frac{\partial \mathbf{h}_i}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} = \begin{pmatrix} \left.\frac{\partial \mathbf{h}_i}{\partial x}\right|_{x=\widehat{x}_{k+1|k}} & \left.\frac{\partial \mathbf{h}_i}{\partial y}\right|_{y=\widehat{y}_{k+1|k}} & \left.\frac{\partial \mathbf{h}_i}{\partial \theta}\right|_{\theta=\widehat{\theta}_{k+1|k}} \end{pmatrix} \tag{5.85}$$

$$= \begin{pmatrix} \frac{\widehat{x}_{k+1|k}-x_{l\,a(i)}}{\sqrt{(\widehat{x}_{k+1|k}-x_{l\,a(i)})^2+(\widehat{y}_{k+1|k}-y_{l\,a(i)})^2}} & \frac{\widehat{y}_{k+1|k}-y_{l\,a(i)}}{\sqrt{(\widehat{x}_{k+1|k}-x_{l\,a(i)})^2+(\widehat{y}_{k+1|k}-y_{l\,a(i)})^2}} & 0 \\ \frac{-(\widehat{y}_{k+1|k}-y_{l\,a(i)})}{(\widehat{x}_{k+1|k}-x_{l\,a(i)})^2+(\widehat{y}_{k+1|k}-y_{l\,a(i)})^2} & \frac{\widehat{x}_{k+1|k}-x_{l\,a(i)}}{(\widehat{x}_{k+1|k}-x_{l\,a(i)})^2+(\widehat{y}_{k+1|k}-y_{l\,a(i)})^2} & -1 \end{pmatrix}. \tag{5.86}$$

Then, it is possible to run the extended Kalman filter. Figure 5.4 shows a typical application of the EKF: the robot's estimated trajectory $\widehat{\mathbf{q}}(t)$ is compared with the actual path.

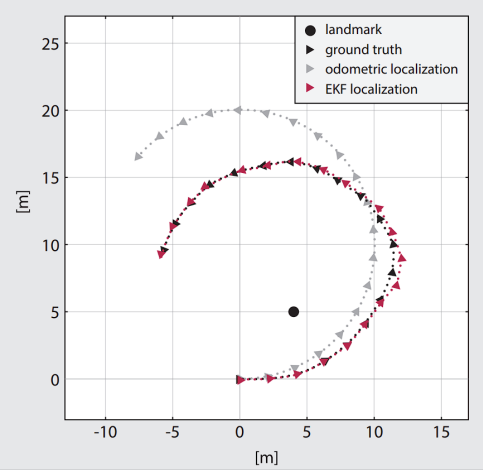


Figure 5.4: Simulation of the extended Kalman filter on a unicycle.

5.4.1 Data Association

Now we drop the assumption that the landmarks can be distinguished from each other: every time we sense a landmark, we cannot tell which of the L landmarks it is, hence the association map a must be estimated as well. The innovation $\boldsymbol{\nu}_k$ in the EKF is the difference between the actual and the predicted measurement; we associate each observation to the landmark that minimizes the innovation magnitude.

At the $(k+1)$-th step, we consider the i-th measurement $\mathbf{y}_{i\,k+1}$ and we compute the **candidate innovation** for the landmark j as follows:

$$\boldsymbol{\nu}_{ij} = \mathbf{y}_{i\,k+1} - \mathbf{h}_i(\widehat{\mathbf{q}}_{k+1|k}, j). \tag{5.87}$$

The smaller $\boldsymbol{\nu}_{ij}$ is, the more likely it is that the i-th measurement corresponds to the j-th landmark. We cannot simply minimize $\|\boldsymbol{\nu}_{ij}\|$, since we also have to consider the *uncertainty* of the measurement. We define the matrix $H_{k+1}(i,j)$ as the Jacobian matrix of the measurement function at time $k+1$, evaluated under the specific hypothesis that the i-th measurement made by the sensor corresponds to the j-th landmark. The already mentioned uncertainty of the measurement is encoded in a matrix S_{ij} defined as follows:

$$S_{ij} = H_{k+1}(i,j) P_{k+1|k} H_{k+1}(i,j)^\top + W_{k+1}(i), \tag{5.88}$$

where $W_{k+1}(i)$ represents the covariance matrix of the measurement noise referring exclusively to the single i-th measurement at time $k+1$. Hence, we minimize the norm of $\boldsymbol{\nu}_{ij}$ weighted with S_{ij}^{-1}, and the landmark associated to the i-th measurement is:

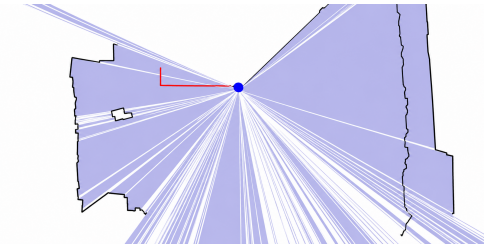
$$j^* = \underset{1\le j\le L}{\arg\min}\, \boldsymbol{\nu}_{ij}^\top S_{ij}^{-1} \boldsymbol{\nu}_{ij}. \tag{5.89}$$

5.4.2 Localization on a Map

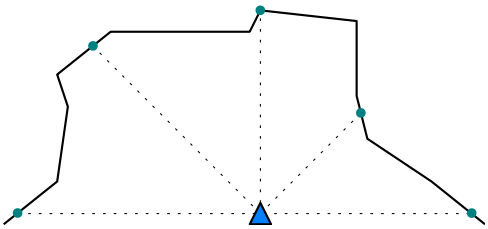
Now, we assume that a metric map $\mathcal{M}$ of the environment is available; in particular, it can be:

- **Line-based map**: A geometric representation that models environmental features, such as walls and corridors, through vector segments and lines, similar to an architectural floor plan.
- **Occupancy grid**: A discrete representation in which space is divided into a grid of cells, each of which indicates whether that specific spatial area is occupied by an obstacle or free.

The map is a global reference known a priori; the EKF uses this map to refine the estimate $\widehat{\mathbf{q}}_k$ by comparing the sensor readings with the readings predicted from the map. We assume that our mobile robot is equipped with a range finder, a sensor that can provide range measurements in all directions. The *whole scan* is used as the output vector, whose components are the range readings in all available directions (usually, a discretization of the angles from 0 to 360 degrees).

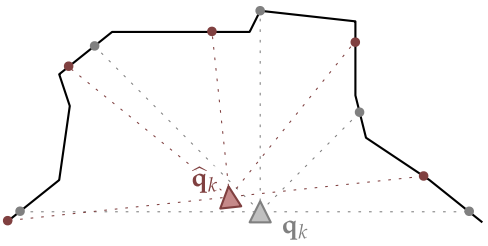


Example: Assume that a range finder scans 5 angles $(0°, 45°, 90°, 135°, 180°)$ at each time step.



The output vector will have 5 components, each containing the distance between the robot and the sensed point at that orientation (hence the bearing is also known). The *innovation* $\boldsymbol{\nu}_{k+1}$ is computed as the difference between the actual scan $\mathbf{y}_{k+1}$ and the predicted scan $\mathbf{h}(\widehat{\mathbf{q}}_{k+1|k}, \mathcal{M})$, which is computed by considering the range values that the robot should read if it were in the configuration $\widehat{\mathbf{q}}_{k+1|k}$ according to the map $\mathcal{M}$:

$$\boldsymbol{\nu}_{k+1} = \mathbf{y}_{k+1} - \mathbf{h}(\widehat{\mathbf{q}}_{k+1|k}, \mathcal{M}). \tag{5.90}$$



At this point we can apply the extended Kalman filter by linearizing the equations and using the innovation just defined. Notice how no feature matching is needed; however, multiple similar features in the environment may cause a severe displacement in the estimate (**aliasing**).

The diagram of the whole localization scheme is shown in Figure 5.5. To summarize all the steps:

- **Odometric prediction**: Use proprioceptive sensors to predict the next configuration, $\widehat{\mathbf{q}}_{k+1|k}$.

- **Output prediction**: Use $\widehat{\mathbf{q}}_{k+1|k}$ to compute the expected sensor readings, $\mathbf{h}(\widehat{\mathbf{q}}_{k+1|k}, \mathcal{M})$, leveraging the known map $\mathcal{M}$ to determine what the robot "should see" at its predicted location.
- **Matching**: Compare the actual sensor readings with the predicted ones to compute the innovation $\boldsymbol{\nu}_{k+1}$.
- **Correction**: Update the robot's configuration estimate using the innovation, which is the difference between the actual and predicted sensor readings. This step uses the Kalman gain matrix:

$$R_{k+1} = P_{k+1|k} H_{k+1}^{\top} (H_{k+1} P_{k+1|k} H_{k+1}^{\top} + W_{k+1})^{-1}$$

to determine how much weight to assign to the predicted configuration $\widehat{\mathbf{q}}_{k+1|k}$ (from odometry) versus the measurements from exteroceptive sensors, based on their respective uncertainties.

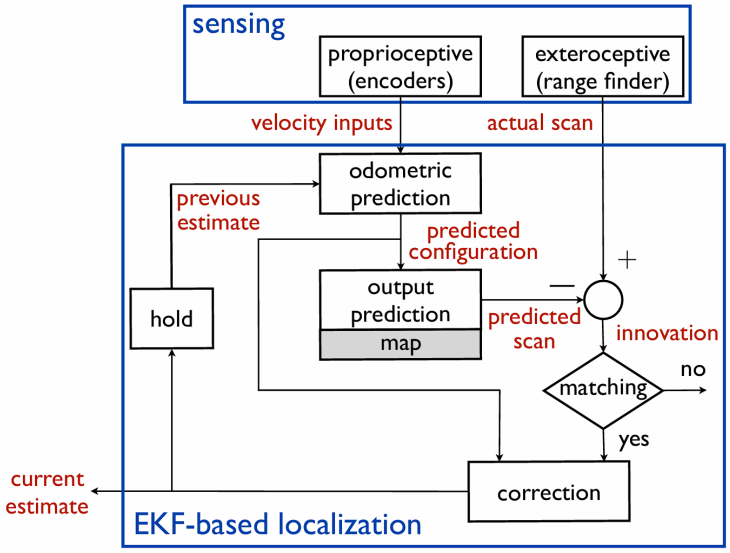


Figure 5.5: Localization diagram.

Summary of EKF localization:

- **state prediction**: $\widehat{\mathbf{q}}_{k+1|k} = f(\widehat{\mathbf{q}}_k, \mathbf{u}_k)$;
- **covariance prediction**: $P_{k+1|k} = F_k P_k F_k^{\top} + V_k$;
- **innovation**: $\boldsymbol{\nu}_{k+1} = \mathbf{y}_{k+1} - \mathbf{h}(\widehat{\mathbf{q}}_{k+1|k})$;
- **Kalman gain**: $R_{k+1} = P_{k+1|k} H_{k+1}^{\top} (H_{k+1} P_{k+1|k} H_{k+1}^{\top} + W_{k+1})^{-1}$;
- **state update**: $\widehat{\mathbf{q}}_{k+1} = \widehat{\mathbf{q}}_{k+1|k} + R_{k+1} \boldsymbol{\nu}_{k+1}$;
- **covariance update**: $P_{k+1} = P_{k+1|k} - R_{k+1} H_{k+1} P_{k+1|k}$.

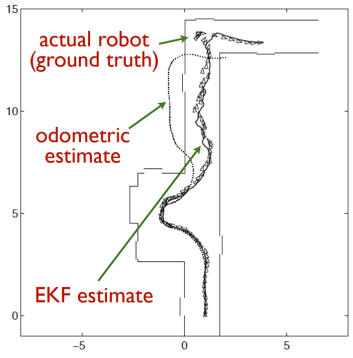


Figure 5.6: Simulation of a robotized wheelchair with high wheel slippage, equipped with 5 ultrasonic sensors that sample at 2 Hz and can sense only in front of the robot.

5.5 SLAM

SLAM is the acronym of **Simultaneous Localization And Map-building**. It is a method that removes the assumption of a known map and aims to use the robot's sensors to build a map and, at the same time, localize the robot. In probabilistic SLAM, the idea is to estimate the map features in addition to the robot configuration; we discuss a simple landmark-based version of the problem, which can be solved using the KF or the EKF.

Observation 5.5.1

In this setup, the state vector includes both the robot configuration and the landmark positions.

To simplify the discussion we work under the following assumptions:

- The robot is an omnidirectional point-robot, whose configuration is then a Cartesian position (no steering angle).
- There are L landmarks distributed in the environment, whose positions are unknown.
- The robot is equipped with a sensor that can see, identify and measure the relative position of all landmarks with respect to itself, i.e., the field of view is unbounded and there is no occlusion.

We denote by $\mathbf{x}$ the extended state vector to be estimated; it contains the Cartesian position (x, y) of the robot and the positions of the L landmarks:

$$\mathbf{x} = \begin{pmatrix} x & y & x_{l1} & y_{l1} & \dots & x_{lL} & y_{lL} \end{pmatrix}^\top \in \mathbb{R}^{2L+2}. \tag{5.91}$$

The equations of motion in $\mathbf{x}$ are easy to define since the landmarks cannot move, hence their positions do not change over time:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + T_s \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_{x,k} \\ u_{y,k} \end{pmatrix} + \begin{pmatrix} v_{x,k} \\ v_{y,k} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \tag{5.92}$$

with $u_{x,k}, u_{y,k}$ being the velocity inputs, T_s the sampling time, and $\mathbf{v}_{xy,k} = \begin{pmatrix} v_{x,k} & v_{y,k} \end{pmatrix}^\top$ a Gaussian white noise with zero mean and covariance matrix $V_{xy,k}$. This is a linear model of the form:

$$\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k + \mathbf{v}_k, \tag{5.93}$$

where $A = I_{2L+2}$, $\mathbf{u}_k = \begin{pmatrix} u_{x,k} & u_{y,k} \end{pmatrix}^\top$, B is the $(2L+2) \times 2$ matrix in the previous equation (including the factor T_s), and $\mathbf{v}_k = \begin{pmatrix} v_{x,k} & v_{y,k} & 0 & \dots & 0 \end{pmatrix}^\top$ is a white Gaussian noise with zero mean and covariance

matrix

$$V_k = \begin{pmatrix} V_{xy,k} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}. \tag{5.94}$$

The i-th measurement contains the relative position of the i-th landmark (which by assumption can be sensed from anywhere) plus the noise:

$$\mathbf{y}_{i,k} = \begin{pmatrix} x_{l\,i} - x_k \\ y_{l\,i} - y_k \end{pmatrix} + \mathbf{w}_{i,k}, \tag{5.95}$$

where $\mathbf{w}_{i,k} \sim \mathcal{N}(\mathbf{0}, W_{i,k})$. The output equation is also linear:

$$\mathbf{y}_{i,k} = C_i \mathbf{x}_k + \mathbf{w}_{i,k}, \tag{5.96}$$

where $C_i = \begin{pmatrix} -I_2 & \mathbf{0} & \dots & I_2 & \dots & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{2\times(2L+2)}$ has the identity block I_2 in the columns corresponding to the i-th landmark. We can stack all measurements to create the output vector:

$$\mathbf{y}_k = C\mathbf{x}_k + \mathbf{w}_k, \tag{5.97}$$

where

$$C = \begin{pmatrix} C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad \mathbf{w}_k = \begin{pmatrix} \mathbf{w}_{1,k} \\ \vdots \\ \mathbf{w}_{L,k} \end{pmatrix}, \tag{5.98}$$

and the covariance of the measurement noise (the noises of different landmarks being independent) is

$$W_k = \begin{pmatrix} W_{1,k} & 0 & \dots & 0 \\ 0 & W_{2,k} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & W_{L,k} \end{pmatrix}. \tag{5.99}$$

At this point, we just apply the Kalman filter (since the model is linear, the EKF reduces to the standard KF).

FORMULA SHEET

Configuration vector

$$\mathbf{q} = \begin{pmatrix} q_1 & \dots & q_n \end{pmatrix}^\top$$

Configuration space of the disk robot / free polyhedral robot

$$\mathcal{C} = \mathbb{R}^2 \times SO(2) = SE(2), \qquad \mathcal{C} = \mathbb{R}^3 \times SO(3) = SE(3)$$

Configuration space of a 2R manipulator

$$\mathcal{C} = SO(2) \times SO(2) = SO(2)^2$$

Properties of a distance

$$\begin{aligned} &d(\mathbf{q}_a, \mathbf{q}_b) \geq 0 \\ &d(\mathbf{q}_a, \mathbf{q}_b) = 0 \iff \mathbf{q}_a = \mathbf{q}_b \\ &d(\mathbf{q}_a, \mathbf{q}_b) = d(\mathbf{q}_b, \mathbf{q}_a) \\ &d(\mathbf{q}_a, \mathbf{q}_b) \leq d(\mathbf{q}_a, \mathbf{q}_c) + d(\mathbf{q}_c, \mathbf{q}_b) \end{aligned}$$

Displacement metric

$$d(\mathbf{q}_a, \mathbf{q}_b) = \max_{\mathbf{p} \in \text{robot}} \|\mathbf{p}(\mathbf{q}_a) - \mathbf{p}(\mathbf{q}_b)\|$$

Control points metric

$$d(\mathbf{q}_a, \mathbf{q}_b) = \max_{1 \leq i \leq c} \|\mathbf{p}_i(\mathbf{q}_a) - \mathbf{p}_i(\mathbf{q}_b)\|$$

Geometric constraint

$$h_i(\mathbf{q}) = 0, \quad i = 1, \dots, k$$

Dimension of the constrained sub-manifold

$$\dim = n - k$$

Kinematic constraint (Pfaffian form)

$$\mathbf{a}^\top(\mathbf{q})\dot{\mathbf{q}} = 0, \qquad A(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0} \iff \dot{\mathbf{q}} \in \mathcal{N}(A(\mathbf{q}))$$

Kinematic constraint implied by a geometric constraint

$$\dot{h}(\mathbf{q}) = \frac{\partial h}{\partial \mathbf{q}}\dot{\mathbf{q}} = (\nabla_{\mathbf{q}} h)^{\top}\dot{\mathbf{q}} = 0$$

Holonomic (integrable) constraint

$$\frac{\partial h}{\partial \mathbf{q}} = \gamma(\mathbf{q})\,\mathbf{a}^{\top}(\mathbf{q}), \quad \gamma(\mathbf{q}) \neq 0$$

Rolling disk constraint

$$\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0$$

Null space basis and dimension

$$\mathcal{N}(A(\mathbf{q})) = \operatorname{span}\{\mathbf{g}_1(\mathbf{q}), \ldots, \mathbf{g}_m(\mathbf{q})\}, \qquad m = n - k$$

Kinematic model

$$\dot{\mathbf{q}} = \sum_{i=1}^{m} u_i \mathbf{g}_i(\mathbf{q}) = G(\mathbf{q})\mathbf{u}$$

Driftless property

$$\mathbf{u} = \mathbf{0} \implies \dot{\mathbf{q}} = \mathbf{0}$$

Unicycle kinematic model

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix} v + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \omega$$

Unicycle pure rolling constraint (matrix form)

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 \end{pmatrix} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = 0$$

Linear velocity of the unicycle

$$v = \pm\sqrt{\dot{x}^2 + \dot{y}^2} = \omega_{\text{roll}}\, r$$

Differential drive velocities

$$v = r\,\frac{\omega_R + \omega_L}{2}, \qquad \omega = (\omega_R - \omega_L)\frac{r}{d}$$

Bicycle: front wheel position

$$x_f = x + l\cos\theta, \qquad y_f = y + l\sin\theta$$

Bicycle kinematic constraints

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 \\ \sin(\theta+\phi) & -\cos(\theta+\phi) & -l\cos\phi & 0 \end{pmatrix} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \mathbf{0}$$

Bicycle kinematic model (rear-wheel drive)

$$\dot{\mathbf{q}} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{l}\tan\phi \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2$$

Bicycle kinematic model (front-wheel drive)

$$\dot{\mathbf{q}} = \begin{pmatrix} \cos\theta\cos\phi \\ \sin\theta\cos\phi \\ \frac{1}{l}\sin\phi \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2$$

Driftless nonlinear system

$$\dot{\mathbf{x}} = \sum_{i=1}^{m} \mathbf{g}_i(\mathbf{x}) u_i$$

Lie bracket

$$[\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}) = \frac{\partial \mathbf{g}_2}{\partial \mathbf{x}} \mathbf{g}_1(\mathbf{x}) - \frac{\partial \mathbf{g}_1}{\partial \mathbf{x}} \mathbf{g}_2(\mathbf{x})$$

Antisymmetry of the Lie bracket

$$[\mathbf{g}_i, \mathbf{g}_j] = -[\mathbf{g}_j, \mathbf{g}_i]$$

Displacement along the Lie bracket

$$\mathbf{x}(4\varepsilon) = \mathbf{x}(0) + \varepsilon^2 [\mathbf{g}_1, \mathbf{g}_2](\mathbf{x}(0)) + O(\varepsilon^3)$$

Input distribution

$$\Delta(\mathbf{x}) = \text{span}\{\mathbf{g}_1(\mathbf{x}), \dots, \mathbf{g}_m(\mathbf{x})\}$$

Accessibility distribution

$$\Delta_{\mathcal{A}}(\mathbf{x}) = \text{span}\{\mathbf{g}_1, \dots, \mathbf{g}_m, \dots, [\mathbf{g}_i, \mathbf{g}_j], \dots, [\mathbf{g}_i, [\mathbf{g}_j, \mathbf{g}_k]], \dots\}(\mathbf{x})$$

Accessibility criterion

$$\text{controllable} \iff \dim \Delta_{\mathcal{A}} = n$$

Unicycle: first order Lie bracket (slide vector field)

$$\mathbf{g}_3 = [\mathbf{g}_1, \mathbf{g}_2] = \begin{pmatrix} \sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$$

Bicycle: first order Lie bracket (wriggle)

$$\mathbf{g}_3 = [\mathbf{g}_1, \mathbf{g}_2] = \begin{pmatrix} 0 \\ 0 \\ -\frac{1}{l\cos^2\phi} \\ 0 \end{pmatrix}$$

Bicycle: second order Lie bracket

$$\mathbf{g}_4 = [\mathbf{g}_1, \mathbf{g}_3] = \begin{pmatrix} -\frac{\sin\theta}{l\cos^2\phi} \\ \frac{\cos\theta}{l\cos^2\phi} \\ 0 \\ 0 \end{pmatrix}$$

Chained-form system

$$\begin{aligned} \dot{z}_1 &= w_1 \\ \dot{z}_2 &= w_2 \\ \dot{z}_3 &= z_2 w_1 \\ &\vdots \\ \dot{z}_n &= z_{n-1} w_1 \end{aligned}$$

Chained-form transformation of the unicycle (coordinates)

$$z_1 = \theta, \qquad z_2 = x\cos\theta + y\sin\theta, \qquad z_3 = x\sin\theta - y\cos\theta$$

Chained-form transformation of the unicycle (inputs)

$$w_1 = \omega, \qquad w_2 = v - z_3\omega$$

Velocity along a path

$$\dot{\mathbf{q}} = \mathbf{q}'\,\dot{s}, \qquad \|\mathbf{q}'\| = 1 \text{ if } s \text{ is the arc length}$$

Admissible tangent vector

$$A(\mathbf{q})\mathbf{q}' = \mathbf{0} \iff \mathbf{q}' \in \mathcal{N}(A)$$

Geometric model

$$\mathbf{q}' = G(\mathbf{q})\tilde{\mathbf{u}}, \qquad \mathbf{u}(t) = \tilde{\mathbf{u}}(s)\,\dot{s}(t)$$

Differential flatness

$$\mathbf{y} = \Psi(\mathbf{x}), \qquad \mathbf{x} = \Phi_x(\mathbf{y}, \dot{\mathbf{y}}, \ldots, \mathbf{y}^{(r_x)}), \qquad \mathbf{u} = \Phi_u(\mathbf{y}, \dot{\mathbf{y}}, \ldots, \mathbf{y}^{(r_u)})$$

Unicycle flat outputs: orientation

$$\theta(s) = \text{atan2}(y', x') + k\pi, \quad k \in \{0, 1\}$$

Unicycle flat outputs: geometric inputs

$$\tilde{v} = \pm\sqrt{(x')^2 + (y')^2}, \qquad \tilde{\omega} = \frac{y''x' - x''y'}{(x')^2 + (y')^2}$$

Bicycle flat outputs: steering angle

$$\phi(s) = \arctan\left(\pm\frac{l\,(x'y'' - y'x'')}{((x')^2 + (y')^2)^{3/2}}\right)$$

Chained-form flat outputs

$$\mathbf{y} = \begin{pmatrix} z_1 \\ z_3 \end{pmatrix}$$

Chained-form reconstruction maps

$$z_2 = \frac{z_3'}{z_1'}, \qquad \tilde{w}_1 = z_1', \qquad \tilde{w}_2 = \frac{z_1' z_3'' - z_1'' z_3'}{(z_1')^2}$$

Cubic polynomial interpolation of the unicycle path

$$x(s) = s^3 x_f - (s-1)^3 x_i + \alpha_x s^2(s-1) + \beta_x s(s-1)^2$$
$$y(s) = s^3 y_f - (s-1)^3 y_i + \alpha_y s^2(s-1) + \beta_y s(s-1)^2$$

Boundary conditions on the Cartesian velocity

$$x'(0) = k\cos\theta_i, \quad x'(1) = k\cos\theta_f$$
$$y'(0) = k\sin\theta_i, \quad y'(1) = k\sin\theta_f$$

Coefficients of the cubic path of the unicycle

$$\alpha_x = k\cos\theta_f - 3x_f, \quad \beta_x = k\cos\theta_i + 3x_i$$
$$\alpha_y = k\sin\theta_f - 3y_f, \quad \beta_y = k\sin\theta_i + 3y_i$$

Path planning for a chained-form system: interpolation

$$z_1(s) = z_{1i} + s(z_{1f} - z_{1i})$$
$$z_3(s) = s^3 z_{3f} - (s-1)^3 z_{3i} + \alpha_3 s^2(s-1) + \beta_3 s(s-1)^2$$

Path planning for a chained-form system: coefficients

$$\alpha_3 = z_{2f}(z_{1f} - z_{1i}) - 3z_{3f}$$
$$\beta_3 = z_{2i}(z_{1f} - z_{1i}) + 3z_{3i}$$

Path planning via parametrized inputs

$$\Delta = z_{1f} - z_{1i}, \qquad \begin{cases} \tilde{w}_1 = \text{sign}(\Delta) \\ \tilde{w}_2 = \sum_{j=0}^{n-2} c_j s^j \end{cases} \qquad s \in [0, |\Delta|]$$

Linear system for the input coefficients

$$D(\Delta) \begin{pmatrix} c_0 \\ \vdots \\ c_{n-2} \end{pmatrix} = \mathbf{d}(\mathbf{z}_i, \mathbf{z}_f, \Delta)$$

Matrices D and $\mathbf{d}$ for 2 inputs and 3 states

$$D = \begin{pmatrix} |\Delta| & \Delta^2/2 \\ \text{sign}(\Delta)\Delta^2/2 & \Delta^3/6 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} z_{2f} - z_{2i} \\ z_{3f} - z_{3i} - z_{2i}\Delta \end{pmatrix}, \quad \det D = -\text{sign}(\Delta)\frac{\Delta^4}{12}$$

Linear timing law

$$s(t) = s_i + \frac{s_f - s_i}{t_f - t_i}(t - t_i), \qquad s(t) = s_i + (s_f - s_i)\frac{t}{T}$$

Trapezoidal timing law (bang-coast-bang)

$$\ddot{s}(t) = \begin{cases} a_{max} & t \in [t_i, t_i + t_m] \\ 0 & t \in (t_i + t_m, t_f - t_m] \\ -a_{max} & t \in (t_f - t_m, t_f] \end{cases}$$

Velocity bounds

$$|v| \leq v_{max}, \qquad |\omega| \leq \omega_{max}$$

Timing law with velocity bounds

$$s(t) = s_i + \frac{t}{k}, \qquad v(t) = \frac{\tilde{v}(s(t))}{k}, \qquad \omega(t) = \frac{\tilde{\omega}(s(t))}{k}$$

Scaling factor of the timing law

$$k = \max\left\{ \frac{\tilde{v}_{peak}}{v_{max}}, \frac{\tilde{\omega}_{peak}}{\omega_{max}} \right\}, \quad \tilde{v}_{peak} = \max_s |\tilde{v}(s)|, \quad \tilde{\omega}_{peak} = \max_s |\tilde{\omega}(s)|$$

Desired trajectory of the unicycle (flatness)

$$\begin{aligned} \theta_d(t) &= \text{atan2}(\dot{y}_d, \dot{x}_d) + k\pi \\ v_d(t) &= \pm\sqrt{\dot{x}_d^2 + \dot{y}_d^2} \\ \omega_d(t) &= \frac{\ddot{y}_d \dot{x}_d - \ddot{x}_d \dot{y}_d}{\dot{x}_d^2 + \dot{y}_d^2} \end{aligned}$$

Tracking error in the rotated frame

$$\mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_d - x \\ y_d - y \\ \theta_d - \theta \end{pmatrix}$$

Error dynamics

$$\begin{aligned} \dot{e}_1 &= v_d \cos e_3 - v + e_2\, \omega \\ \dot{e}_2 &= v_d \sin e_3 - e_1\, \omega \\ \dot{e}_3 &= \omega_d - \omega \end{aligned}$$

Approximate linearization

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}, \quad A = \left.\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{x}}\right|_{\mathbf{0},\mathbf{0}}, \quad B = \left.\frac{\partial \boldsymbol{\eta}}{\partial \mathbf{u}}\right|_{\mathbf{0},\mathbf{0}}$$

Closed loop with linear state feedback

$$\mathbf{u} = K\mathbf{x} \implies \dot{\mathbf{x}} = (A + BK)\mathbf{x}$$

Kalman controllability matrix

$$\begin{pmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{pmatrix}$$

Input transformation for the error dynamics

$$w_1 = v_d \cos e_3 - v, \qquad w_2 = \omega_d - \omega$$

Error dynamics with transformed inputs

$$\dot{\mathbf{e}} = \begin{pmatrix} \omega_d e_2 \\ -\omega_d e_1 + v_d \sin e_3 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & -e_2 \\ 0 & e_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

Linearized error dynamics

$$A(t) = \begin{pmatrix} 0 & \omega_d & 0 \\ -\omega_d & 0 & v_d \\ 0 & 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Linear feedback gain matrix

$$\mathbf{w} = K\mathbf{e}, \qquad K = \begin{pmatrix} -k_1 & 0 & 0 \\ 0 & -k_2 & -k_3 \end{pmatrix}$$

Closed-loop error dynamics

$$\dot{\mathbf{e}} = A_{cl}(t)\mathbf{e} = \begin{pmatrix} -k_1 & \omega_d & 0 \\ -\omega_d & 0 & v_d \\ 0 & -k_2 & -k_3 \end{pmatrix} \mathbf{e}$$

Gains for approximate linearization

$$k_1 = k_3 = 2\zeta a, \qquad k_2 = \frac{a^2 - \omega_d^2}{v_d}, \qquad a > 0,\ \zeta \in (0,1)$$

Characteristic polynomial of the closed loop

$$p(\lambda) = (\lambda + 2\zeta a)(\lambda^2 + 2\zeta a\lambda + a^2)$$

Trajectory tracking control law (approximate linearization)

$$\begin{aligned} v &= v_d \cos e_3 + k_1 e_1 \\ \omega &= \omega_d + k_2 e_2 + k_3 e_3 \end{aligned}$$

Trajectory tracking control law (global stability)

$$\begin{aligned} w_1 &= -k_1(v_d, \omega_d)\, e_1 \\ w_2 &= -k_2\, v_d \frac{\sin e_3}{e_3}\, e_2 - k_3(v_d, \omega_d)\, e_3 \end{aligned}$$

Input-output linearization: decoupling matrix

$$\dot{\mathbf{y}} = \frac{\partial h}{\partial \mathbf{x}} G(\mathbf{x})\mathbf{u} = T(\mathbf{x})\mathbf{u}$$

Input-output linearization: linearizing feedback

$$\mathbf{u} = T^{-1}(\mathbf{x})\mathbf{w} \implies \dot{\mathbf{y}} = \mathbf{w}$$

Input-output linearization: output error dynamics

$$\mathbf{e} = \mathbf{y}_d - \mathbf{y}, \qquad \mathbf{w} = \dot{\mathbf{y}}_d + K\mathbf{e}, \qquad \dot{\mathbf{e}} = -K\mathbf{e}$$

Input-output linearization: control law

$$\mathbf{u} = T^{-1}(\mathbf{x})(\dot{\mathbf{y}}_d + K\mathbf{e})$$

Static feedback: displaced output point

$$\begin{pmatrix} x_B \\ y_B \end{pmatrix} = \begin{pmatrix} x + b\cos\theta \\ y + b\sin\theta \end{pmatrix}$$

Static feedback: decoupling matrix

$$\begin{pmatrix} \dot{x}_B \\ \dot{y}_B \end{pmatrix} = \begin{pmatrix} \cos\theta & -b\sin\theta \\ \sin\theta & b\cos\theta \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} = T(\theta) \begin{pmatrix} v \\ \omega \end{pmatrix}, \qquad \det T = b$$

Static feedback: linearizing input transformation

$$\begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\dfrac{\sin\theta}{b} & \dfrac{\cos\theta}{b} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

Static feedback: linearized system and zero dynamics

$$\dot{x}_B = w_1, \qquad \dot{y}_B = w_2, \qquad \dot{\theta} = \frac{w_2\cos\theta - w_1\sin\theta}{b}$$

Static feedback: control law

$$\begin{aligned} w_1 &= \dot{x}_d + k_x(x_d - x_B) \\ w_2 &= \dot{y}_d + k_y(y_d - y_B) \end{aligned}$$

Dynamic feedback: dynamic extension

$$\dot{v} = a$$

Dynamic feedback: second derivative of the output

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} \cos\theta & -v\sin\theta \\ \sin\theta & v\cos\theta \end{pmatrix} \begin{pmatrix} a \\ \omega \end{pmatrix} = T(v,\theta) \begin{pmatrix} a \\ \omega \end{pmatrix}, \qquad \det T = v$$

Dynamic feedback: linearizing input transformation

$$\begin{pmatrix} a \\ \omega \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\dfrac{\sin\theta}{v} & \dfrac{\cos\theta}{v} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

Dynamic feedback: control law

$$\begin{aligned} w_1 &= \ddot{x}_d + k_{p,x}(x_d - x) + k_{d,x}(\dot{x}_d - \dot{x}) \\ w_2 &= \ddot{y}_d + k_{p,y}(y_d - y) + k_{d,y}(\dot{y}_d - \dot{y}) \end{aligned}$$

Lyapunov stability theorem

$$\dot{V} < 0 \implies \text{asymptotically stable}, \qquad \dot{V} \leq 0 \implies \text{stable}$$

Barbalat's lemma

$$\ddot{V} \text{ bounded} \implies \lim_{t\to\infty} \dot{V}(t) = 0$$

Cartesian error and projection on the sagittal axis

$$\mathbf{e}_p = \begin{pmatrix} -x \\ -y \end{pmatrix}, \qquad \mathbf{e}_p^\top \mathbf{n} = -x\cos\theta - y\sin\theta, \qquad \mathbf{n} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Pointing error

$$\gamma = \text{atan2}(y, x) + \pi - \theta$$

Regulation of the Cartesian position: control law

$$\begin{aligned} v &= -k_1(x\cos\theta + y\sin\theta) = k_1 \mathbf{e}_p^\top \mathbf{n} \\ \omega &= k_2\gamma \end{aligned}$$

Regulation of the Cartesian position: Lyapunov function

$$V = \frac{1}{2}(x^2 + y^2), \qquad \dot{V} = -k_1(x\cos\theta + y\sin\theta)^2 = -k_1(\mathbf{e}_p^\top \mathbf{n})^2$$

Polar coordinates

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \\ \gamma &= \text{atan2}(y, x) + \pi - \theta \\ \delta &= \gamma + \theta \end{aligned}$$

Unicycle kinematic model in polar coordinates

$$\begin{aligned} \dot{\rho} &= -v\cos\gamma \\ \dot{\gamma} &= \frac{\sin\gamma}{\rho}v - \omega \\ \dot{\delta} &= \frac{\sin\gamma}{\rho}v \end{aligned}$$

Regulation of the whole configuration: Lyapunov function

$$V = \frac{1}{2}(\rho^2 + \gamma^2 + \delta^2), \qquad \dot{V} = -k_1\rho^2\cos^2\gamma - k_2\gamma^2$$

Regulation of the whole configuration: control law

$$\begin{aligned} v &= k_1\rho\cos\gamma \\ \omega &= k_2\gamma + k_1\frac{\sin\gamma\cos\gamma}{\gamma}(\gamma + \delta) \end{aligned}$$

Kinematic model of a mobile manipulator

$$\dot{\mathbf{q}} = \begin{pmatrix} \dot{\mathbf{q}}_b \\ \dot{\mathbf{q}}_m \end{pmatrix} = \begin{pmatrix} G_b(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \begin{pmatrix} \mathbf{u}_b \\ \mathbf{u}_m \end{pmatrix} = G(\mathbf{q})\mathbf{u}$$

End-effector differential kinematics

$$\dot{\mathbf{y}}_e = J_b\dot{\mathbf{q}}_b + J_m\dot{\mathbf{q}}_m, \qquad J_b = \frac{\partial k_e}{\partial \mathbf{q}_b}, \quad J_m = \frac{\partial k_e}{\partial \mathbf{q}_m}$$

Jacobian of the mobile manipulator

$$\dot{\mathbf{y}}_e = \begin{pmatrix} J_b(\mathbf{q})G_b(\mathbf{q}_b) & J_m(\mathbf{q}) \end{pmatrix} \begin{pmatrix} \mathbf{u}_b \\ \mathbf{u}_m \end{pmatrix} = J_{mm}(\mathbf{q})\mathbf{u}$$

2R planar manipulator on a unicycle: end-effector position

$$\begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} x + d\cos\theta + a_1\cos(\theta + q_1) + a_2\cos(\theta + q_1 + q_2) \\ y + d\sin\theta + a_1\sin(\theta + q_1) + a_2\sin(\theta + q_1 + q_2) \end{pmatrix}$$

2R planar manipulator on a unicycle: Jacobians

$$J_b = \begin{pmatrix} 1 & 0 & -ds_\theta - a_1 s_{\theta 1} - a_2 s_{\theta 12} \\ 0 & 1 & dc_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} \end{pmatrix}$$
$$J_m = \begin{pmatrix} -a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_2 s_{\theta 12} \\ a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_2 c_{\theta 12} \end{pmatrix}$$

2R planar manipulator on a unicycle: mobile manipulator Jacobian

$$J_{mm} = \begin{pmatrix} c_\theta & -ds_\theta - a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_1 s_{\theta 1} - a_2 s_{\theta 12} & -a_2 s_{\theta 12} \\ s_\theta & dc_\theta + a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_1 c_{\theta 1} + a_2 c_{\theta 12} & a_2 c_{\theta 12} \end{pmatrix}$$

Pseudo-inverse (full row rank)

$$J_{mm}^{\#} = J_{mm}^\top (J_{mm} J_{mm}^\top)^{-1}$$

Inverse differential kinematics (minimum norm)

$$\mathbf{u} = J_{mm}^{\#} \dot{\mathbf{y}}_d$$

Null-space projector

$$\Pi = I - J_{mm}^{\#} J_{mm}, \qquad J_{mm}\Pi\mathbf{u}_N = \mathbf{0}$$

Inverse differential kinematics with null-space term

$$\mathbf{u} = J_{mm}^{\#}\dot{\mathbf{y}}_d + (I - J_{mm}^{\#} J_{mm})\mathbf{u}_N$$

Time derivative of the performance criterion

$$\dot{H}(\mathbf{q}) = (\nabla H(\mathbf{q}))^\top G(\mathbf{q})\mathbf{u}$$

Null-space direction compatible with the constraints

$$\mathbf{u}_N = \pm\alpha G^\top(\mathbf{q})\nabla H(\mathbf{q}) = \pm\alpha \begin{pmatrix} G_b^\top(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \nabla H(\mathbf{q})$$

Mobile manipulator control law

$$\mathbf{u} = J_{mm}^{\#}\left(\dot{\mathbf{y}}_d + K(\mathbf{y}_d - \mathbf{y}_e)\right) \pm \alpha(I - J_{mm}^{\#} J_{mm}) \begin{pmatrix} G_b^\top(\mathbf{q}_b) & \mathcal{O} \\ \mathcal{O} & I \end{pmatrix} \nabla H(\mathbf{q})$$

Euler integration (general)

$$\mathbf{q}_{k+1} = \mathbf{q}_k + f(\mathbf{q}_k, \mathbf{u}_k)T_s$$

Euler integration (unicycle)

$$x_{k+1} = x_k + v_k T_s \cos\theta_k$$
$$y_{k+1} = y_k + v_k T_s \sin\theta_k$$
$$\theta_{k+1} = \theta_k + \omega_k T_s$$

Runge-Kutta integration (unicycle)

$$\begin{aligned} x_{k+1} &= x_k + v_k T_s \cos\left(\theta_k + \frac{\omega_k T_s}{2}\right) \\ y_{k+1} &= y_k + v_k T_s \sin\left(\theta_k + \frac{\omega_k T_s}{2}\right) \\ \theta_{k+1} &= \theta_k + \omega_k T_s \end{aligned}$$

Exact integration (unicycle)

$$\begin{aligned} x_{k+1} &= x_k + \frac{v_k}{\omega_k}\left(\sin(\theta_k + \omega_k T_s) - \sin\theta_k\right) \\ y_{k+1} &= y_k - \frac{v_k}{\omega_k}\left(\cos(\theta_k + \omega_k T_s) - \cos\theta_k\right) \\ \theta_{k+1} &= \theta_k + \omega_k T_s \end{aligned}$$

Radius of the circular arc

$$R = \frac{v_k}{\omega_k}$$

Odometry from encoders (differential drive)

$$v_k T_s = \Delta s = \frac{r}{2}(\Delta\phi_R + \Delta\phi_L), \qquad \omega_k T_s = \Delta\theta = \frac{r}{d}(\Delta\phi_R - \Delta\phi_L)$$

Expected value

$$\bar{\mathbf{X}} = \mathbb{E}(\mathbf{X}) = \int_{\mathbb{R}^n} \mathbf{x} f_X(\mathbf{x}) d\mathbf{x}$$

Covariance matrix

$$P_X = \mathbb{E}\left((\mathbf{X} - \bar{\mathbf{X}})(\mathbf{X} - \bar{\mathbf{X}})^\top\right)$$

Multivariate Gaussian distribution

$$f_X(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det P_X}} \exp\left(-\frac{1}{2}(\mathbf{x} - \bar{\mathbf{X}})^\top P_X^{-1}(\mathbf{x} - \bar{\mathbf{X}})\right)$$

Linear discrete-time system

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k, \qquad \mathbf{y}_k = C_k \mathbf{x}_k$$

Innovation

$$\boldsymbol{\nu}_{k+1} = \mathbf{y}_{k+1} - C_{k+1}\widehat{\mathbf{x}}_{k+1|k}$$

Zero-noise Kalman filter (two-step observer)

$$\begin{aligned} \widehat{\mathbf{x}}_{k+1|k} &= A_k \widehat{\mathbf{x}}_k + B_k \mathbf{u}_k \\ \widehat{\mathbf{x}}_{k+1} &= \widehat{\mathbf{x}}_{k+1|k} + C_{k+1}^{\#} \boldsymbol{\nu}_{k+1} \end{aligned}$$

System with noise on the dynamics

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k + \mathbf{v}_k, \qquad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, V_k)$$

Covariance prediction

$$P_{k+1|k} = A_k P_k A_k^\top + V_k$$

Maximum likelihood correction (weighted pseudo-inverse)

$$\Delta\mathbf{x} = C_{k+1,M}^{\#}\boldsymbol{\nu}_{k+1} = P_{k+1|k}C_{k+1}^\top(C_{k+1}P_{k+1|k}C_{k+1}^\top)^{-1}\boldsymbol{\nu}_{k+1}$$

Kalman filter with noise on the dynamics only

$$\begin{aligned}
\widehat{\mathbf{x}}_{k+1|k} &= A_k\widehat{\mathbf{x}}_k + B_k\mathbf{u}_k \\
P_{k+1|k} &= A_k P_k A_k^\top + V_k \\
\widehat{\mathbf{x}}_{k+1} &= \widehat{\mathbf{x}}_{k+1|k} + C_{k+1,M}^{\#}\boldsymbol{\nu}_{k+1} \\
P_{k+1} &= P_{k+1|k} - C_{k+1,M}^{\#}C_{k+1}P_{k+1|k}
\end{aligned}$$

System with noise on dynamics and observations

$$\mathbf{x}_{k+1} = A_k\mathbf{x}_k + B_k\mathbf{u}_k + \mathbf{v}_k, \qquad \mathbf{y}_k = C_k\mathbf{x}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, W_k)$$

Kalman gain matrix

$$R_{k+1} = P_{k+1|k}C_{k+1}^\top(C_{k+1}P_{k+1|k}C_{k+1}^\top + W_{k+1})^{-1}$$

Kalman filter

$$\begin{aligned}
\widehat{\mathbf{x}}_{k+1|k} &= A_k\widehat{\mathbf{x}}_k + B_k\mathbf{u}_k \\
P_{k+1|k} &= A_k P_k A_k^\top + V_k \\
R_{k+1} &= P_{k+1|k}C_{k+1}^\top(C_{k+1}P_{k+1|k}C_{k+1}^\top + W_{k+1})^{-1} \\
\widehat{\mathbf{x}}_{k+1} &= \widehat{\mathbf{x}}_{k+1|k} + R_{k+1}\boldsymbol{\nu}_{k+1} \\
P_{k+1} &= P_{k+1|k} - R_{k+1}C_{k+1}P_{k+1|k}
\end{aligned}$$

Optimality of the Kalman filter

$$\widehat{\mathbf{x}}_{k+1} = \arg\min\ \mathbb{E}\left(\|\mathbf{x}_{k+1} - \widehat{\mathbf{x}}_{k+1}\|^2\right)$$

Nonlinear system for the extended Kalman filter

$$\mathbf{x}_{k+1} = f_k(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k, \qquad \mathbf{y}_k = h_k(\mathbf{x}_k) + \mathbf{w}_k$$

Jacobians of the extended Kalman filter

$$F_k = \left.\frac{\partial f_k}{\partial \mathbf{x}}\right|_{\mathbf{x}=\widehat{\mathbf{x}}_k,\,\mathbf{u}=\mathbf{u}_k}, \qquad H_{k+1} = \left.\frac{\partial h_{k+1}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\widehat{\mathbf{x}}_{k+1|k}}$$

Extended Kalman filter

$$\begin{aligned}
\widehat{\mathbf{x}}_{k+1|k} &= f_k(\widehat{\mathbf{x}}_k, \mathbf{u}_k) \\
P_{k+1|k} &= F_k P_k F_k^\top + V_k \\
\boldsymbol{\nu}_{k+1} &= \mathbf{y}_{k+1} - h_{k+1}(\widehat{\mathbf{x}}_{k+1|k}) \\
R_{k+1} &= P_{k+1|k}H_{k+1}^\top(H_{k+1}P_{k+1|k}H_{k+1}^\top + W_{k+1})^{-1} \\
\widehat{\mathbf{x}}_{k+1} &= \widehat{\mathbf{x}}_{k+1|k} + R_{k+1}\boldsymbol{\nu}_{k+1} \\
P_{k+1} &= P_{k+1|k} - R_{k+1}H_{k+1}P_{k+1|k}
\end{aligned}$$

Unicycle model with noise (landmark-based localization)

$$\begin{aligned} x_{k+1} &= x_k + v_k T_s \cos\theta_k + v_{1k} \\ y_{k+1} &= y_k + v_k T_s \sin\theta_k + v_{2k} \\ \theta_{k+1} &= \theta_k + \omega_k T_s + v_{3k} \end{aligned}$$

Landmark measurement (range and bearing)

$$\mathbf{h}_i(\mathbf{q}_k, a(i)) = \begin{pmatrix} \sqrt{(x_k - x_{l\,a(i)})^2 + (y_k - y_{l\,a(i)})^2} \\ \text{atan2}\left(y_{l\,a(i)} - y_k,\, x_{l\,a(i)} - x_k\right) - \theta_k \end{pmatrix}$$

Measurement vector

$$\mathbf{y}_k = \begin{pmatrix} \mathbf{h}_1(\mathbf{q}_k, a(1)) \\ \vdots \\ \mathbf{h}_{L_k}(\mathbf{q}_k, a(L_k)) \end{pmatrix} + \begin{pmatrix} \mathbf{w}_{1k} \\ \vdots \\ \mathbf{w}_{L_k k} \end{pmatrix}$$

Jacobian of the unicycle dynamics

$$F_k = \begin{pmatrix} 1 & 0 & -v_k T_s \sin\widehat{\theta}_k \\ 0 & 1 & v_k T_s \cos\widehat{\theta}_k \\ 0 & 0 & 1 \end{pmatrix}$$

Jacobian of the measurement function

$$\left.\frac{\partial \mathbf{h}_i}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} = \begin{pmatrix} \dfrac{\widehat{x}_{k+1|k} - x_{l\,a(i)}}{\sqrt{(\widehat{x}_{k+1|k} - x_{l\,a(i)})^2 + (\widehat{y}_{k+1|k} - y_{l\,a(i)})^2}} & \dfrac{\widehat{y}_{k+1|k} - y_{l\,a(i)}}{\sqrt{(\widehat{x}_{k+1|k} - x_{l\,a(i)})^2 + (\widehat{y}_{k+1|k} - y_{l\,a(i)})^2}} & 0 \\ \dfrac{-(\widehat{y}_{k+1|k} - y_{l\,a(i)})}{(\widehat{x}_{k+1|k} - x_{l\,a(i)})^2 + (\widehat{y}_{k+1|k} - y_{l\,a(i)})^2} & \dfrac{\widehat{x}_{k+1|k} - x_{l\,a(i)}}{(\widehat{x}_{k+1|k} - x_{l\,a(i)})^2 + (\widehat{y}_{k+1|k} - y_{l\,a(i)})^2} & -1 \end{pmatrix}$$

Stacked Jacobian of the measurements

$$H_{k+1} = \begin{pmatrix} \left.\dfrac{\partial \mathbf{h}_1}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} \\ \vdots \\ \left.\dfrac{\partial \mathbf{h}_{L_k}}{\partial \mathbf{q}}\right|_{\mathbf{q}=\widehat{\mathbf{q}}_{k+1|k}} \end{pmatrix} \in \mathbb{R}^{2L_k \times 3}$$

Candidate innovation (data association)

$$\boldsymbol{\nu}_{ij} = \mathbf{y}_{i\,k+1} - \mathbf{h}_i(\widehat{\mathbf{q}}_{k+1|k}, j)$$

Innovation covariance (data association)

$$S_{ij} = H_{k+1}(i,j) P_{k+1|k} H_{k+1}(i,j)^\top + W_{k+1}(i)$$

Data association criterion

$$j^* = \underset{1 \le j \le L}{\arg\min}\, \boldsymbol{\nu}_{ij}^\top S_{ij}^{-1} \boldsymbol{\nu}_{ij}$$

Innovation for localization on a map

$$\boldsymbol{\nu}_{k+1} = \mathbf{y}_{k+1} - \mathbf{h}(\widehat{\mathbf{q}}_{k+1|k}, \mathcal{M})$$

SLAM: extended state vector

$$\mathbf{x} = \begin{pmatrix} x & y & x_{l\,1} & y_{l\,1} & \dots & x_{l\,L} & y_{l\,L} \end{pmatrix}^\top \in \mathbb{R}^{2L+2}$$

SLAM: motion model

$$\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k + \mathbf{v}_k, \qquad A = I_{2L+2}, \qquad B = T_s \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}$$

SLAM: process noise covariance

$$V_k = \begin{pmatrix} V_{xy,k} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

SLAM: measurement of the i-th landmark

$$\mathbf{y}_{i,k} = \begin{pmatrix} x_{l\,i} - x_k \\ y_{l\,i} - y_k \end{pmatrix} + \mathbf{w}_{i,k} = C_i\mathbf{x}_k + \mathbf{w}_{i,k}, \qquad C_i = \begin{pmatrix} -I_2 & \mathbf{0} & \dots & I_2 & \dots & \mathbf{0} \end{pmatrix}$$

SLAM: stacked output equation

$$\mathbf{y}_k = C\mathbf{x}_k + \mathbf{w}_k, \qquad C = \begin{pmatrix} C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad \mathbf{w}_k = \begin{pmatrix} \mathbf{w}_{1,k} \\ \vdots \\ \mathbf{w}_{L,k} \end{pmatrix}$$

SLAM: measurement noise covariance

$$W_k = \begin{pmatrix} W_{1,k} & 0 & \dots & 0 \\ 0 & W_{2,k} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & W_{L,k} \end{pmatrix}$$